

# **AI AND FEA BASED DESIGN AND ANALYSIS OF EFFICIENT EV MOTOR**

## **A PROJECT REPORT**

*submitted by*

**CB.EN.U4ELC22009      DHARANI MALAR V.  
CB.EN.U4ELC22020      KARTHIGHA S.**

*in partial fulfillment for the award of the degree  
of*

## **BACHELOR OF TECHNOLOGY IN ELECTRICAL AND COMPUTER ENGINEERING**



**AMRITA SCHOOL OF ENGINEERING, COIMBATORE**

**AMRITA VISHWA VIDYAPEETHAM  
COIMBATORE- 641112  
March 2026**

**AMRITA VISHWA VIDYAPEETHAM**  
**AMRITA SCHOOL OF ENGINEERING, COIMBATORE,**  
**641112**



**BONAFIDE CERTIFICATE**

This is to certify that the project report entitled, **AI AND FEA BASED DESIGN AND ANALYSIS OF EFFICIENT EV MOTOR** submitted by

**CB.EN.U4ELC22009 DHARANI MALAR V.**

**CB.EN.U4ELC22020 KARTHIGA S.**

In partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in ELECTRICAL & COMPUTER ENGINEERING** is a Bonafide record of the work carried out under my guidance and supervision at Amrita School of Engineering, Coimbatore.

Dr. R Resmi.  
Assistant Professor (Selection Grade),  
Department of EEE  
Amrita School of Engineering, Coimbatore.

Dr. Balamurugan S.  
CHAIRPERSON  
Department of EEE,  
Amrita School of Engineering, Coimbatore

This project report was evaluated by us on 23.03.2026.

INTERNAL EXAMINER

EXTERNAL EXAMINER

## **ACKNOWLEDGEMENT**

We would like to express our sincere gratitude to Dr. R. Resmi, Assistant Professor, Department of Electrical and Electronics Engineering, Amrita School of Engineering, for her invaluable guidance and support throughout the course of our project.

We are also grateful to Dr. Balamurugan, Chairperson, Department of Electrical and Electronics Engineering, for providing us with the opportunity and encouragement to successfully carry out this project.

Finally, we express our gratitude to the staff of the Power Electronics Lab, Department of Electrical and Electronics Engineering, Amrita Vishwa Vidyapeetham, for providing the laboratory facilities and resources that contributed to the successful completion of this project.

## ABSTRACT

The increasing demand for sustainable electric vehicles (EVs) requires motors that provide high efficiency, reduced torque ripple, and cost-effectiveness while minimizing dependence on rare-earth materials. This project focuses on the design and optimization of a Synchronous Reluctance Motor (SynRM) for EV propulsion using a hybrid Artificial Intelligence (AI) and Finite Element Analysis (FEA) approach. The baseline motor geometry is developed using ANSYS RMxprt and validated through 2D/3D FEA in ANSYS Maxwell to evaluate electromagnetic and thermal performance. A Random Forest (RF) regression model is trained using datasets generated from FEA simulations to serve as a surrogate model for predicting key performance metrics such as average torque, torque ripple, efficiency, and total harmonic distortion (THD), and is integrated with a Genetic Algorithm (GA) to optimize the rotor geometry with the objective of maximizing torque and efficiency while minimizing torque ripple and THD. Quantitative analysis shows that the baseline Synchronous Reluctance Motor exhibits a torque ripple of 7.5%, while the Induction Motor and Brushless DC Motor show torque ripples of 4.57% and 15.34%, respectively. After applying the RF–GA optimization, the torque ripple of the SynRM is reduced to 5.2%, with the Random Forest model predicting a value of 5.12%, demonstrating good agreement between predicted and calculated results. The proposed RF–GA optimization approach significantly reduces computational time compared to conventional iterative FEA-based methods while improving torque characteristics and electromagnetic performance, highlighting the optimized SynRM as a promising cost-effective and magnet-free alternative to Permanent Magnet Synchronous Motors (PMSMs) and Induction Motors (IMs) for next-generation EV applications.

# TABLE OF CONTENTS

| CHAPTER N0 | TITLE                                                     | PAGE N0 |
|------------|-----------------------------------------------------------|---------|
|            | LIST OF FIGURES                                           | i       |
|            | LIST OF TABLES                                            | iii     |
|            | LIST OF SYMBOLS                                           | v       |
|            | LIST OF ABBREVIATIONS                                     | vi      |
| 1          | INTRODUCTION                                              | 1       |
| 1.1        | PREFACE                                                   | 1       |
| 1.2        | OBJECTIVE                                                 | 1       |
| 1.3        | ORGANISATION OF THE PROJECT                               | 2       |
| 2          | LITERATURE REVIEW                                         | 3       |
| 2.1        | INTRODUCTION                                              | 3       |
| 2.2        | LITERATURE SURVEY                                         | 3       |
| 2.3        | OPTIMIZATION METHODS IN MOTOR DESIGN                      | 5       |
|            | 2.3.1 RANDOM FOREST (RF) SURROGATE MODELING               |         |
|            | 2.3.2 GENETIC ALGORITHM (GA) FOR ROTOR OPTIMIZATION       | 5       |
| 2.4        | COMBINED RF–GA OPTIMIZATION TECHNIQUES                    | 8       |
| 2.5        | RESEARCH GAP                                              | 9       |
| 2.6        | DRAWBACKS OF CONVENTIONAL MOTORS AND MOTIVATION FOR SYNRM |         |
| 2.7        | SUMMARY                                                   | 11      |
| 3          | METHODOLOGY                                               | 11      |
| 3.1        | INTRODUCTION                                              | 12      |
| 3.2        | DESIGN OF INDUCTION MOTOR                                 | 12      |
|            | 3.2.1 INDUCTION MOTOR MODEL DESIGN                        | 12      |
| 3.3        | DESIGN OF BRUSHLESS DC MOTOR                              | 19      |
|            | 3.3.1 BRUSHLESS DC MOTOR MODEL DESIGN                     |         |
| 3.4        | DESIGN OF SYNRM MOTOR                                     |         |
|            | 3.4.1 SYNRM RMXPR T DESIGN                                | 25      |
|            | 3.4.2 MATHEMATICAL MODELLING OF SYNRM MOTOR               | 29      |
| 3.5        | SUMMARY                                                   | 36      |
| 4          | 2D MAXWELL DESIGN OF MOTOR                                | 36      |
| 4.1        | INTRODUCTION                                              | 36      |
| 4.2        | INDUCTION MOTOR 2D MAXWELL DESIGN                         | 37      |
|            | 4.2.1 ANALYSIS AND RESULTS                                | 40      |
|            | 4.2.2 MAGNETIC FLUX DENSITY DISTRIBUTION                  | 41      |
|            | 4.2.3 MAGNETIC FLUX LINE DISTRIBUTION                     | 42      |
|            | 4.3.4 AIR GAP FLUX CHARACTERISTICS                        | 43      |
|            | 4.2.5 OBSERVATIONS                                        | 43      |
| 4.3        | BRUSHLESS DC MOTOR 2D MAXWELL DESIGN                      | 44      |
|            | 4.3.1 ANALYSIS AND RESULTS                                |         |

|     |                                                                      |    |
|-----|----------------------------------------------------------------------|----|
|     | 4.3.2 MAGNETIC FLUX DENSITY DISTRIBUTION                             | 47 |
|     | 4.3.3 MAGNETIC FLUX LINE DISTRIBUTION                                | 48 |
|     | 4.3.4 AIR GAP FLUX CHARACTERISTICS                                   | 49 |
|     | 4.3.5 OBSERVATIONS                                                   | 49 |
| 4.4 | SYNCHRONOUS RELUCTANCE MOTOR 2D MAXWELL DESIGN                       | 50 |
|     | 4.4.1 ANALYSIS AND RESULTS                                           | 51 |
|     | 4.4.2 MAGNETIC FLUX DENSITY DISTRIBUTION                             | 53 |
|     | 4.4.3 MAGNETIC FLUX LINE DISTRIBUTION                                | 54 |
|     | 4.4.4 AIR GAP FLUX CHARACTERISTICS                                   | 55 |
|     | 4.4.5 OBSERVATIONS                                                   | 55 |
| 4.5 | TORQUE RIPPLE CALCULATION AND COMPARISON BETWEEN IM,BLDC AND SYNRM   | 56 |
|     | 4.5.1 TORQUE RIPPLE CALCULATION                                      | 56 |
|     | 4.5.2 TORQUE RIPPLE COMPARISON                                       | 58 |
| 4.6 | DATASET GENERATION                                                   | 59 |
|     | 4.6.1 INPUT PARAMETERS                                               | 59 |
|     | 4.6.2 INPUT PARAMETERS EXTRACTION LATIN HYPERCUBE SAMPLING           | 62 |
|     | 4.6.3 RANGE OF INPUT PARAMETERS TAKEN USING LATIN HYPERCUBE SAMPLING | 62 |
|     | 4.6.4 OUTPUT PARAMETERS                                              | 64 |
|     | 4.6.5 OUTPUT PARAMETERS EXTRACTION                                   | 64 |
| 4.7 | SUMMARY                                                              | 65 |
| 5   | IMPLEMENTATION OF RF AND GA                                          | 65 |
| 5.1 | INTRODUCTION                                                         | 67 |
| 5.2 | RANDOM FOREST BASED SURROGATE MODEL                                  | 67 |
|     | 5.2.1 PURPOSE OF USING RANDOM FOREST                                 | 67 |
|     | 5.2.2 LIBRARIES AND TOOLS USED                                       | 68 |
|     | 5.2.3 DATASET DESCRIPTION                                            | 68 |
|     | 5.2.4 RANDOM FOREST MODEL TRAINING-HYPERPARAMETERS AND DESCRIPTION   | 68 |
|     | 5.2.5 RANDOM FOREST MODEL VALIDATION AND PERFORMANCE EVALUATION      | 70 |
|     | 5.2.6 FEATURE IMPORTANCE                                             | 71 |
| 5.3 | GENETIC ALGORITHM BASED OPTIMIZATION                                 | 73 |
|     | 5.3.1 GENETIC ALGORITHM BASED OPTIMIZATION OF SYNRM                  | 73 |
|     | 5.3.2 DESIGN VARIABLES AND SEARCH SPACE                              | 73 |
|     | 5.3.3 FITNESS FUNCTION AND CONSTRAINTS                               | 73 |
|     | 5.3.4 GENETIC ALGORITHM HYPERPARAMETERS                              | 74 |
|     | 5.3.5 OPTIMIZATION PROCEDURE                                         | 74 |
|     | 5.3.6 FITNESS FUNCTION FORMULATION AND GENETIC OPERATION             | 74 |

|     |                                                                                 |    |
|-----|---------------------------------------------------------------------------------|----|
|     | 5.3.7 EVOLUTIONARY PROCESS AND OPTIMAL<br>DESIGN EXTRACTION                     | 75 |
|     | 5.3.8 OPTIMIZED ROTOR DESIGN RESULTS                                            | 75 |
|     | 5.3.9 VALIDATION OF OPTIMIZED ROTOR<br>USING TIME-DOMAIN SIMULATION             | 76 |
|     | 5.3.10 TORQUE RIPPLE CALCULATION                                                | 77 |
|     | 5.3.11 COMPARISON BETWEEN RANDOM<br>FOREST PREDICTION AND SIMULATION<br>RESULTS | 78 |
| 5.4 | SUMMARY                                                                         | 79 |
| 6   | VALIDATION OF OPTIMIZED SYNRM USING FINITE<br>ELEMENT SIMULATION                | 80 |
| 6.1 | INTRODUCTION                                                                    | 80 |
| 6.2 | RMXPRT DESIGN OF OPTIMIZED SYNRM                                                | 80 |
| 6.3 | 2D MAXWELL DESIGN OF OPTIMIZED SYNRM                                            | 82 |
|     | 6.3.1 ANALYSIS AND RESULTS                                                      | 83 |
| 6.4 | OPTIMIZED SYNRM MAGNETIC FIELD ANALYSIS                                         | 83 |
|     | 6.4.1 MAGNETIC FLUX DENSITY DISTRIBUTION                                        | 84 |
|     | 6.4.2 MAGNETIC FLUX LINE DISTRIBUTION                                           | 84 |
|     | 6.4.3 AIR GAP FLUX CHARACTERISTICS                                              | 85 |
|     | 6.4.4 OBSERVATIONS                                                              | 85 |
| 6.5 | SUMMARY                                                                         | 86 |
| 7   | CONCLUSION                                                                      | 87 |

## LIST OF FIGURES

| <b>FIGURE<br/>NO.</b> | <b>FIGURE CAPTION</b>                                        | <b>PAGE<br/>NO.</b> |
|-----------------------|--------------------------------------------------------------|---------------------|
| 3.1                   | Block Diagram of the Proposed Model                          | 12                  |
| 3.2                   | Designed RMxpvt Induction Motor Model                        | 15                  |
| 3.3                   | Efficiency vs Speed graph                                    | 16                  |
| 3.4                   | Torque vs Speed graph                                        | 16                  |
| 3.5                   | Efficiency vs Output power graph                             | 17                  |
| 3.6                   | Torque vs Output Power                                       | 18                  |
| 3.7                   | Designed RMxpvt BLDC Motor Model                             | 21                  |
| 3.8                   | Efficiency vs Speed                                          | 22                  |
| 3.9                   | Output power vs Speed                                        | 22                  |
| 3.10                  | Output Torque vs Speed                                       | 23                  |
| 3.11                  | Winding Currents under Load                                  | 24                  |
| 3.12                  | Designed SynRM Motor Model                                   | 27                  |
| 3.13                  | Efficiency vs Torque angle graph for Designed<br>SynRM Motor | 28                  |
| 3.14                  | SIMULINK Modeling and Simulation of a SynRM<br>Motor         | 32                  |
| 3.15                  | Dq currents Subsystem                                        | 33                  |
| 3.16                  | Electromagnetic Torque Subsystem                             | 33                  |
| 3.17                  | Mechanical System Subsystem                                  | 34                  |
| 3.18                  | Speed of Rotor (rps) vs Time (sec)                           | 34                  |
| 4.1                   | Induction Motor 2D Maxwell Design                            | 37                  |
| 4.2                   | Torque vs Time                                               | 37                  |
| 4.3                   | Current vs Time                                              | 38                  |
| 4.4                   | Flux linkage vs Time                                         | 39                  |
| 4.5                   | Induced voltage vs Time                                      | 39                  |



|      |                                                                |    |
|------|----------------------------------------------------------------|----|
| 4.6  | Magnetic Flux Density Distribution plot for IM                 | 41 |
| 4.7  | Magnetic Flux Line Distribution for IM                         | 42 |
| 4.8  | Brushless DC Motor 2D Maxwell Design                           | 43 |
| 4.9  | Torque vs Time                                                 | 44 |
| 4.10 | Flux linkage vs Time                                           | 45 |
| 4.11 | Induced voltage vs Time                                        | 45 |
| 4.12 | Current vs Time                                                | 46 |
| 4.13 | Magnetic Flux Density Distribution plot for BLDC               | 48 |
| 4.14 | Magnetic Flux Line Distribution for BLDC                       | 49 |
| 4.15 | Synchronous Reluctance Motor 2D Maxwell Design                 | 50 |
| 4.16 | Torque vs Time                                                 | 51 |
| 4.17 | Flux linkage vs Time                                           | 51 |
| 4.18 | Induced voltage vs Time                                        | 52 |
| 4.19 | Current vs Time                                                | 52 |
| 4.20 | Magnetic Flux Density Distribution plot for SynRM              | 54 |
| 4.21 | Magnetic Flux Line Distribution for SynRM                      | 55 |
| 5.1  | Feature Importance - Average Torque                            | 72 |
| 5.2  | Feature Importance - Torque Ripple                             | 72 |
| 5.3  | Torque vs Time graph for Optimized Motor                       | 77 |
| 6.1  | RMxpert Design of Optimized SynRM Motor                        | 81 |
| 6.2  | 2D Maxwell Design of SynRM Motor                               | 84 |
| 6.3  | Torque vs Time graph                                           | 84 |
| 6.4  | Magnetic Flux Density Distribution plot for<br>Optimized SynRM | 86 |
| 6.5  | Magnetic Flux Line Distribution plot for Optimized<br>SynRM    | 87 |

## LIST OF TABLES

| FIGURE<br>NO | TABLE TITLE                               | PAGE NO. |
|--------------|-------------------------------------------|----------|
| 3.1          | General Data of the Induction Motor used  | 13       |
| 3.2          | Rotor properties of Induction Motor       | 13       |
| 3.3          | Stator properties of Induction Motor      | 14       |
| 3.4          | Rated Magnetic Data of IM Motor           | 18       |
| 3.5          | No Load Operation Data of IM Motor        | 19       |
| 3.6          | General Data of the BLDC Motor used       | 19       |
| 3.7          | Rotor properties of BLDC Motor            | 20       |
| 3.8          | Stator properties of BLDC Motor           | 20       |
| 3.9          | Full Load Operation of BLDC Motor         | 24       |
| 3.10         | General Data of the SynRM used            | 25       |
| 3.11         | Rotor properties of SynRM Motor           | 26       |
| 3.12         | Stator properties of SynRM Motor          | 26       |
| 3.13         | Full Load Operation of SynRM Motor        | 28       |
| 3.14         | No Load Magnetic Variables of SynRM Motor | 28       |
| 4.1          | Torque Ripple Comparison                  | 58       |
| 4.2          | Range of Input parameters                 | 62       |

## LIST OF TABLES

| <b>FIGURE<br/>NO</b> | <b>TABLE TITLE</b>                                                   | <b>PAGE NO.</b> |
|----------------------|----------------------------------------------------------------------|-----------------|
| 5.1                  | Train-Test Split Configuration                                       | 69              |
| 5.2                  | Random Forest Regressor Hyperparameters                              | 69              |
| 5.3                  | Random Forest Performance Evaluation                                 | 70              |
| 5.4                  | Genetic Algorithm Hyperparameters                                    | 74              |
| 5.5                  | Optimized Rotor Design Parameters                                    | 76              |
| 5.6                  | Comparison between Random forest predicted and<br>Simulation results | 78              |
| 6.1                  | Rotor properties of SynRM Motor                                      | 81              |
| 6.2                  | Stator properties of SynRM Motor                                     | 82              |
| 6.3                  | Optimized Rotor Design Parameters                                    | 83              |

## LIST OF SYMBOLS

| SYMBOL     | DEFINITION                        |
|------------|-----------------------------------|
| $L_d$      | d-axis inductance                 |
| $L_q$      | q-axis inductance                 |
| $R_s$      | Stator resistance                 |
| $p$        | Number of rotor pole pairs        |
| $T_e$      | Rotor torque                      |
| $T_L$      | Load torque                       |
| $\omega_m$ | Rotor mechanical speed<br>(rad/s) |
| $J$        | Rotor Inertia                     |

## LIST OF ABBREVIATIONS

| ABBREVIATION | EXPANSION                                  |
|--------------|--------------------------------------------|
| EV           | Electric Vehicle                           |
| SynRM        | Synchronous Reluctance Motor               |
| PMSM         | Permanent Magnet Synchronous Motor         |
| BLDC         | Brushless Direct Current Motor             |
| IM           | Induction Motor                            |
| FEA          | Finite Element Analysis                    |
| RF           | Random Forest                              |
| GA           | Genetic Algorithm                          |
| THD          | Total Harmonic Distortion                  |
| AI           | Artificial Intelligence                    |
| EMF          | Electromotive Force                        |
| RMxpert      | Rapid Machine Expert (ANSYS design module) |

## Chapter 1

# INTRODUCTION

### 1.1 PREFACE

The rapid adoption of electric vehicles (EVs) has accelerated the demand for efficient, reliable, and cost-effective electric motors. Traditional EV motors, such as Permanent Magnet Synchronous Motors (PMSMs) and Brushless DC Motors (BLDCs), often rely heavily on rare-earth permanent magnets, leading to high manufacturing costs and supply chain challenges. The Synchronous Reluctance Motor (SynRM) has emerged as a promising alternative due to its magnet-free rotor structure, inherent robustness, high efficiency, and lower production cost.

This project integrates Finite Element Analysis (FEA) with Artificial Intelligence (AI) techniques to develop an optimized SynRM design suitable for EV propulsion. The proposed approach combines Random Forest (RF) surrogate modeling and a Genetic Algorithm (GA) to optimize rotor geometry for improved torque density, reduced torque ripple, and enhanced efficiency.

### 1.2 OBJECTIVE

The primary objective of this project is to design and optimize a Synchronous Reluctance Motor for electric vehicle applications using an AI-assisted FEA-based methodology. Specific objectives include:

- To model the initial SynRM using ANSYS RMxprt and perform electromagnetic analysis using ANSYS Maxwell.
- To develop a Random Forest surrogate model capable of accurately predicting performance parameters such as torque, efficiency, torque ripple, and THD.
- To implement a Genetic Algorithm that optimizes rotor geometry based on the surrogate model.
- To compare the optimized SynRM with standard PMSM and IM benchmarks in terms of efficiency, torque ripple, and cost-effectiveness.

## 1.3 ORGANISATION OF THE PROJECT

The report is structured as follows:

**Chapter 2 – Literature Review:** Discusses existing EV motor technologies, recent advancements in SynRM design, and the integration of AI-based optimization techniques.

**Chapter 3 – Proposed Solution and Methodology:** Describes the RF–GA- based optimization approach and outlines the modeling and simulation process.

**Chapter 4 – Results and Discussion:** Presents the simulation results, optimized design parameters, and performance comparison.

**Chapter 5 – Conclusion and Future Work:** Summarizes key findings and proposes directions for future research.

## Chapter 2

# LITERATURE REVIEW

## 2.1 INTRODUCTION

Electric motor technology has advanced significantly as a result of the shift to electric mobility, especially for traction applications in electric vehicles (EVs). Low production costs, high torque density, and energy efficiency have emerged as crucial performance standards. The current market is dominated by conventional motors like Brushless DC (BLDC) motors and Permanent Magnet Synchronous Motors (PMSMs), but their reliance on rare-earth magnets raises questions about sustainability and cost. Synchronous Reluctance Motors (SynRMs), on the other hand, have drawn interest as a magnet-free substitute that promises similar efficiency with less reliance on materials.

## 2.2 LITERATURE SURVEY

Research on SynRM design and optimization for electric vehicle applications has gained significant attention in recent years. A comparative analysis between a standard 5 H.P SynRM and an FEA-optimized model demonstrated that finite element analysis (FEA) significantly enhances efficiency and torque performance, providing a baseline for evaluating SynRM performance in this study (Khude et al., 2024). A Random Forest (RF) surrogate model integrated with a Genetic Algorithm (GA) has also been proposed for optimal rotor design of PMa-SynRM, achieving improvements in torque density and efficiency while reducing computational requirements (Kwon and Lim, 2023). Similarly, a comparative study of PMSM, BLDC, SynRM, and PMa-SynRM for two-wheeler EV applications highlighted the cost-effectiveness and competitive performance of SynRM, supporting its selection in this work (Gore and Ugale, 2022).

The importance of rotor design has been emphasized in several studies. Investigations on rotor flux barrier configurations showed that barrier geometry significantly influences efficiency and torque characteristics, establishing it as a key design parameter for SynRM optimization (Jędryczka et al., 2022). From a control perspective, comparisons between



Field-Oriented Control (FOC) and Direct Torque Control (DTC) techniques revealed their influence on torque ripple and dynamic performance (Sheikh et al., 2025). In addition, a cascaded P-I-I controller has been proposed to minimize torque ripple in SynRM drives (Abou-ElSoud et al., 2022). While these works focus on control strategies, the present study addresses torque ripple reduction through design-stage optimization rather than post-design control methods.

Several studies have also contributed to design benchmarks and optimization strategies. The development of a high-torque-density SynRM for EV applications provided important baseline parameters for motor design (Park et al., 2021). Multi-objective optimization methods have been proposed for designing low-noise and high-efficiency SynRMs, influencing the optimization approach used in this research (Lee and Kim, 2023). Flux control strategies have also been explored to reduce power losses, emphasizing loss minimization as a key performance index (Zhao et al., 2022). Comparative studies of PMSM and SynRM drives for compact EV platforms further confirmed the cost and efficiency advantages of SynRM (Nguyen et al., 2024). In addition, SynRM performance analysis using ANSYS Maxwell with parameters obtained from RMxpert has served as a useful reference for establishing initial simulation conditions in this study (Yildiz and Polat, 2023).

## **2.3 OPTIMIZATION METHODS IN MOTOR DESIGN**

Air-gap length, stator slotting, and rotor geometry are examples of interdependent design parameters that affect motor performance. Conventional optimization based on finite element analysis (FEA) is precise but time-consuming and computationally costly. In order to overcome this, recent studies have used artificial intelligence and data-driven surrogate models to expedite computation while effectively exploring design spaces (Cheng et al., 2024).

### **2.3.1 RANDOM FOREST (RF) SURROGATE MODELING**

A machine learning technique called Random Forest combines multiple decision trees to estimate outputs based on input features. Using datasets produced by FEA simulations, Random Forest models can be trained in motor design applications to forecast crucial

performance metrics like torque, torque ripple, efficiency, and total harmonic distortion (THD). Random Forest has been widely used as a surrogate model to replace repeated full-scale simulations due to its ability to handle nonlinear and high-dimensional data. While M. Cheng, X. Zhao, W. Qiu, and S. Niu reported that data-driven surrogate models can significantly reduce computational cost while maintaining prediction accuracy, M.-S. Kwon and D.-K. Lim demonstrated the use of Random Forest-based surrogate modeling for optimization of electric vehicle motors.

### **2.3.2 GENETIC ALGORITHM (GA) FOR ROTOR OPTIMIZATION**

Natural selection is the basis for population-based evolutionary optimization methods known as genetic algorithms. These algorithms explore a wide range of potential solutions by using operations like selection, crossover, and mutation. Genetic algorithms are widely used in electric vehicle motor design to optimize rotor geometry, increasing torque density and efficiency while decreasing torque ripple and related losses. While M. El-Neram, M. Afifi, H. Rezk, and M. Ibrahim presented successful multi-objective rotor optimization using GA-based techniques, leading to increased efficiency and decreased losses, A. Boglietti, A. Cavagnino, and M. Lazzari showed the efficacy of Genetic Algorithms for geometrical optimization of electric motor drives. In the design of electric motors, complex and nonlinear multi-objective optimization problems are particularly well-suited for genetic algorithms.

## **2.4 COMBINED RF-GA OPTIMIZATION TECHNIQUES**

In recent years, there has been a trend of using machine learning techniques in conjunction with evolutionary optimization algorithms for electric motor design problems, especially for Synchronous Reluctance Motors (SynRM). M.-S. Kwon and D.-K. Lim showed that using Random Forest as a surrogate modeling method with Genetic Algorithms for optimization problems of EV motors is efficient in terms of balancing prediction accuracy with computational cost. This method has been of interest to many researchers due to its potential for accelerating the design process without compromising prediction accuracy.

In this method, a large set of data is created by employing finite element analysis (FEA)-based software such as ANSYS Maxwell or RMxpert. This data set contains design variables such as dimensions of the rotor barrier, slot dimensions, length of the air gap, and pole configuration. Additionally, performance parameters such as average torque, torque ripple, efficiency, and total harmonic distortion (THD) are also included. Cheng, Zhao, Qiu, and Niu found that surrogate models created by RF can accurately predict motor performance by eliminating the need to run simulations.

Once the surrogate model has high prediction accuracy, a Genetic Algorithm is used to search the design space. El-Nemr, Afifi, Rezk, and Ibrahim demonstrated that the GA can optimize a multi-objective problem in designing motors by maximizing torque and efficiency while minimizing torque ripple and harmonic effects. The RF model is used to evaluate the design space in a short time, and the Genetic Algorithm uses global search techniques such as selection, crossover, and mutation.

Studies have shown that this RF-GA optimization approach can significantly reduce the overall design cycle time, torque density, and smoothness in torque characteristics with optimized rotor barrier and slot configurations. In another study by Kwon and Lim, they were able to show that with RF-GA optimization, it is possible to perform rapid optimization at multiple operating points pertinent to electric vehicle driving conditions, thus showing its viability in motor design optimization for electric vehicle motors. The synergy between RF and GA transforms electric motor design from a traditional trial and error process into an intelligent optimization process.

## **2.5 RESEARCH GAP**

Although the integration of Random Forest (RF) surrogate models with Genetic Algorithm (GA) optimization has demonstrated promising results in electric motor design, several research challenges remain insufficiently addressed, particularly in the context of electric vehicle (EV)-oriented Synchronous Reluctance Motor (SynRM) applications.

**1.Limited Operating Range Consideration:** Most existing studies validate optimized motor designs under a single operating condition, such as a fixed speed or load. However, electric vehicles operate over a wide range of speeds and torque demands, making single-point optimization inadequate for real-world EV applications.

**2.Incomplete Multi-objective Integration:** While parameters such as torque density and efficiency are commonly optimized, other critical factors including torque ripple, total harmonic distortion (THD), thermal constraints, and mechanical stresses are often considered separately rather than within a unified multi-objective optimization framework.

**3.Data Generation and Computational Burden:** Although surrogate models reduce optimization time, they still rely on high-quality datasets generated through finite element analysis (FEA). The generation of large and diverse datasets that accurately represent real-world operating conditions remains computationally expensive and time-consuming.

**4.Limited Generalization Across Motor Variants:** Most reported optimization models are developed for a specific motor rating or geometry. Manufacturing tolerances, material property variations, and small deviations in rotor barrier dimensions are rarely incorporated, which may lead to performance discrepancies when applied to different motor sizes or production variations.

**5.Lack of Comprehensive Validation:** Optimized designs are often evaluated only through electromagnetic performance metrics. Validation through thermal analysis, noise–vibration–harshness (NVH) assessment, and long-term reliability studies—critical for automotive applications—is frequently missing.

To address these gaps, this project proposes a robust, multi-objective RF–GA optimization framework integrated with FEA-based modeling for the rotor geometry optimization of SynRMs intended for EV propulsion. The proposed approach aims to enhance torque density, minimize torque ripple, and improve efficiency while considering computational feasibility and practical industrial design constraints, thereby bridging the gap between academic research and real-world EV motor design.

## 2.6 DRAWBACKS OF CONVENTIONAL MOTORS AND MOTIVATION FOR SYNRM

Despite their widespread use in electric vehicle (EV) applications, conventional motors such as Induction Motors (IM) and Brushless DC (BLDC) motors exhibit certain limitations that motivate the adoption of Synchronous Reluctance Motors (SynRMs).

**Induction Motors (IM):** Induction motors are robust and relatively cost-effective; however, they generally suffer from lower efficiency and power factor compared to synchronous machines, particularly under partial-load operating conditions commonly encountered in EV applications. Rotor copper losses result in additional heat generation, leading to reduced overall efficiency and increased cooling requirements. Moreover, induction motors typically exhibit lower torque density, which limits their suitability for compact and lightweight EV propulsion systems.

**Brushless DC Motors (BLDC):** BLDC motors offer high torque density and good efficiency; however, their dependence on permanent magnets increases overall cost and exposes the motor design to rare-earth material price fluctuations and supply risks. In addition, BLDC motors exhibit higher torque ripple due to trapezoidal back-EMF characteristics and electronic commutation, which can lead to increased acoustic noise and reduced ride comfort in EV applications.

**Motivation for Using Synchronous Reluctance Motors (SynRM):** Synchronous Reluctance Motors address several of the limitations associated with conventional motor technologies by employing a magnet-free rotor structure, thereby eliminating rare-earth material dependency and reducing cost. SynRMs provide high efficiency comparable to Permanent Magnet Synchronous Motors (PMSMs), improved thermal robustness, and reduced rotor losses. With optimized rotor flux barrier design, SynRMs can achieve competitive torque density and reduced torque ripple, making them a promising, reliable, and sustainable alternative for electric vehicle propulsion.

## 2.7 SUMMARY

Synchronous Reluctance Motors (SynRMs) are emerging as a promising magnet-free alternative to PMSM and BLDC motors for EV applications. Although PMSMs and BLDCs dominate due to high efficiency and torque density, their dependence on rare-earth magnets raises concerns about cost and sustainability. SynRMs offer comparable performance with reduced material dependency. Finite Element Analysis (FEA) is widely used to enhance SynRM performance, particularly through optimized rotor design, with design-stage improvements proving more effective than control strategies in reducing torque ripple and losses.

Recent research focuses on reducing the computational cost of FEA using AI-based techniques. The integration of Random Forest (RF) models with Genetic Algorithms (GA) enables efficient multi-objective optimization, improving torque density, efficiency, and design speed. However, existing studies still face challenges such as limited operating condition analysis, high data generation effort, and lack of validation for thermal and NVH effects. This work proposes an RF–GA-based FEA optimization framework to address these gaps and improve SynRM performance for EV applications.

# METHODOLOGY

### 3.1 INTRODUCTION

The primary objective of this project is to design and optimize a Synchronous Reluctance Motor (SynRM) for electric vehicle propulsion using a hybrid Artificial Intelligence (AI) and Finite Element Analysis (FEA) framework. Initially, three different motor types—Induction Motor (IM), Brushless DC Motor (BLDC), and Synchronous Reluctance Motor (SynRM)—are designed using the ANSYS RMxprt model to analyze and compare their fundamental performance characteristics. The baseline SynRM design is then selected for further analysis and optimization. Traditional motor design approaches rely heavily on repeated FEA simulations, which are computationally expensive and time-consuming, especially when optimizing multiple performance parameters such as average torque, torque ripple, efficiency, and total harmonic distortion (THD). This limitation restricts the ability to explore a wide design space and slows down the development of high-performance motor configurations.

To address these challenges, the proposed solution integrates a Random Forest (RF) surrogate model with a Genetic Algorithm (GA) optimization strategy. The RF model is trained using datasets generated from FEA simulations to accurately predict motor performance without requiring full-scale simulations for every design variation, thereby significantly reducing computational overhead. The GA then performs global multi-objective optimization on the rotor geometry parameters using the RF predictions as the fitness evaluation. This hybrid AI–FEA approach accelerates the design process and enables efficient exploration of a broader design space. As a result, the optimized SynRM design achieves improved torque characteristics, enhanced efficiency, and reduced torque ripple, making it suitable for electric vehicle applications. Figure 3.1 shows the block diagram of the proposed methodology.

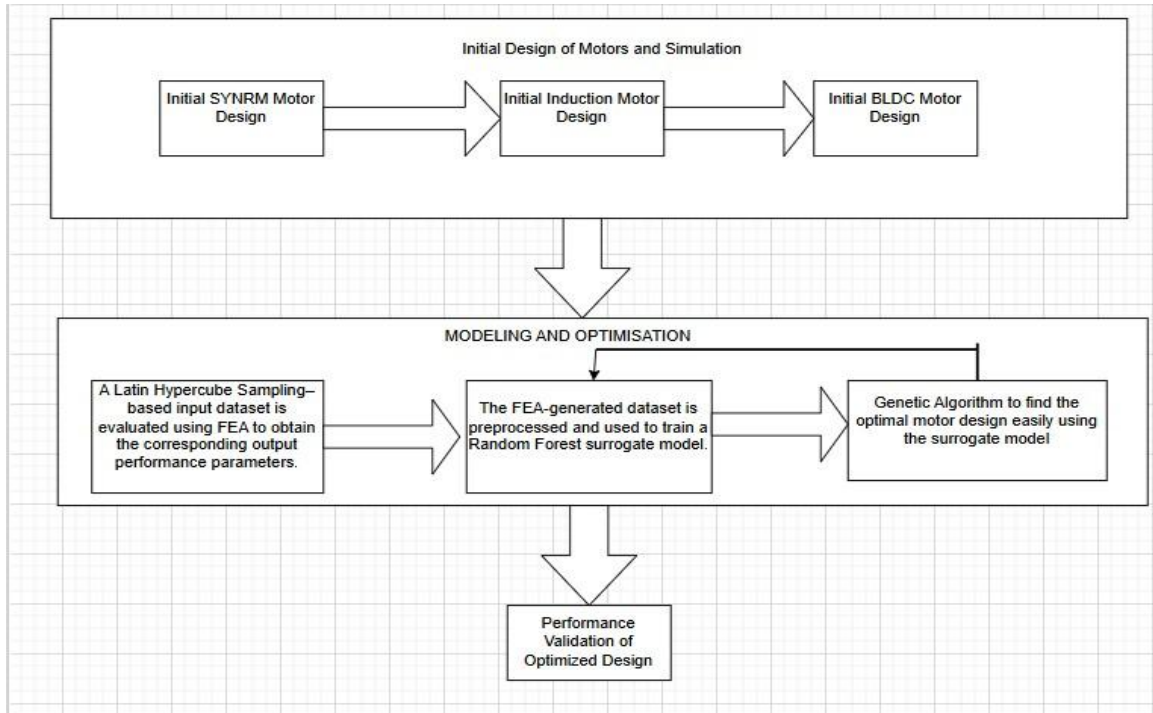


Figure 3.1 Block Diagram of Proposed model

## 3.2 DESIGN OF INDUCTION MOTOR

### 3.2.1 INDUCTION MOTOR MODEL DESIGN

#### Design Parameters

The preliminary design of the Induction Motor (IM) was carried out using RMxpert in ANSYS before proceeding to detailed finite element analysis (FEA) in ANSYS Maxwell. RMxpert enabled fast evaluation of core design parameters, predicted motor performance, and generated a Maxwell-ready model for advanced electromagnetic simulations.



Table 3.1 General Data of the Induction Motor used

| Parameter                  | Value       |
|----------------------------|-------------|
| Operation Type             | Motor       |
| Source Type                | AC          |
| Rated Output Power (kW)    | 1.1         |
| Rated Power Factor         | 0.8         |
| Capacitive Power Factor    | No          |
| Frequency (Hz)             | 50          |
| Rated Voltage (V)          | 380         |
| Load Type                  | Const Power |
| Rated Speed (rpm)          | 1500        |
| Operating Temperature (°C) | 75          |

Table 3.2. Rotor properties of IM Motor

| Name            | Value  | Unit | Evaluated Value. | Description                        |
|-----------------|--------|------|------------------|------------------------------------|
| Outer Diameter  | 76     | mm   | 76mm             | Outer diameter of the stator core  |
| Inner Diameter  | 32     | mm   | 32mm             | Inner diameter of the stator core  |
| Length          | 250    | mm   | 250mm            | Length of the stator core          |
| Stacking Factor | 0.95   |      |                  | Stacking factor of the stator core |
| Steel Type      | D21_50 |      |                  | Steel type of the stator core      |
| Skew Width      | 1      |      | 1                | Skew width measured in slot number |
| Number of Slots | 28     |      |                  | Number of slots of the stator core |

Table 3.3. Stator properties of IM Motor

|                       | Value  | Unit | Evaluation | Description                                                    |
|-----------------------|--------|------|------------|----------------------------------------------------------------|
| Outer Diameter        | 140    | mm   | 140 mm     | Outer diameter of the iron core                                |
| Inner Diameter        | 78     | mm   | 78 mm      | Inner diameter of iron core                                    |
| Length                | 250    | mm   | 250 mm     | Length of the iron core                                        |
| Stacking Factor       | 0.95   |      |            | Stacking factor of the iron core                               |
| Steel Type            | D21_50 |      |            | Steel type of the iron core                                    |
| No of Slots           | 36     |      |            | Number of slots of the stator core                             |
| Slot Type             | 1      |      |            | Slot type of the stator core                                   |
| Lamination Sectors    | 3      |      |            | Number of lamination sectors                                   |
| Press Board Thickness | 0      |      |            | Magnetic press board thickness, 0 for non-magnetic press board |
| Skew Width            | 0      | deg  | 0 deg      | Skew width measured in slot number                             |

### Induction Motor Model:

An Induction Motor (IM) (figure 3.2) is one of the most widely used AC machines due to its simple construction, robustness, and low maintenance. Operating on the principle of electromagnetic induction, it induces rotor currents through the rotating magnetic field of the stator without direct electrical connection. Classified into squirrel cage and wound rotor types, IMs are extensively applied in industries, commercial systems, and electric vehicles. It is chosen for this study as a reference machine because of its widespread use, reliable performance, and cost-effectiveness, making it an ideal baseline for comparison with SynRM and PMSM.

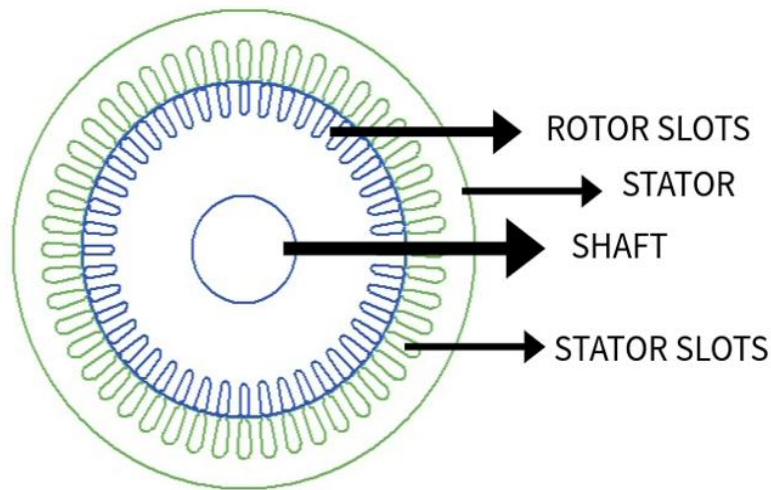


Figure 3.2 Designed RMxpert Induction Motor Model

### Analysis and Results:

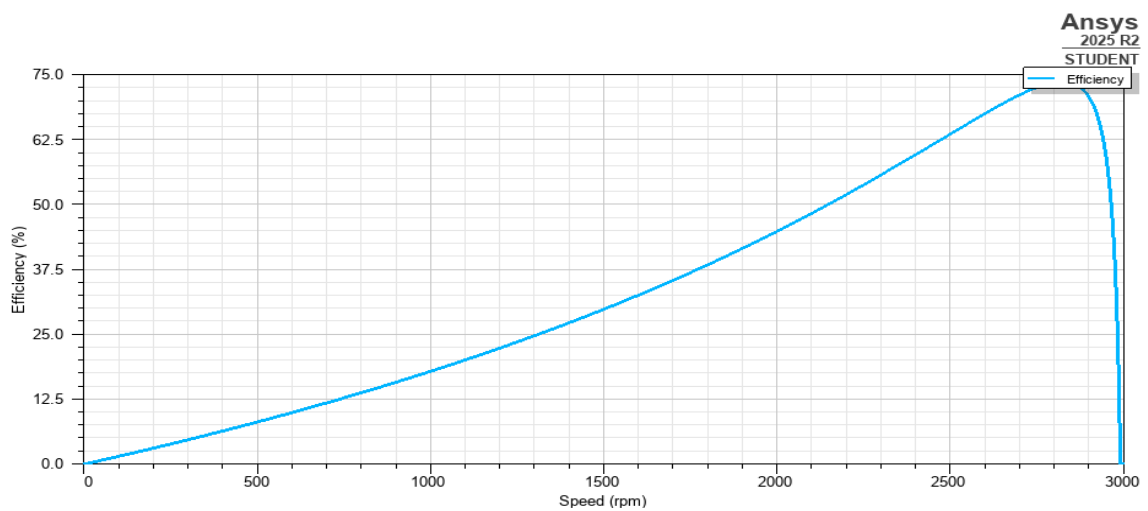


Figure 3.3 Efficiency vs Speed graph

**Efficiency vs Speed (Figure 3.3):** Efficiency varies significantly with motor speed, as shown in Figure 3.3. At low speeds, the efficiency is nearly zero and then increases almost linearly as the speed rises. The motor achieves its maximum efficiency of about 75% at around 2800 rpm, which is close to its rated operating speed, indicating optimal performance in this region. Beyond this point, as the motor speed approaches the no-load condition, the efficiency decreases sharply due to increased losses and reduced load on the motor.

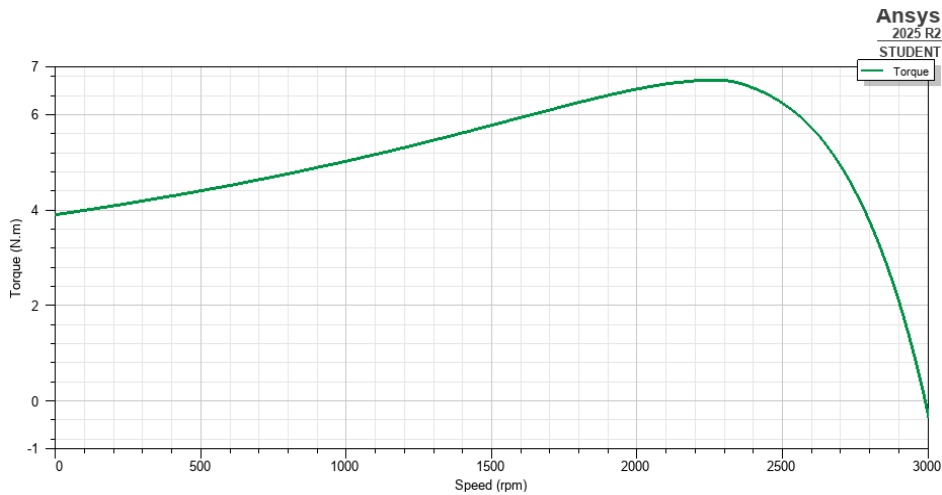


Figure 3.4 Torque vs Speed graph

**Torque vs Speed (Figure 3.4):** The torque–speed characteristic of the motor, illustrated in Figure 3.4, follows a typical motor behavior. At standstill, the motor develops a starting torque of approximately 4 N·m, which enables it to initiate motion under load. As the speed increases, the torque rises steadily and reaches a maximum or breakdown torque of about 7 N·m at around 2500 rpm, representing the motor’s maximum load-handling capability. Beyond this point, the torque decreases rapidly with further increase in speed and eventually drops to zero as the motor approaches its synchronous speed.

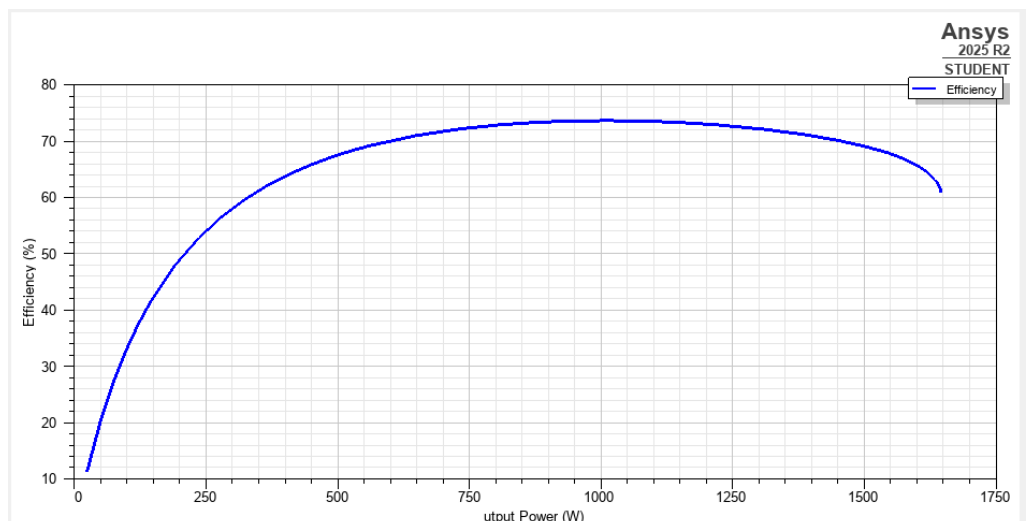


Figure 3.5 Efficiency vs Output Power graph

**Efficiency vs Output Power (Figure 3.5):** The motor's efficiency as a function of output power, shown in Figure 3.6, is relatively low at light loads. As the output power increases, efficiency rises and reaches a peak of approximately 58% when the power is between 1000 W and 1200 W, indicating the motor's optimal operating range. Beyond this point, efficiency shows a slight decrease with further loading, reflecting increased losses at higher power levels. For output powers above 1000 W, the efficiency tends to saturate at a value of about 0.68, suggesting stable performance under higher loads.

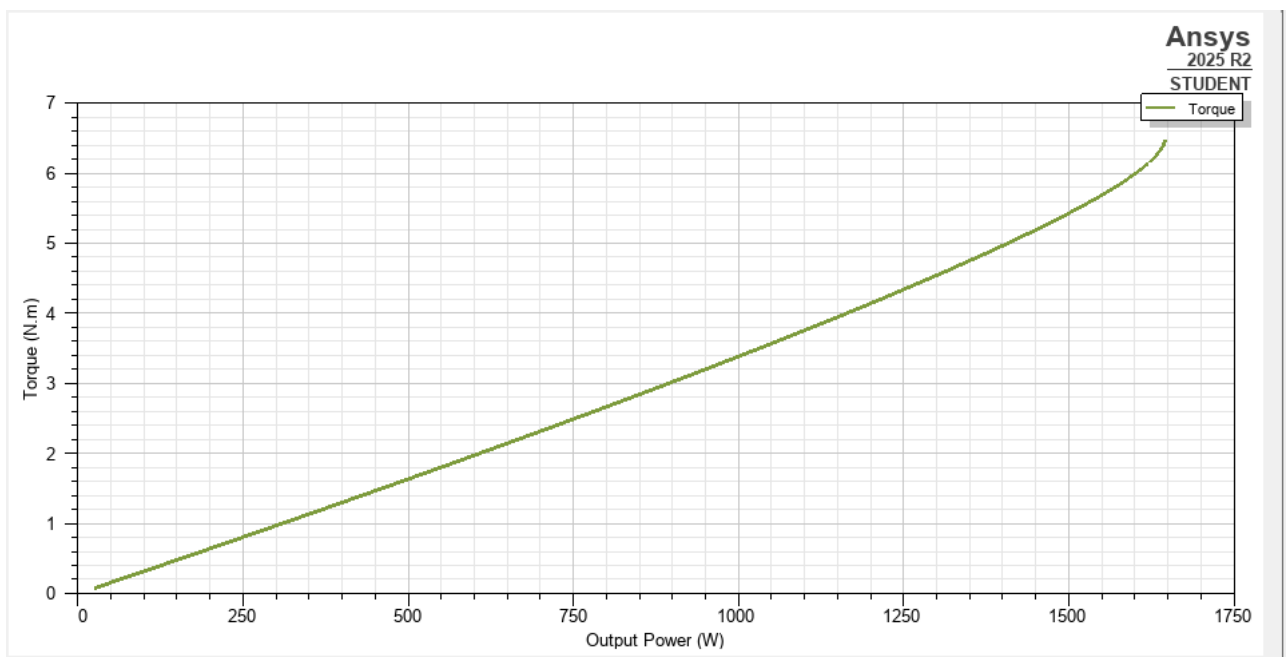


Figure 3.6 Torque vs Output power graph

**Torque vs Output Power (Figure 3.13):** The relationship between torque and output power, depicted in Figure 3.13, is directly proportional. As the motor delivers more output power, the torque increases linearly, indicating that higher power demands result in a proportional rise in the torque generated by the motor.

Table 3.4. Rated Magnetic Data of IM Motor

| Name                          | Value    | Units |
|-------------------------------|----------|-------|
| Stator-Teeth Flux Density     | 0.497871 | tesla |
| Rotor-Teeth Flux Density      | 1.14776  | tesla |
| Stator-Yoke Flux Density      | 1.0956   | tesla |
| Rotor-Yoke Flux Density       | 0.43297  | tesla |
| Air-Gap Flux Density          | 0.260489 | tesla |
| Stator-Teeth Ampere Turns     | 3.11222  | A.T   |
| Rotor-Teeth Ampere Turns      | 7.69401  | A.T   |
| Stator-Yoke Ampere Turns      | 44.1161  | A.T   |
| Rotor-Yoke Ampere Turns       | 1.98243  | A.T   |
| Air-Gap Ampere Turns          | 214.511  | A.T   |
| Stator Yoke Correction Factor | 0.645693 |       |
| Rotor Yoke Correction Factor  | 0.673304 |       |
| Teeth Saturation Factor       | 1.05038  |       |
| Total Saturation Factor       | 1.26528  |       |
| Back EMF Factor               | 0.889803 |       |

Table 3.5. No Load operation Data of IM Motor

| Name                         | Value      | Units |
|------------------------------|------------|-------|
| No-Load Stator Phase Current | 1321.91    | mA    |
| No-Load Iron-Core Loss       | 63716.9    | mW    |
| No-Load Input Power          | 199318     | mW    |
| No-Load Power Factor         | 0.229086   |       |
| No-Load Slip                 | 0.00401922 |       |
| No-Load Shaft Speed          | 2987.94    | rpm   |
| Stator Resistance            | 4.62044    | ohm   |
| Stator Leakage Reactance     | 5.66475    | ohm   |
| Rotor Resistance             | 4.54592    | ohm   |
| Rotor Leakage Reactance      | 271.693    | ohm   |

### 3.3 DESIGN OF BRUSHLESS DC MOTOR

#### 3.3.1 BRUSHLESS DC MOTOR MODEL DESIGN

##### Design Parameters

The preliminary design of the Brushless DC Motor (BLDC) was carried out using RMxpert in ANSYS before proceeding to detailed finite element analysis (FEA) in ANSYS Maxwell. RMxpert enabled fast evaluation of core design parameters, predicted motor performance, and generated a Maxwell-ready model for advanced electromagnetic simulations.

Table 3.6. General Data for the designed BLDC motor

| Parameter                  | Value       |
|----------------------------|-------------|
| Operation Type             | Motor       |
| Source Type                | DC          |
| Rated Output Power (Kw)    | 1.1         |
| Rated Power Factor         | 0.8         |
| Capacitive Power Factor    | No          |
| Frequency (Hz)             | 50          |
| Rated Voltage (V)          | 380         |
| Load Type                  | Const Power |
| Rated Speed (rpm)          | 1500        |
| Operating Temperature (°C) | 75          |

Table 3.7. Rotor properties of BLDC Motor

| Name            | Value   | Unit | Evaluated Value | Description                       |
|-----------------|---------|------|-----------------|-----------------------------------|
| Outer Diameter  | 74      | mm   | 74 mm           | Outer diameter of the rotor core  |
| Inner Diameter  | 26      | mm   | 26 mm           | Inner diameter of the rotor core  |
| Length          | 65      | mm   | 65 mm           | Length of the rotor core          |
| Stacking Factor | 0.95    |      |                 | Stacking factor of the rotor core |
| Steel Type      | M19_24G |      |                 | Steel type of rotor core          |
| Pole Type       | 1       |      |                 | Pole type of rotor                |

Table 3.8. Stator properties of BLDC Motor

|                 | Value   | Unit | Evaluation | Description                        |
|-----------------|---------|------|------------|------------------------------------|
| Outer Diameter  | 120     | mm   | 120 mm     | Outer diameter of the iron core    |
| Inner Diameter  | 75      | mm   | 75 mm      | Inner diameter of iron core        |
| Length          | 65      | mm   | 65 mm      | Length of the iron core            |
| Stacking Factor | 0.95    |      |            | Stacking factor of the iron core   |
| Steel Type      | M19_24G |      |            | Steel type of the iron core        |
| Number of Slots | 24      |      |            | Number of slots of the stator core |
| Slot Type       | 2       |      |            | Slot type of the stator core       |
| Skew Width      | 0       | deg  | 0 deg      | Skew width measured in degrees     |

#### **BLDC Motor Model:**

A Brushless DC (BLDC) motor (figure 3.7) is an electronically commutated motor that operates using direct current (DC) while achieving performance similar to an AC motor through electronic control techniques. Unlike conventional brushed DC motors, BLDC motors do not contain mechanical brushes. Instead, an electronic controller regulates the switching of current in the stator windings to produce a rotating magnetic field that drives the rotor. The rotor generally consists of permanent magnets, whereas the stator contains multiple coils. Because there are no brushes, friction and mechanical wear are minimized, resulting in higher efficiency, longer service life, reduced maintenance, and improved speed–torque performance. Due to these advantages, BLDC motors are widely used in applications such as electric vehicles, drones, industrial automation systems, robotics, and household appliances.



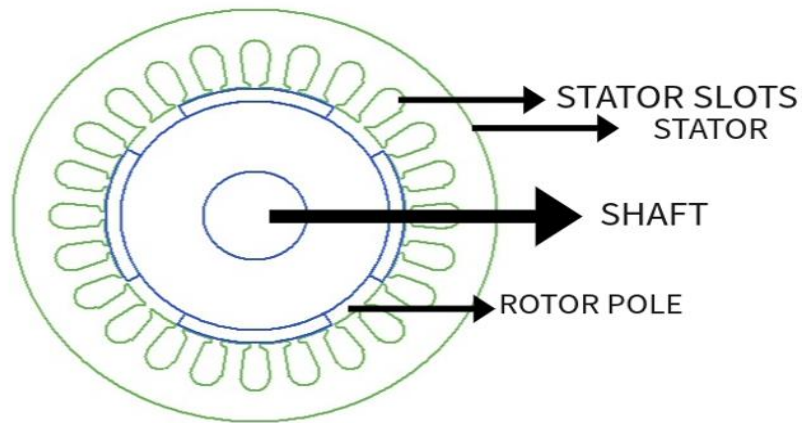


Figure 3.7 Designed RMxprt BLDC Motor Model

### Analysis and Results:

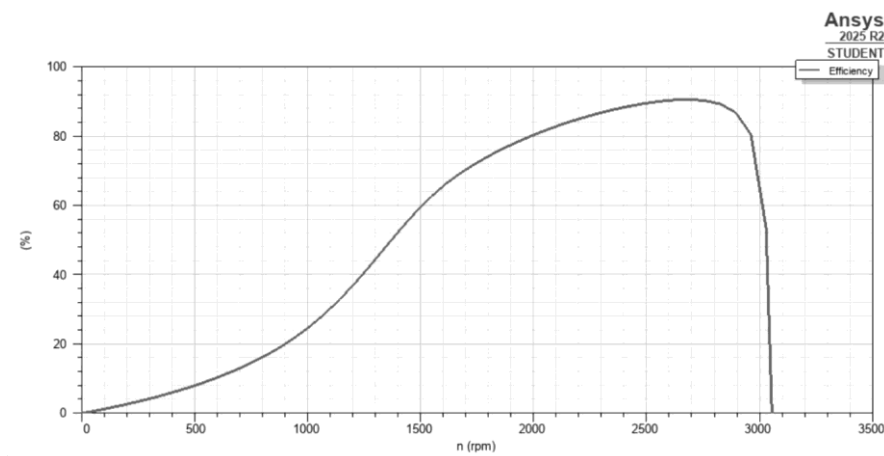


Figure 3.8 Efficiency vs Speed

**Efficiency vs Speed (Figure 3.8):** The efficiency of the BLDC motor is very low at low speeds due to high copper losses resulting from large current draw during startup. As the speed increases beyond approximately 500 rpm, the efficiency rises steeply and reaches a peak value of around 90% at approximately 2500 rpm. At very high speeds above 3000 rpm, the efficiency drops sharply, which can be attributed to increased core losses and reduced torque output.

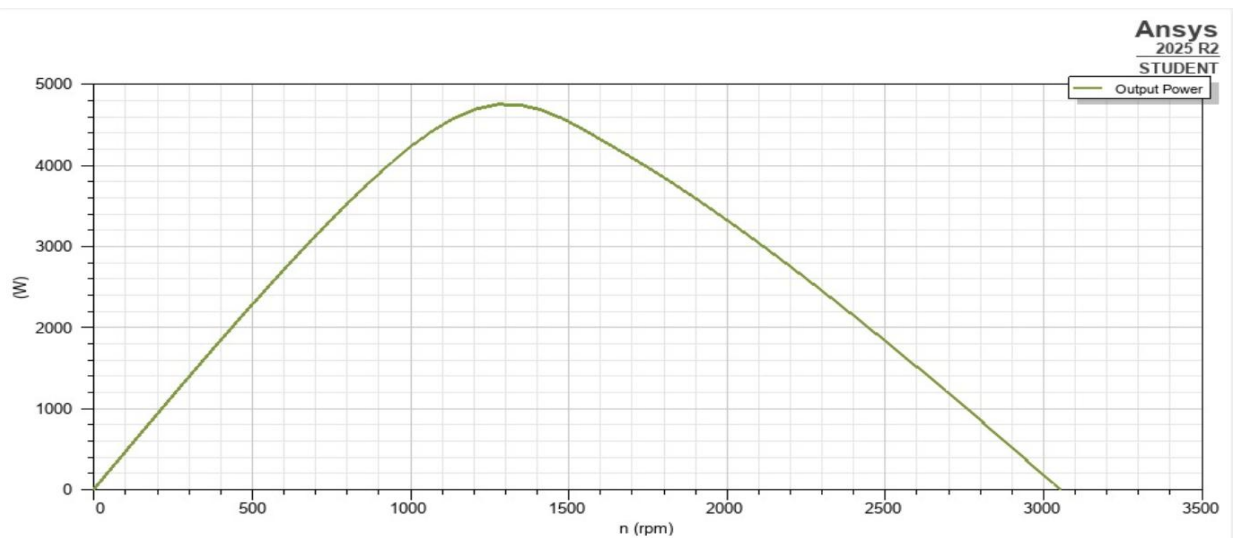


Figure 3.9 Output power vs Speed

**Output Power vs Speed (Figure 3.9):** The output power of the BLDC motor increases sharply from standstill and reaches a maximum value of approximately 4800 W at around 1500 rpm. Beyond this speed, the output power decreases steadily as the motor speed increases, which is consistent with the corresponding torque–speed characteristics of the machine. This behavior indicates that the motor delivers its maximum power in the mid-speed operating region.

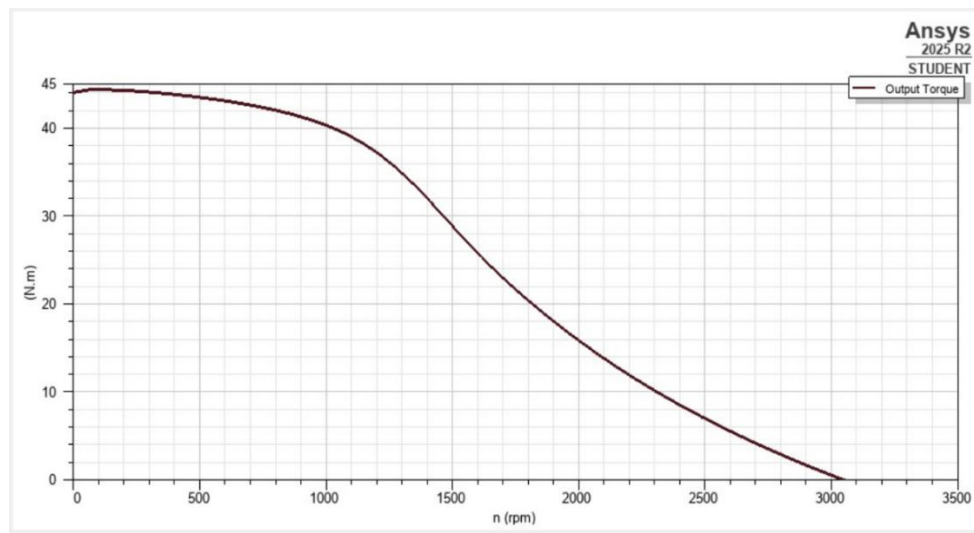


Figure 3.10 Output torque vs speed

**Output Torque vs Speed (Figure 3.10):** The output torque of the BLDC motor remains nearly constant, in the range of approximately 42–44 Nm, up to a speed of about 1000 rpm. Beyond this operating point, the torque gradually decreases as the speed increases, indicating a transition from the constant torque region to the constant power region of operation. This behavior is characteristic of BLDC motors and reflects the influence of back EMF and voltage limitations at higher speeds.

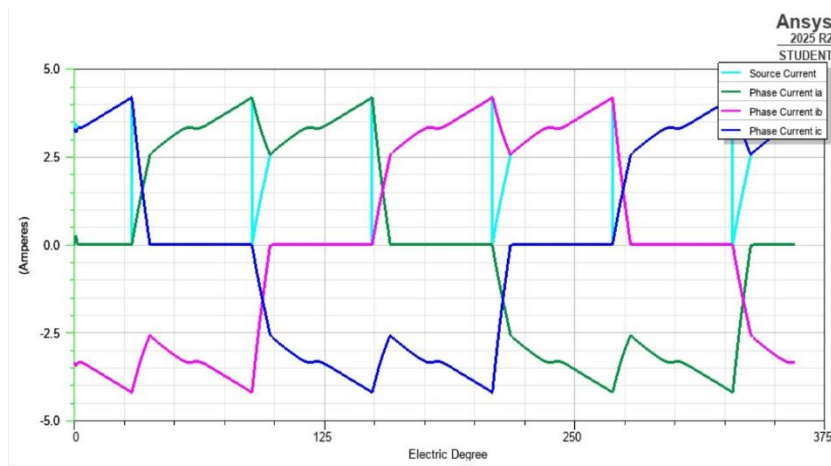


Figure 3.11 Winding Currents under Load

**Winding Currents under Load (Figure 3.11):** Under load conditions, the winding phase currents of the BLDC motor are balanced and exhibit the expected trapezoidal or quasi-rectangular waveform patterns characteristic of BLDC operation. The observed peak current values of approximately  $\pm 5$  A confirm proper load sharing among the phases and effective current control. The alignment of the current waveforms with the back-EMF profile indicates suitable commutation and efficient torque production. These current characteristics further demonstrate stable motor operation under load and validate the correctness of the RMxpert-based BLDC mode.

Table 3.9. Full Load Operation of BLDC Motor

| Name                        | Value   | Units                           |
|-----------------------------|---------|---------------------------------|
| Average Input Current       | 3207.5  | mA                              |
| RMS Armature Current        | 2759.5  | mA                              |
| Armature Thermal Load       | 59.4201 | A <sup>2</sup> /mm <sup>3</sup> |
| Specific Electric Loading   | 14054   | A/meter                         |
| Armature Current Density    | 4227980 | A/m <sup>2</sup>                |
| Frictional and Windage Loss | 21802.7 | mW                              |
| Iron-Core Loss              | 41457.8 | mW                              |
| Armature Copper Loss        | 41576.1 | mW                              |
| Transistor Loss             | 12975.5 | mW                              |
| Diode Loss                  | 591.951 | mW                              |
| Total Loss                  | 118404  | mW                              |
| Output Power                | 1100450 | mW                              |
| Input Power                 | 1218850 | mW                              |
| Efficiency                  | 90.2856 | %                               |
| Rated Speed                 | 2725.34 | rpm                             |
| Rated Torque                | 3.85585 | NewtonMeter                     |
| Locked-Rotor Torque         | 43.9604 | NewtonMeter                     |
| Locked-Rotor Current        | 103123  | mA                              |
| Maximum Output Power        | 4743180 | mW                              |

### 3.4 DESIGN OF SYNRM MOTOR

#### 3.4.1 SYNRM RMXprt DESIGN

##### Design Parameters

In RMXprt, the following key inputs were defined for the motor design. The rated power and voltage were set according to the requirements of the EV application. The stator dimensions, including outer diameter, slot number, slot type, and stack length, were chosen based on standard SynRM configurations. For the rotor geometry, multiple flux barriers were designed to maximize saliency, i.e., the difference between  $L_d$  and  $L_q$ . Additionally, the winding data was specified as a three-phase distributed winding with a star connection to ensure balanced operation and efficiency.

Table 3.10. General Data of the SynRM Motor

| Parameter                  | Value       |
|----------------------------|-------------|
| Operation Type             | Motor       |
| Source Type                | AC          |
| Rated Output Power (kW)    | 1.1         |
| Rated Power Factor         | 0.8         |
| Capacitive Power Factor    | No          |
| Frequency (Hz)             | 50          |
| Rated Voltage (V)          | 380         |
| Load Type                  | Const Power |
| Rated Speed (rpm)          | 1500        |
| Operating Temperature (°C) | 75          |

Table 3.11 Rotor parameters of SynRM Motor

| Name            | Value   | Unit | Evaluated Value | Description                      |
|-----------------|---------|------|-----------------|----------------------------------|
| Outer Diameter  | 84      | mm   | 84              | Outer diameter of the iron core  |
| Inner Diameter  | 30      | mm   | 30              | Inner diameter of the iron core  |
| Length          | 90      | mm   | 90              | Length of the iron core          |
| Stacking Factor | 0.95    |      |                 | Stacking factor of the iron core |
| Steel Type      | M19_24G |      |                 | Steel type of the iron core      |
| Pole Type       | 4       |      |                 | Pole type of the iron core       |

Table 3.12 Stator properties of SynRM Motor

|                       | Value   | Unit | Evaluation | Description                                             |
|-----------------------|---------|------|------------|---------------------------------------------------------|
| Outer Diameter        | 140     | mm   | 140 mm     | Outer diameter of the iron core                         |
| Inner Diameter        | 85      | mm   | 85 mm      | Inner diameter of iron core                             |
| Length                | 90      | mm   | 90 mm      | Length of the iron core                                 |
| Stacking Factor       | 0.95    |      |            | Stacking factor of the iron core                        |
| Steel Type            | M19_24G |      |            | Steel type of the iron core                             |
| Press Board Thickness | 0       | mm   |            | One side thickness of the press boards at two core ends |
| Magnetic Press Board  |         |      |            | Pole press board is made of magnetic material           |
| Skew Width            | 0       | deg  | 0 deg      | Skew width measured in degrees                          |

### Synrm Motor Model:

The preliminary design of the Synchronous Reluctance Motor (SynRM) was carried out using RMXprt, a dedicated motor design module in ANSYS, before proceeding to detailed finite element analysis (FEA) in ANSYS Maxwell. RMXprt provides a parametric and analytical approach for fast prototyping of rotating electrical machines, enabling the designer to evaluate core parameters, predict performance, and export a ready-to-simulate Maxwell model.

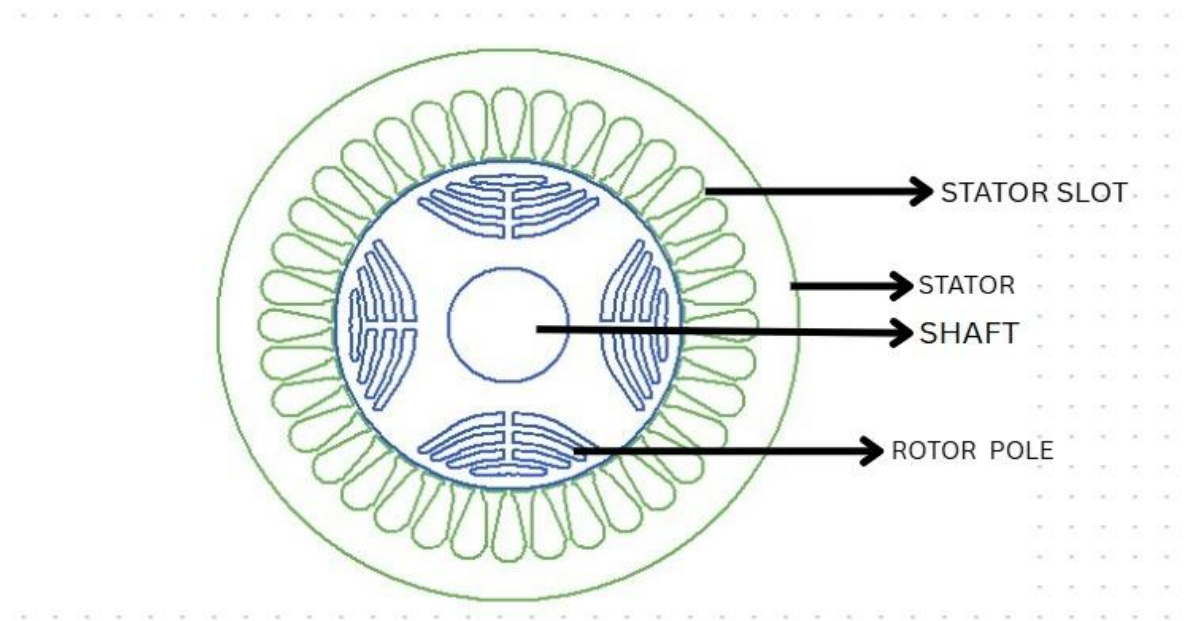


Figure 3.12 Designed SynRM Motor Model

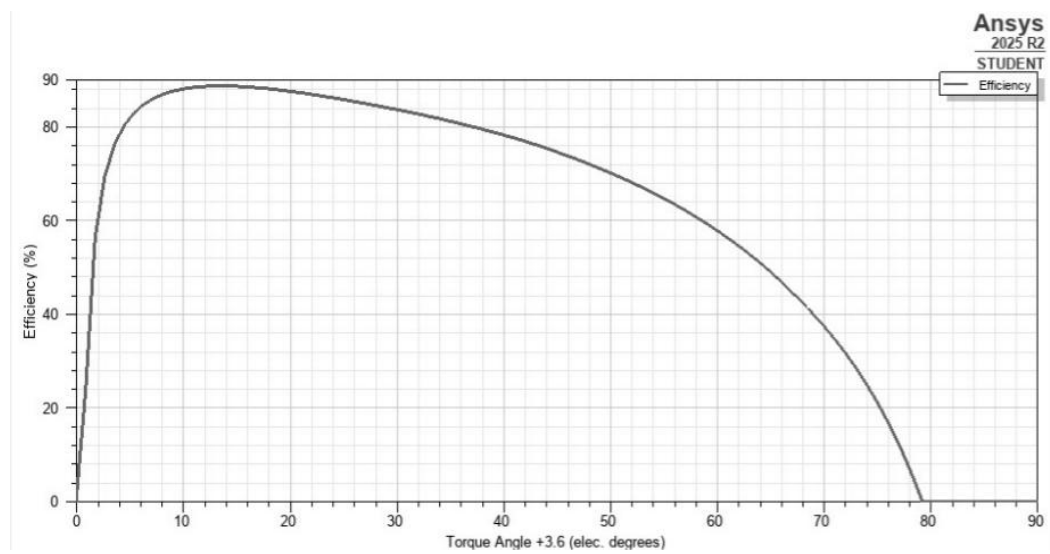


Figure 3.13 Efficiency vs Torque angle graph for the Designed SynRM Motor

Table 3.13. Full Load Operation of SynRM Motor

| Name                        | Value    | Units                           |
|-----------------------------|----------|---------------------------------|
| RMS Armature Current        | 5386.77  | mA                              |
| Armature Thermal Load       | 229.353  | A <sup>2</sup> /mm <sup>3</sup> |
| Specific Electric Loading   | 35064.3  | A/meter                         |
| Armature Current Density    | 6540920  | A/m <sup>2</sup>                |
| Frictional and Windage Loss | 0        | mW                              |
| Iron-Core Loss              | 0        | mW                              |
| Armature Copper Loss        | 194963   | mW                              |
| Transistor Loss             | 0        | mW                              |
| Diode Loss                  | 0        | mW                              |
| Total Loss                  | 194963   | mW                              |
| Output Power                | 1100150  | mW                              |
| Input Power                 | 1295110  | mW                              |
| Power Factor                | 0.365276 |                                 |
| Efficiency                  | 84.9462  | %                               |
| Torque Angle                | 2.89428  | deg                             |
| Rated Speed                 | 1500     | rpm                             |
| Rated Torque                | 7.00375  | NewtonMeter                     |

Table 3.14. No Load Magnetic Variables of SynRM Motor

| Name                          | Value    | Units |
|-------------------------------|----------|-------|
| Stator Tooth Flux Density     | 2.05576  | tesla |
| Stator Yoke Flux Density      | 0.585421 | tesla |
| Rotor Top-Tooth Flux Density  | 0.530924 | tesla |
| Rotor Yoke Flux Density       | 0.484254 | tesla |
| Air-Gap Flux Density          | 0.581622 | tesla |
| Stator Tooth Ampere Turns     | 156.126  | A.T   |
| Stator Yoke Ampere Turns      | 24.737   | A.T   |
| Rotor Top-Tooth Ampere Turns  | 9.20407  | A.T   |
| Rotor Yoke Ampere Turns       | 4.58842  | A.T   |
| Air-Gap Ampere Turns          | 270.902  | A.T   |
| Total Ampere Turn Drop        | 465.558  | A.T   |
| Saturation Factor             | 1.71854  |       |
| Stator Yoke Correction Factor | 0.646762 |       |
| Rotor Yoke Correction Factor  | 1        |       |

In the initial region (0°–10° torque angle), the efficiency rises sharply from 0% to around 85–90%. This occurs because, at very small torque angles, the motor begins producing torque with minimal losses. The rapid increase in efficiency indicates that even a small torque angle generates significant mechanical output relative to the electrical input.



At around  $10^\circ$  torque angle, the efficiency reaches its maximum of approximately 89–90%. This point represents the optimal operating condition for the motor, where losses such as stator copper, rotor iron, and friction are minimal relative to the output torque. For this reason, designers often target this torque angle for rated operation.

Beyond  $10^\circ$ , in the range of  $10^\circ$ – $80^\circ$ , the efficiency gradually decreases as the torque angle increases. At larger angles, rotor current and flux linkage are no longer optimally aligned, which leads to increased stator and rotor losses. Near  $80^\circ$ , the efficiency falls close to 0%, corresponding to the pull-out or maximum torque angle. Beyond this point, the motor cannot produce useful torque efficiently and may become unstable.

### 3.4.2 MATHEMATICAL MODELLING OF SYNCHRONOUS RELUCTANCE MOTOR

#### i) Electrical System ( $D_q$ Currents)

The purpose of this block is to convert three-phase stator quantities (abc) into the d–q reference frame using Park’s transformation. This transformation simplifies the analysis and control of synchronous machines by representing sinusoidal quantities as DC components in a rotating frame of reference. It allows easier calculation of torque, flux, and other parameters essential for motor performance evaluation.

The inputs to this block include stator voltages (  $V_d, V_q$ ), rotor speed ( $\omega_r$ ), stator resistance ( $R_s$ ), and the d-axis and q-axis inductances (  $L_d, L_q$ ). These parameters provide the necessary electrical and mechanical characteristics for accurate transformation.

The outputs are the D–Q axis currents ( $i_d, i_q$ ), which play a crucial role in torque calculation. These currents reflect how electrical inputs interact with the rotor’s magnetic field and are directly used in control strategies for optimizing motor efficiency and dynamic response.

$$di_d/d_t = \left(\frac{1}{L_d}\right) (V_d + p\omega L_q i_q - R_s i_d) \quad (3.1)$$

$$di_q/d_t = (1/L_q)(V_q - p\omega L_d i_d - R_s i_q) \quad (3.2)$$

## ii) Electromagnetic Torque

The purpose of this block is to compute the electromagnetic torque generated by the Synchronous Reluctance Motor (SynRM). Torque calculation is a critical step as it links the electrical behavior of the machine with its mechanical performance.

The inputs required for this computation include the D–Q axis currents ( $i_d$ ,  $i_q$ ), the number of pole pairs ( $p$ ), and the inductances along the d-axis ( $L_d$ ) and q-axis ( $L_q$ ). These parameters define the interaction between the stator currents and the rotor's magnetic saliency, which is the key principle behind torque production in SynRMs.

The output of this block is the electromagnetic torque ( $T_e$ ), which is then fed into the mechanical system for further analysis. This torque output represents the motor's ability to drive mechanical loads and serves as the primary input for evaluating system-level performance, including speed, efficiency, and stability.

$$T = \frac{3}{2}p(L_d i_d i_q - L_q i_q i_d) \quad (3.3)$$

## iii) Mechanical System

The purpose of this block is to simulate the mechanical behavior of the rotor when subjected to the applied electromagnetic torque and external load. This block essentially models the dynamics of the rotor, taking into account inertia and damping effects, to determine how the machine responds under various operating conditions.

The key outputs are the rotor speed ( $\omega_m$ ) and rotor position. These variables are critical because they are continuously fed back into the d–q reference frame subsystem, ensuring accurate synchronization between the electrical and mechanical domains of the motor model.

$$\frac{d\omega_m}{dt} = \left(\frac{1}{J}\right) (T_e - T_L - B\omega_m) \quad (3.4)$$

$$\omega_e = p\omega_m \quad (3.5)$$

$$\frac{d\theta_m}{dt} = \omega_m \quad (3.6)$$

**Parameters:**

$L_d$  – d axis inductance = 0.0031 H

$L_q$  – q axis inductance = 0.004H

$R_s$  – stator resistance = 0.004H

$P$ : number of rotor pole pairs = 2

$T_e$ : Rotor torque

$T_L$ : Load torque

$w_m$ : Rotor mechanical speed rad/s

$J$ : Rotor inertia =  $0.47 \times 10^{-4} \text{kgm}^2$

**SIMULINK Modeling of a SynRM Motor:**

The Simulink model of the Synchronous Reluctance Motor (SynRM) is structured into interconnected subsystems to accurately represent the electrical and mechanical behavior of the motor. The block diagram primarily consists of three major functional blocks: D–Q Current Control Subsystem, Electromagnetic Torque Subsystem, and Mechanical Dynamics Subsystem.

**Working Principle:**

The model starts by applying the defined voltages to the d–q current block, which computes the instantaneous current response. These currents then produce torque based on the rotor saliency. The generated torque drives the mechanical system, which produces speed and position feedback. This feedback alters the current dynamics in real time, ensuring a closed-loop representation of actual motor behavior.

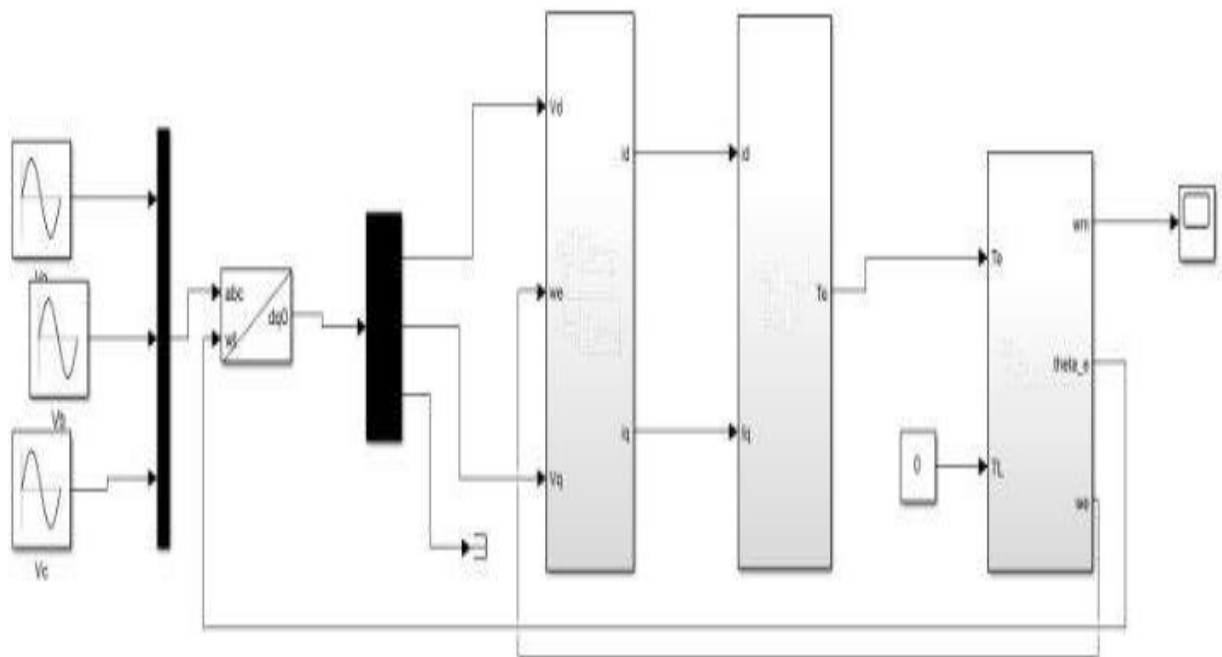


Figure 3.14 SIMULINK Modeling and Simulation of a SynRM Motor

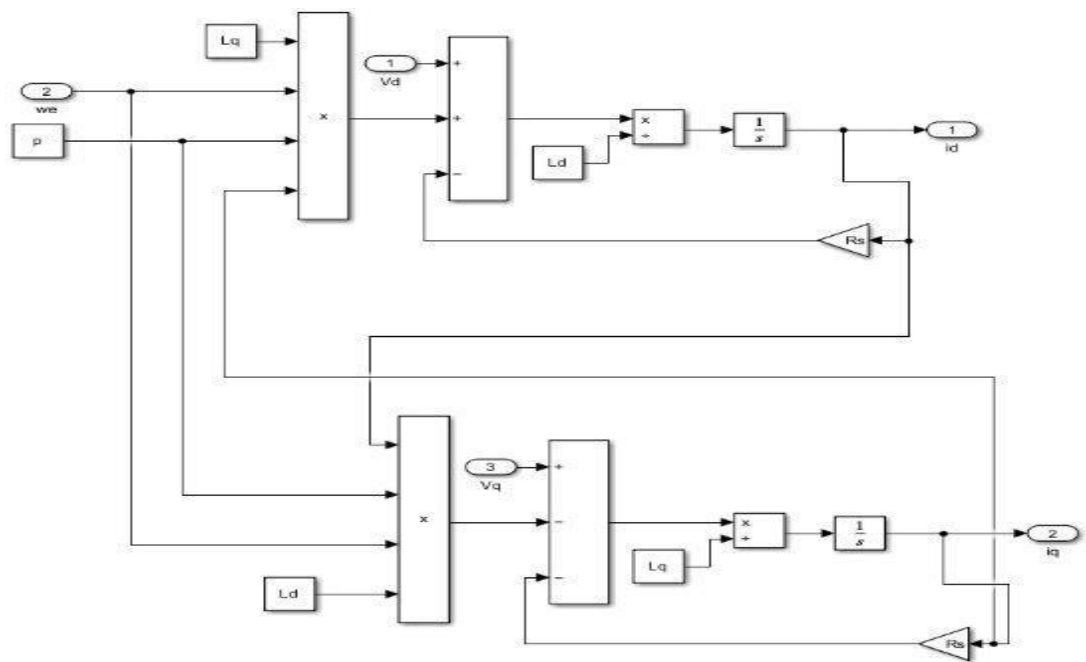


Figure 3.15 d q currents Subsystem

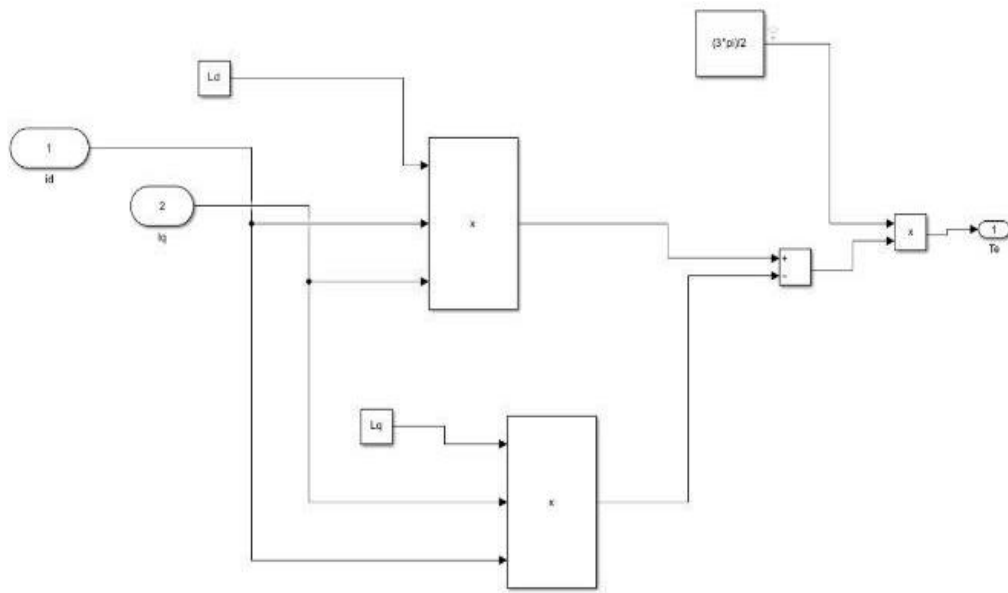


Figure 3.16 Electromagnetic Torque Subsystem

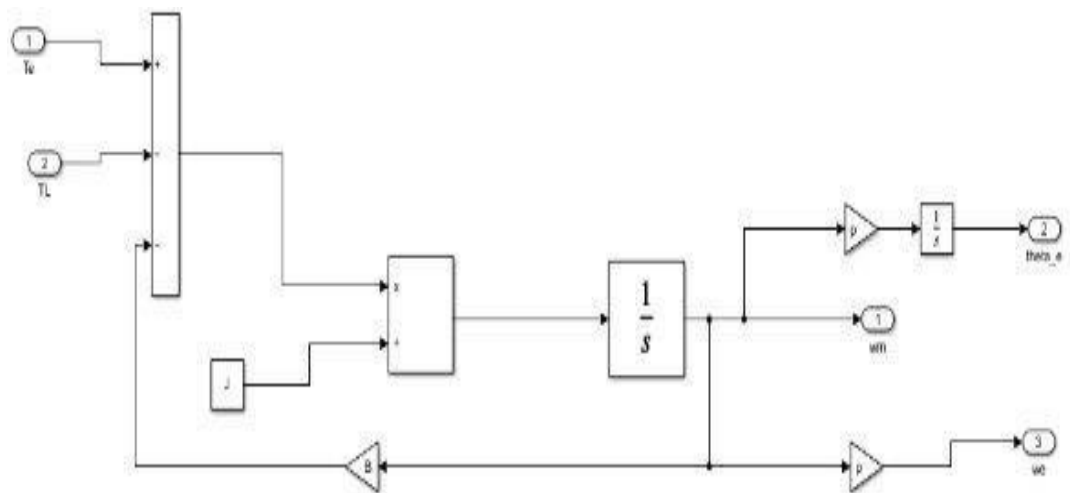


Figure 3.17 Mechanical System Subsystem

### Output Graph:

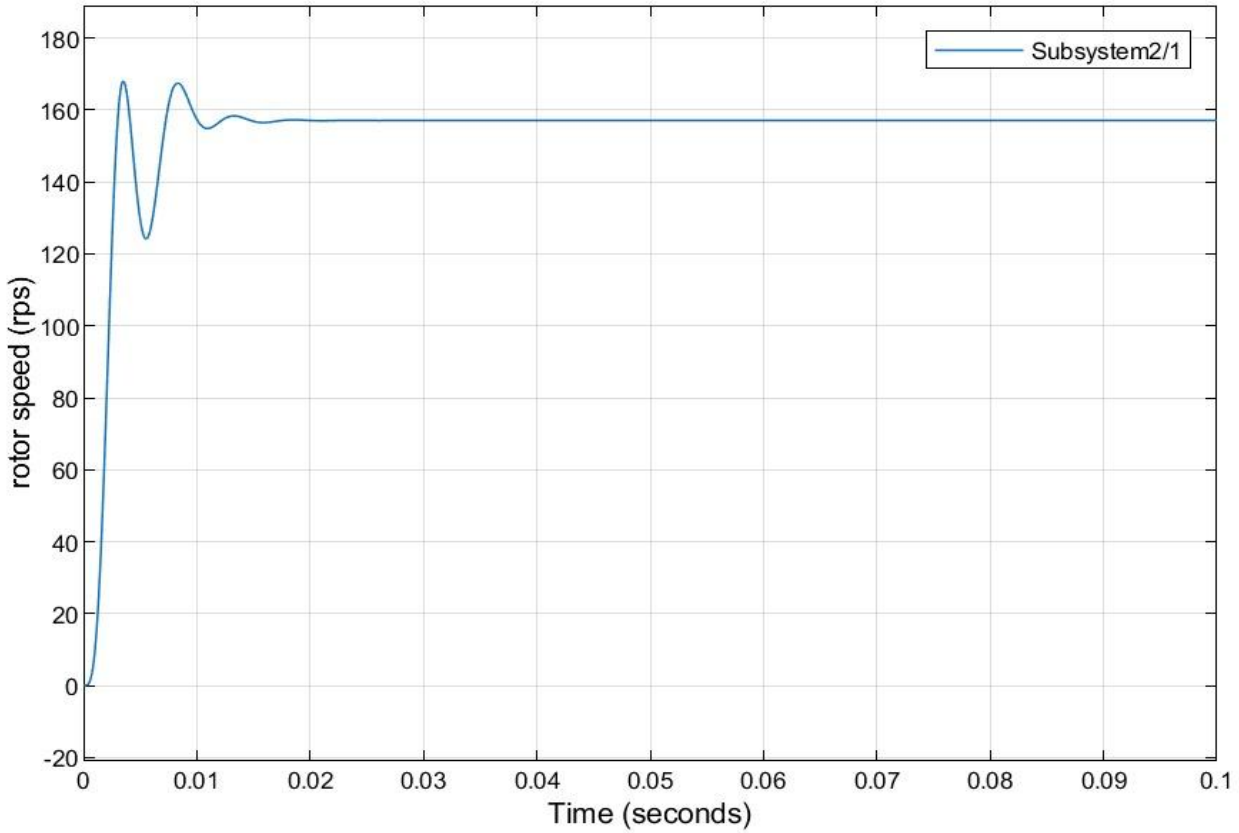


Figure 3.18 Speed of Rotor (rps) vs Time (sec) plot

The torque response of the SynRM shows noticeable oscillations during startup and sudden load variations. These oscillations are primarily caused by rotor saliency, due to the difference between  $L_d$  and  $L_q$ , and the cross-coupling effect between the d-q axes. Although the torque initially fluctuates, the oscillations decay as the rotor speed stabilizes and the d-q currents reach their reference values. Importantly, the optimized RF-GA design significantly reduces torque ripple compared to the base model, leading to smoother torque production and improved rotor performance.

The current response in both the d-axis and q-axis also exhibits transient overshoot and oscillations at the beginning. This behavior arises from the strong electromagnetic coupling between the axes and slight delays in rotor position feedback. While these oscillations gradually settle, they can cause higher copper losses and additional thermal stress if prolonged. Eventually, the currents converge to their steady-state values, supporting stable torque generation and

efficient operation of the machine.

The speed response demonstrates a slight overshoot and underdamped behavior before reaching steady-state operation. This dynamic behavior is influenced by the interaction between the electromagnetic torque, the rotor inertia ( $J$ ), transient load variations, and mechanical damping ( $B$ ). Overall, the dynamic oscillations observed in torque, current, and speed are transient phenomena influenced by both electromagnetic and mechanical characteristics. The optimized design not only reduces the magnitude and duration of these oscillations but also enhances overall stability, reduces energy losses, and ensures smoother system performance.

### **3.5 SUMMARY**

In this chapter, the methodology adopted to design and optimize a Synchronous Reluctance Motor (SynRM) for electric vehicle drive systems using a hybrid AI-FEA approach is discussed. In this regard, an Induction Motor, BLDC Motor, and SynRM are first designed in ANSYS RMxprt and compared in terms of efficiency, torque, and performance. Subsequently, SynRM is chosen to be further analyzed. The SynRM is then modeled in RMxprt and analyzed in Maxwell. In addition to this, a mathematical model is created in MATLAB Simulink to analyze torque, current, and speed response. To optimize motor performance and reduce computation time, a Random Forest-GA optimization algorithm is utilized to optimize rotor parameters to achieve better efficiency, torque smoothness, and stability for electric vehicle motor drive systems.

# 2D MAXWELL DESIGN OF MOTORS

## 4.1 INTRODUCTION

This chapter presents the electromagnetic modeling and analysis of three different electric motors: the Induction Motor (IM), Brushless DC Motor (BLDC), and Synchronous Reluctance Motor (SynRM) using ANSYS Maxwell 2D. The motor geometries initially developed in ANSYS RMXprt are exported to the Maxwell environment to perform detailed electromagnetic simulations under rated operating conditions. The 2D finite element model enables accurate analysis of the electromagnetic behavior of the motors, including torque production, flux distribution, induced voltages, and current characteristics.

For each motor, transient electromagnetic analysis is carried out to obtain important performance parameters such as torque–time characteristics, phase currents, flux linkages, and induced voltages. These results help evaluate the dynamic performance and steady-state behavior of the motors. In addition to waveform analysis, magnetic flux density contour plots and flux line distributions are obtained to visualize the magnetic field distribution within the stator, rotor, and air-gap regions. These contour plots provide insights into magnetic saturation, flux concentration, and the effectiveness of the magnetic circuit design.

The results obtained from the simulations are used to analyze and compare the electromagnetic characteristics of the three motor types. This analysis provides a deeper understanding of their operating principles and performance behavior, which serves as a basis for further evaluation and optimization of the Synchronous Reluctance Motor (SynRM) for electric vehicle applications.

## 4.2 INDUCTION MOTOR 2D MAXWELL DESIGN

The induction motor (IM) is designed and analyzed using ANSYS Maxwell 2D to study its electromagnetic performance under rated operating conditions. A two-dimensional cross-sectional model of the motor is created, including the stator core, squirrel-cage rotor, air-gap, and windings. The stator and rotor geometries are defined based on standard IM design practices, and appropriate material properties are assigned to accurately represent electromagnetic behavior.



Transient electromagnetic analysis is performed to compute the time-varying torque produced due to the interaction between the rotating stator magnetic field and induced rotor currents. The electromagnetic torque is calculated using Maxwell's stress tensor method, and the average torque is obtained from the steady-state region of the torque waveform. Torque ripple is evaluated using an RMS-based approach, which captures torque fluctuations caused by slot harmonics and supply harmonics. The results show that the induction motor exhibits smooth torque characteristics with low torque ripple, highlighting the inherent damping effect of rotor currents and the robustness of IM operation as analyzed using ANSYS Maxwell 2D.

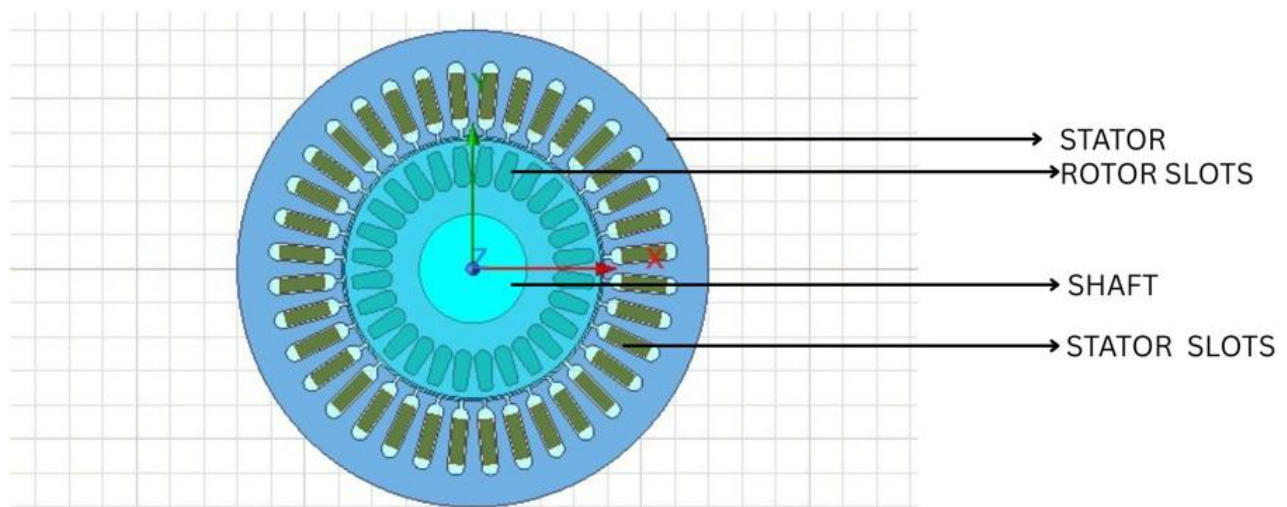


Figure 4.1 Induction Motor 2D Maxwell Design

#### 4.2.1 ANALYSIS AND RESULTS

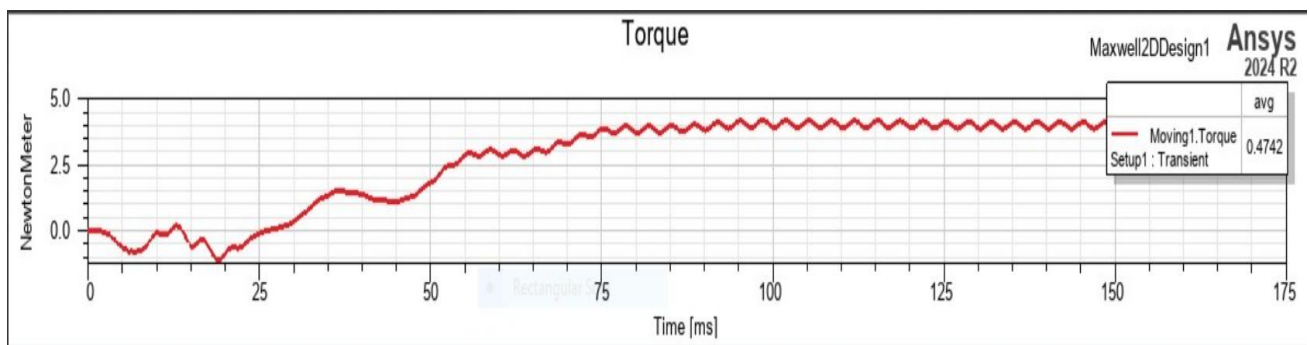


Figure 4.2 Torque vs. Time

The electromagnetic torque waveform (Figure 4.1) of the induction motor shows a smooth build-up during the startup phase, followed by a stable steady-state torque. Initial torque oscillations are observed due to transient rotor current establishment and variations in slip as the motor accelerates. Once steady-state operation is achieved, the torque remains nearly constant with only

very small periodic fluctuations, indicating the presence of low torque ripple. This smooth torque profile is primarily attributed to rotor current damping, which effectively suppresses torque pulsations inherent in the machine. When compared to synchronous reluctance and BLDC motors, the induction motor exhibits the lowest torque ripple, thereby confirming its superior torque smoothness. Such torque characteristics make induction motors well suited for applications requiring reduced vibration, lower acoustic noise, and reliable continuous operation.

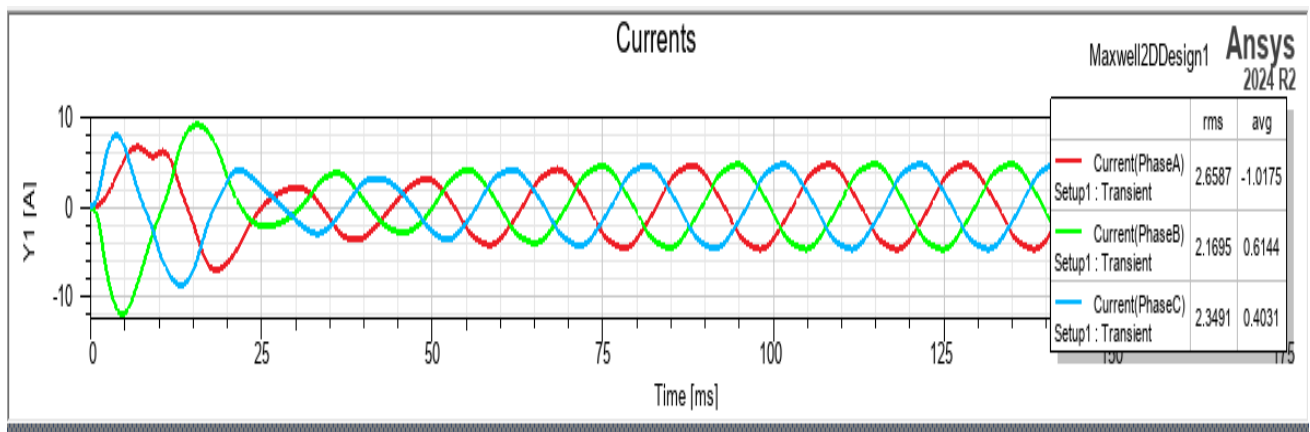


Figure 4.3 Current vs. time

In Figure 4.3 three-phase stator currents of the induction motor are balanced, sinusoidal in nature, and displaced by  $120^\circ$ , indicating proper three-phase excitation and correct stator winding configuration. A higher current magnitude is observed during the startup phase due to inrush current, which gradually decreases as the motor accelerates and reaches steady-state operation. In steady-state, the stator currents settle into smooth sinusoidal waveforms with minimal distortion, reflecting stable electromagnetic performance. The sinusoidal current profile confirms reduced harmonic content and efficient energy conversion within the motor. Rotor current induction provides inherent electromagnetic damping, which contributes to reduced torque ripple and smoother overall operation. These current characteristics further demonstrate the robustness of induction motors under steady operating conditions and varying load demands.

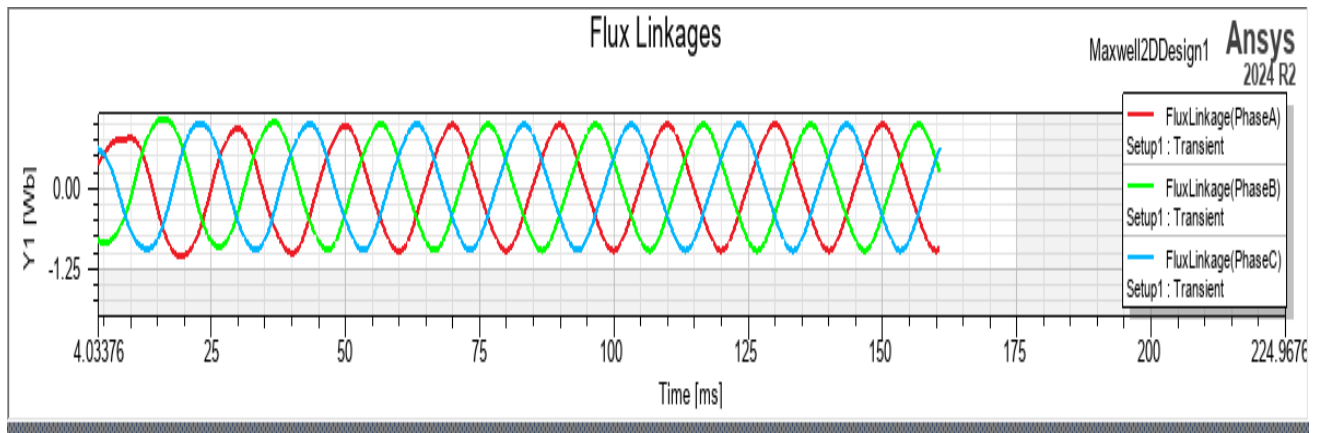


Figure 4.4 Flux linkage vs. time

In Figure 4.4 the flux linkages of all three phases are sinusoidal in nature and remain well balanced, maintaining a  $120^\circ$  phase displacement, which indicates proper magnetic symmetry in the induction motor. The smooth flux linkage waveforms reflect a uniform air-gap flux distribution and stable magnetic field rotation. The absence of flat-topping or sharp distortions in the waveforms suggests low magnetic saturation in both the stator and rotor cores, ensuring efficient magnetic utilization. These characteristics confirm effective electromagnetic coupling between the stator and rotor during operation. The sinusoidal flux linkage directly contributes to smooth torque production with minimal pulsations. Additionally, such flux behavior helps reduce core losses and enhances the overall efficiency of the induction motor.

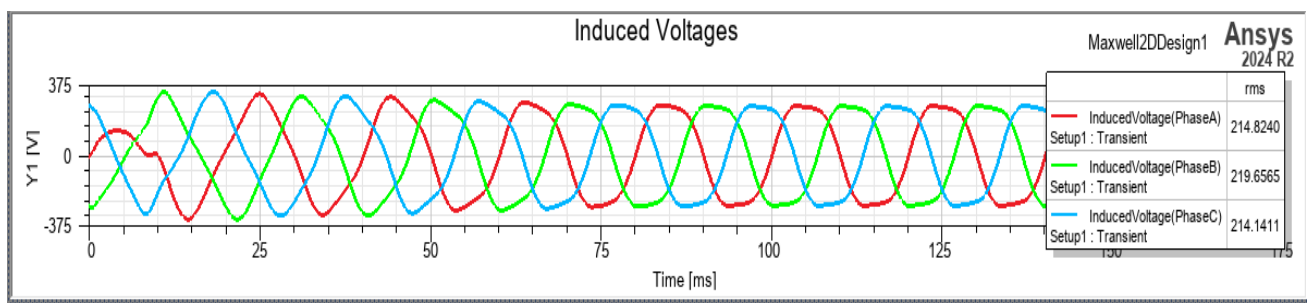


Figure 4.5 Induced voltage vs. time

In Figure 4.5 the induced phase voltages of the induction motor exhibit clean sinusoidal waveforms with equal magnitudes across all three phases, indicating balanced electromagnetic operation. A consistent  $120^\circ$  phase displacement is maintained, confirming correct stator winding symmetry and proper phase alignment. The minimal voltage distortion observed in the waveforms suggests low harmonic content and a uniform magnetic field distribution within the machine. These balanced induced voltages further validate the robustness and accuracy of the

induction motor design. The smooth voltage waveforms contribute to reduced electromagnetic noise and lower electrical losses during operation. Additionally, such voltage characteristics support improved power quality and enhance the reliability of the motor under steady-state conditions. The absence of abrupt voltage transitions also indicates smooth flux variation, which minimizes insulation stress and improves long-term durability of the stator windings.

### **Overall Performance Interpretation**

The induction motor demonstrates excellent torque smoothness with minimal torque ripple due to its inherent electromagnetic damping mechanism. The presence of balanced sinusoidal stator currents, flux linkages, and induced phase voltages indicates stable and reliable steady-state operation. When compared to synchronous reluctance and BLDC motors, the induction motor offers superior torque quality, although with comparatively lower torque density. The simulation results obtained using ANSYS Maxwell 2D accurately capture the characteristic behavior of the induction motor under rated operating conditions. These results confirm the suitability of induction motors for applications requiring smooth operation, reduced vibration, and high operational reliability.

#### **4.2.2 MAGNETIC FLUX DENSITY DISTRIBUTION:**

The magnetic flux density distribution in Fig 4.6 of the induction motor was analyzed using ANSYS Maxwell. The MAG plot illustrates the variation of magnetic flux density within the stator, rotor, and air-gap regions of the machine.

The maximum magnetic flux density obtained from the simulation is approximately 2.1–2.6 T, which occurs mainly in the stator teeth and stator yoke. This is expected since the stator teeth carry the majority of the magnetic flux produced by the stator windings. The rotor core shows comparatively lower flux density because the rotor magnetic field is induced by the stator's rotating magnetic field.

The air-gap region exhibits relatively uniform flux density, indicating a well-distributed magnetic field across the air gap. This uniform distribution is essential for producing smooth electromagnetic torque and minimizing torque ripple.

The simulation results also indicate that the flux follows the intended magnetic path through the stator teeth, across the air gap, through the rotor core, and back through the stator yoke, forming a closed magnetic circuit.

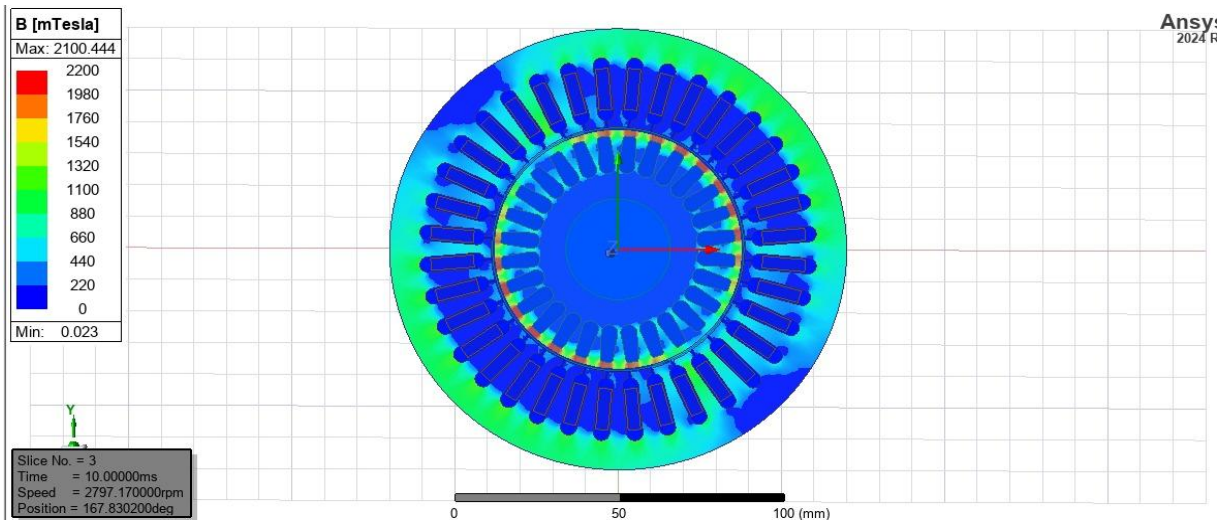


Figure 4.6. Magnetic Flux Density Distribution plot for IM

### 4.2.3 MAGNETIC FLUX LINE DISTRIBUTION

The flux line plot shown in Fig. 4.7 represents the distribution of magnetic vector potential and visualizes the path of magnetic field lines inside the motor. These flux lines form continuous closed loops between the stator and rotor, indicating effective magnetic coupling between the two components. The magnetic field is produced by the stator windings, creating a rotating magnetic field within the machine. This rotating field induces currents in the rotor bars, which generate their own magnetic field that interacts with the stator field to produce electromagnetic torque. The flux lines appear smooth and uniformly distributed across the air-gap region, suggesting balanced magnetic loading and minimal distortion of the magnetic field within the machine.

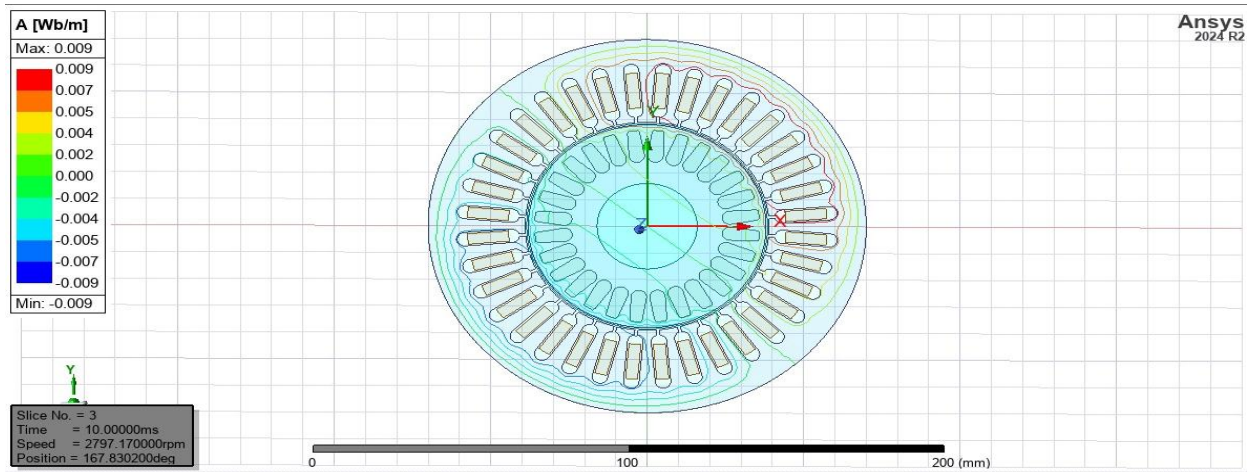


Figure 4.7. Magnetic Flux Line Distribution plot for IM

#### 4.2.4 AIR GAP FLUX CHARACTERISTICS

The air gap is an important region in the machine since it significantly influences torque generation. In an induction motor, the magnetic field produced by the stator crosses the air gap and interacts with the rotor, inducing currents in the rotor conductors. The simulation results indicate that the air-gap flux distribution is fairly uniform, which is beneficial for stable motor performance. A uniform air-gap flux helps reduce electromagnetic force fluctuations and contributes to smoother torque production.

#### 4.2.5 OBSERVATIONS

From the magnetic field analysis of the induction motor, it can be noted that the magnetic flux is mainly concentrated in the stator teeth and stator yoke, showing that these regions carry a large portion of the machine's magnetic flux. The rotor experiences induced magnetic flux due to the rotating magnetic field created by the stator windings, enabling electromagnetic interaction between the stator and rotor. The air-gap flux distribution appears relatively uniform, indicating effective magnetic coupling and stable machine operation. In addition, the flux lines form closed magnetic paths between the stator and rotor, confirming the presence of a proper magnetic circuit. These observations indicate that the induction motor demonstrates an efficient magnetic flux distribution, supporting effective electromagnetic energy conversion.

### 4.3 BRUSHLESS DC MOTOR 2D MAXWELL DESIGN

The brushless DC (BLDC) motor is designed and analyzed using ANSYS Maxwell 2D to evaluate its electromagnetic torque characteristics and torque ripple behavior. A two-dimensional cross-sectional model of the motor is developed, incorporating the stator core, permanent magnets on the rotor, air-gap, and concentrated windings. Appropriate magnetic material properties and excitation conditions corresponding to electronic commutation are assigned to accurately represent BLDC motor operation. Transient electromagnetic analysis is carried out to compute the time-varying electromagnetic torque produced due to the interaction between stator currents and rotor permanent magnet flux. The electromagnetic torque is calculated using Maxwell's stress tensor method, and the average torque is obtained from the steady-state portion of the torque waveform. Torque ripple is evaluated using an RMS-based formulation, capturing torque pulsations caused by trapezoidal back-EMF, phase current switching, and cogging torque. The simulation results indicate that the BLDC motor exhibits higher torque ripple compared to induction and synchronous reluctance motors, which is consistent with the inherent electronic commutation and permanent magnet interaction effects.

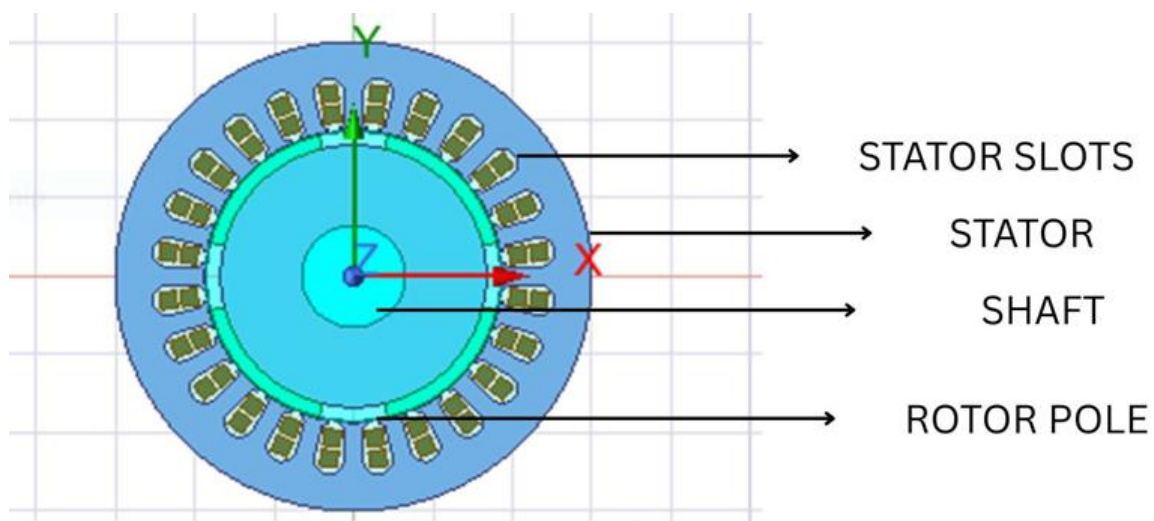


Figure 4.8 Brushless DC Motor 2D Maxwell Design



### 4.3.1 ANALYSIS AND RESULTS

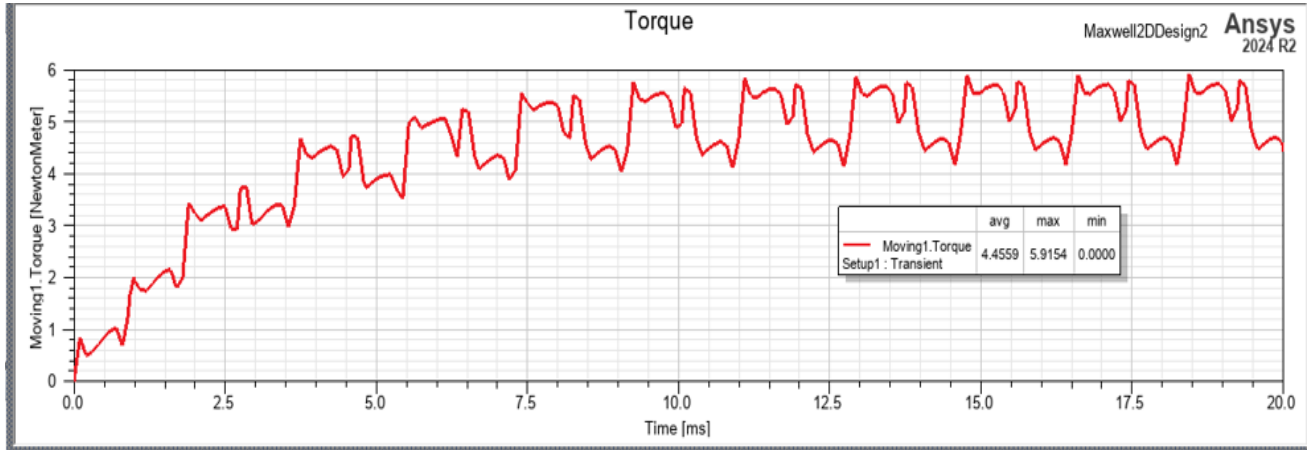


Figure 4.9 Torque vs. Time

The electromagnetic torque waveform shown in Fig. 4.9 displays noticeable ripple characteristics, which are typical for BLDC motors because of electronic commutation. During the starting stage, the torque gradually increases as the motor accelerates and eventually reaches a steady operating condition. After reaching steady state, the torque fluctuates periodically around the average torque value, indicating the presence of torque ripple.

These torque variations mainly occur due to the trapezoidal back-EMF profile, overlap of phase commutation, and cogging torque caused by the interaction between the rotor permanent magnets and the stator slots. In addition, the discrete switching of inverter phases during commutation results in non-uniform torque generation. Excessive torque ripple may lead to increased vibration and acoustic noise, which can negatively influence the overall performance and reliability of the motor drive system if not properly controlled. The obtained results confirm that BLDC motors generally exhibit higher torque ripple when compared with Induction Motors (IM) and Synchronous Reluctance Motors (SynRM).



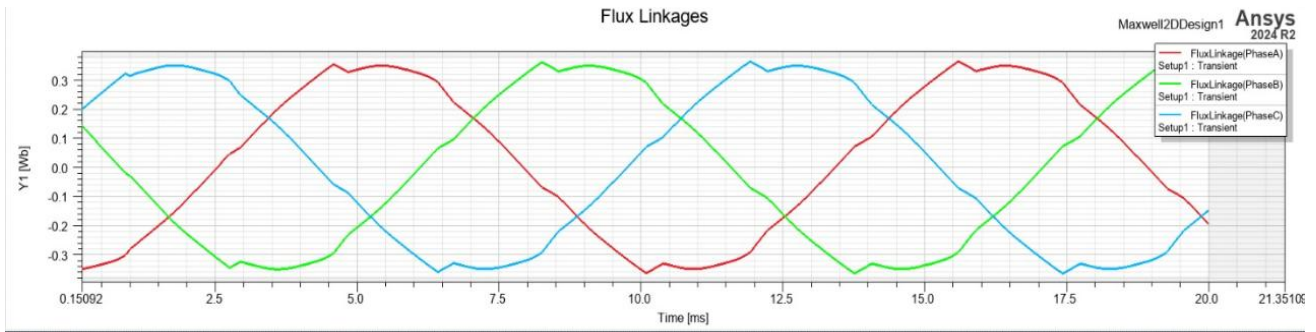


Figure 4.10 Flux linkage vs. time

The three-phase flux linkage waveforms (Figure 4.10) are balanced but non-sinusoidal in nature, reflecting the trapezoidal operating characteristics of the BLDC motor. Each phase exhibits a nearly flat-top profile over a portion of the electrical cycle, which is a typical feature of BLDC machines designed for trapezoidal back-EMF operation. The  $120^\circ$  phase displacement between the three phases is clearly maintained, confirming the correct winding arrangement and symmetrical phase distribution. Minor distortions observed in the waveforms indicate the influence of rotor magnet shape and stator slotting effects on the flux distribution. These flux linkage characteristics play a significant role in torque production and directly contribute to torque pulsations during commutation intervals. Furthermore, any deviation from an ideal trapezoidal flux profile can increase commutation ripple and affect the overall smoothness of motor operation. The consistency of flux linkage magnitude across phases also indicates uniform magnetic loading, which is essential for stable performance and reduced phase imbalance.

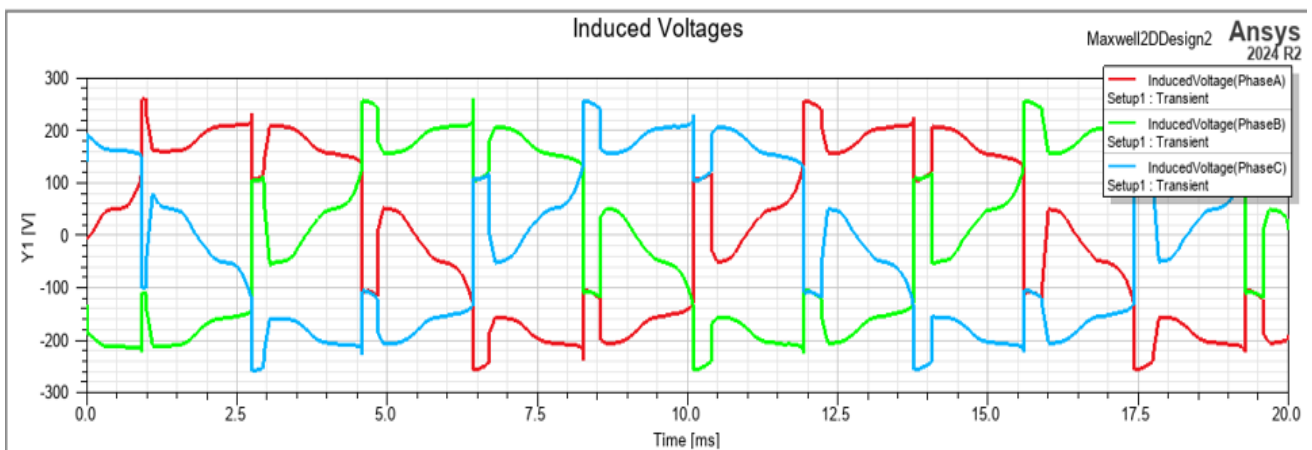


Figure 4.11 Induced voltage vs. time

The induced phase voltages exhibit a trapezoidal waveform (Figure 4.11), which is a defining

characteristic of BLDC motors operating under electronic commutation. Sharp transitions are clearly observed at the commutation points, corresponding to the switching instants of the inverter during electronic commutation. The phase voltages are balanced in magnitude and maintain a  $120^\circ$  phase displacement, confirming symmetrical winding configuration and proper phase alignment. The presence of abrupt voltage changes introduces harmonic components in the voltage waveforms, which further contribute to torque ripple and increased acoustic noise during operation. These voltage characteristics validate the accuracy of the electromagnetic modeling of the BLDC motor in Maxwell 2D. Additionally, the close agreement between the simulated voltage profiles and theoretical BLDC characteristics confirms the effectiveness of the design parameters and boundary conditions used in the simulation.

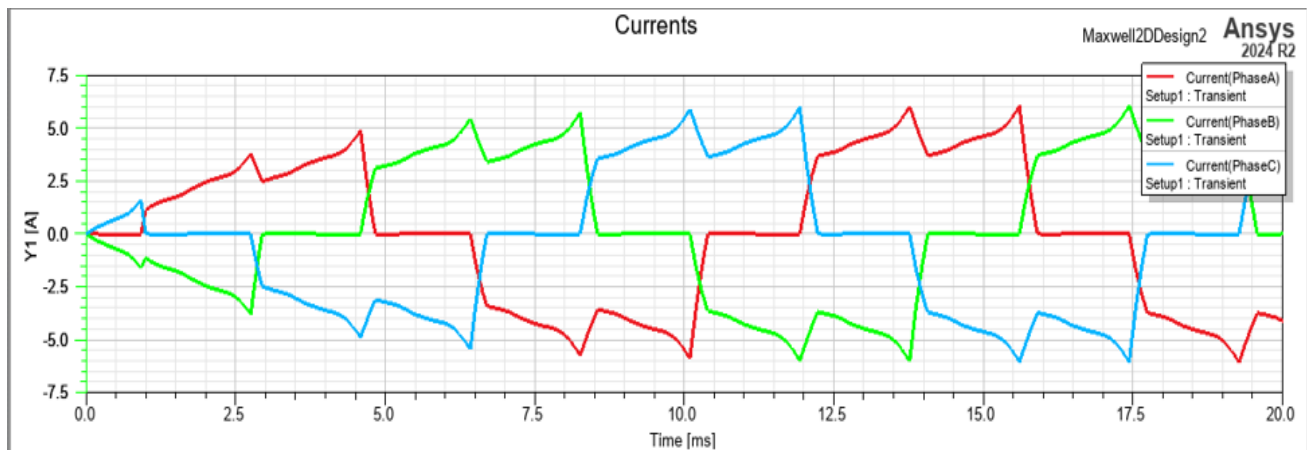


Figure 4.12 Current vs. time

In Figure 4.12 the stator phase currents exhibit a quasi-rectangular, non-sinusoidal waveform, which is typical of BLDC motor current excitation under electronic commutation. Each phase conducts for approximately  $120^\circ$  of electrical duration, consistent with six-step commutation strategy employed in BLDC drives. Clear current transitions between phases are observed, indicating precise commutation events and proper switching action of the inverter. The distortion present in the current waveforms contributes directly to torque ripple as well as increased switching and conduction losses. Balanced current magnitudes across all phases confirm the correctness of the commutation logic and ensure stable motor operation. Furthermore, controlled current overlap during commutation helps maintain continuous torque production while minimizing current discontinuities.

## **Overall Performance Interpretation**

The BLDC motor demonstrates comparatively high torque ripple when compared to induction motors and synchronous reluctance motors, primarily due to electronic commutation and the trapezoidal nature of its back-EMF. The interaction between the rotor permanent magnets and stator slots leads to cogging torque, which further increases torque pulsations during operation. Although BLDC motors offer high torque density and fast dynamic response, their torque smoothness remains inferior in the absence of advanced control and modulation techniques. The simulation results obtained using ANSYS Maxwell 2D effectively capture these inherent characteristics of BLDC motors and closely align with theoretical expectations. These findings highlight the importance of employing improved control strategies, such as current shaping or advanced PWM techniques, to mitigate torque ripple and enhance overall drive performance.

### **4.3.2 MAGNETIC FLUX DENSITY DISTRIBUTION**

The magnetic flux density distribution of the Brushless DC (BLDC) motor shown in Fig. 4.13 was analyzed using ANSYS Maxwell. The MAG contour plot presents the distribution of magnetic flux density across the stator, rotor, and air-gap regions of the motor.

The maximum magnetic flux density obtained from the simulation is approximately 1.7–1.8 T, mainly concentrated in the stator teeth and stator yoke. This occurs because the magnetic flux produced by the rotor permanent magnets flows through the stator teeth and returns through the stator yoke, thereby forming a closed magnetic circuit.

The rotor consists of permanent magnets that act as the primary source of magnetic flux in the BLDC motor. These magnets generate a strong magnetic field that interacts with the stator windings to produce electromagnetic torque. The air-gap region exhibits moderate flux density levels, indicating effective magnetic interaction between the rotor magnets and the stator windings. This interaction is responsible for torque generation in the motor.

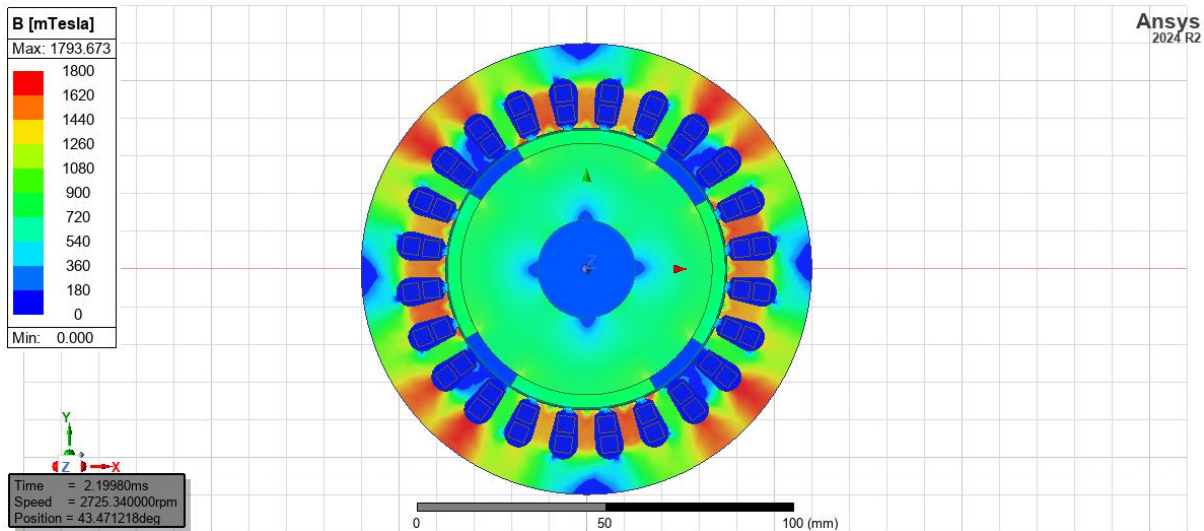


Figure 4.13 Magnetic Flux Density Distribution plot for BLDC

### 4.3.3 MAGNETIC FLUX LINE DISTRIBUTION

The magnetic flux line in Fig 4.14 plot represents the magnetic vector potential distribution inside the BLDC motor. The flux lines originate from the permanent magnets located on the rotor, cross the air gap, pass through the stator teeth, and return through the stator yoke, forming closed magnetic loops.

The direction and shape of the flux lines indicate the proper alignment of magnetic poles and effective coupling between the rotor and stator. The flux paths clearly demonstrate the transfer of magnetic flux from the rotor magnets into the stator magnetic circuit.

However, compared to the induction motor, the flux lines in the BLDC motor tend to concentrate more strongly near the stator teeth and magnet edges due to the presence of permanent magnets. This localized flux concentration is typical in permanent magnet machines.

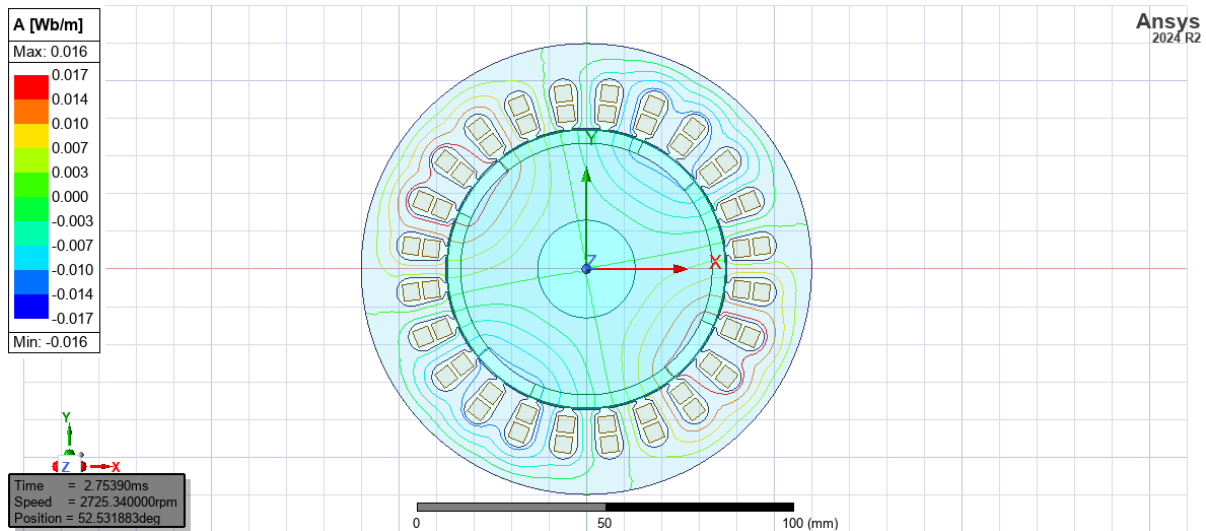


Figure 4.14 Magnetic Flux Line Distribution plot for BLDC

#### 4.3.4 AIR GAP FLUX CHARACTERISTICS

The air-gap flux in a BLDC motor is produced directly by the permanent magnets in the rotor. The simulation results show that the magnetic field crosses the air gap and interacts with the stator windings to generate torque. The air-gap flux density is relatively strong and varies along the air gap due to the discrete magnet poles on the rotor. This variation in magnetic field distribution can lead to torque pulsations, commonly referred to as torque ripple.

#### 4.3.5 OBSERVATIONS

From the magnetic field analysis of the BLDC motor, it can be observed that the magnetic flux is primarily generated by the permanent magnets located on the rotor. The stator teeth experience higher magnetic flux density due to the concentration of magnetic flux in these regions. The air-gap flux distribution varies along the rotor circumference because of the presence of discrete magnet poles on the rotor. Additionally, the magnetic field lines form closed loops between the rotor magnets and the stator iron, establishing a continuous magnetic path. These observations indicate proper magnetic coupling between the stator windings and the rotor magnets, which enables efficient electromagnetic interaction and effective torque production in the BLDC motor.

## 4.4 SYNCHRONOUS RELUCTANCE MOTOR 2D MAXWELL DESIGN

The synchronous reluctance motor (SynRM) is designed and analyzed using ANSYS Maxwell 2D to evaluate its electromagnetic performance. A two-dimensional cross-sectional model of the motor is developed, incorporating the stator geometry, rotor laminations, air-gap, and flux-barrier structure. The stator is kept fixed, while the rotor flux-barrier geometry is parameterized using key design variables such as barrier height, barrier width, fillet radius, bottom barrier radius, first barrier bottom width, and number of barriers per pole. These parameters control the magnetic saliency of the motor by influencing the d-axis and q-axis flux paths. Transient electromagnetic analysis is performed under rated operating conditions, and the time-varying electromagnetic torque is computed using Maxwell's stress tensor method. From the steady-state torque waveform, the average electromagnetic torque is extracted, and torque ripple is evaluated using an RMS-based formulation. The simulation results demonstrate that the designed SynRM achieves moderate torque ripple and satisfactory torque performance, validating the effectiveness of the flux-barrier rotor design using ANSYS Maxwell 2D.

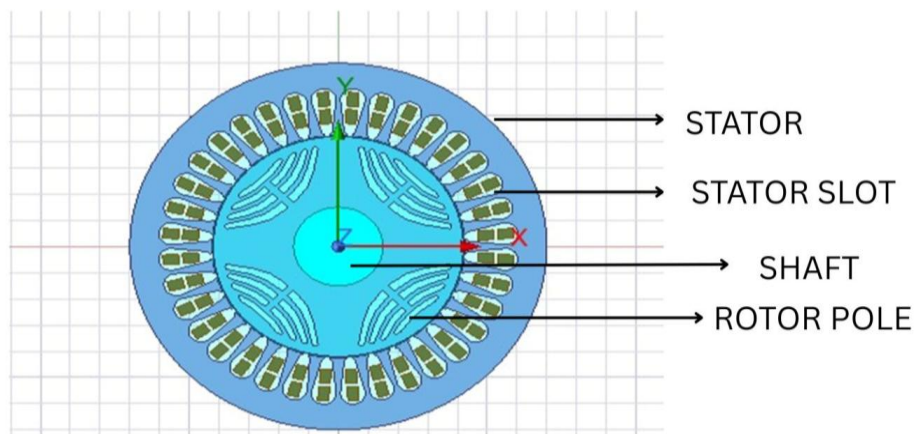


Figure 4.15 Synchronous Reluctance Motor 2D Maxwell Design

#### 4.4.1 ANALYSIS AND RESULTS

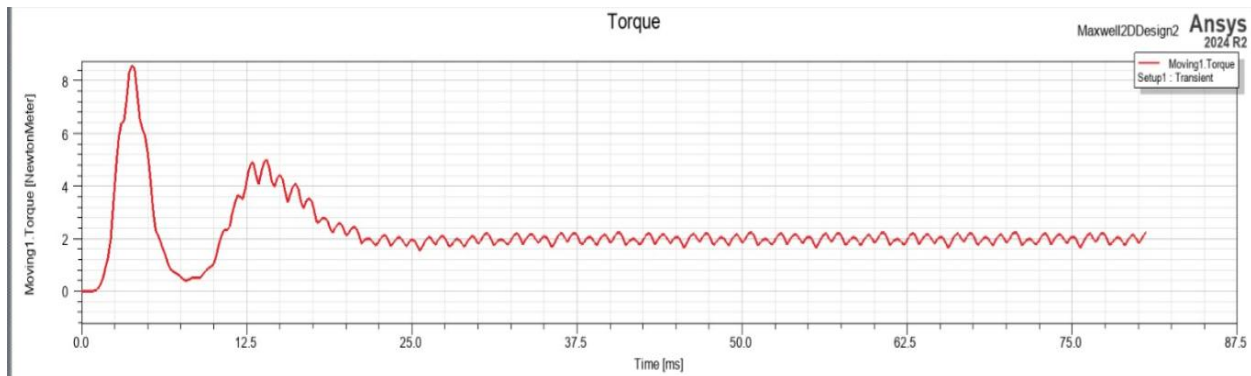


Figure 4.16 Torque vs. Time

In Figure 4.16 the electromagnetic torque waveform of the synchronous reluctance motor exhibits an initial transient peak during the startup phase, followed by a stable steady-state region. After approximately 20 ms, the motor reaches steady-state operation with a nearly constant average torque. Small periodic oscillations are observed around the mean torque value, indicating moderate torque ripple, which is inherent to SynRM operation due to air-gap permeance variation caused by the presence of flux barriers in the rotor. The absence of deep torque dips or negative torque regions confirms stable synchronous operation of the motor. This torque behavior highlights the effectiveness of the rotor saliency in producing reluctance torque while maintaining operational stability.



Figure 4.17 Flux linkage vs. time

The three-phase flux linkage waveforms (Figure 4.17) of the synchronous reluctance motor are balanced and sinusoidal in nature, maintaining a clear 120° phase displacement, which indicates proper stator winding configuration and magnetic symmetry. The absence of significant waveform distortion suggests low magnetic saturation in both the stator and rotor cores, ensuring efficient magnetic operation. The smooth flux linkage profiles further confirm effective d–q axis



flux separation achieved by the flux-barrier rotor design, which is a key characteristic of SynRM operation. Minor amplitude variations observed across the phases are expected due to saliency-based torque production inherent to synchronous reluctance motors. Such flux linkage behavior supports stable torque generation while maintaining good efficiency under steady-state operating conditions.



Figure 4.18 Induced voltage vs. time

In Figure 4.18 the induced phase voltages of all three phases in the synchronous reluctance motor are sinusoidal and well balanced, exhibiting equal magnitudes with a consistent  $120^\circ$  phase displacement. This behavior indicates a uniform air-gap flux distribution and effective electromagnetic coupling between the stator and rotor. The smooth nature of the voltage waveforms confirms reduced harmonic content, thereby validating the effectiveness of the rotor flux-barrier geometry in shaping the magnetic field. Balanced induced voltages further imply good electromagnetic symmetry in the designed SynRM. Such voltage characteristics contribute to improved power quality and reduced electromagnetic noise during steady-state operation.



Figure 4.19 Current vs. time

In Figure 4.19 the stator phase currents of the synchronous reluctance motor are balanced and sinusoidal during steady-state operation, indicating proper current excitation and stable electromagnetic behavior. Initial transient current peaks are observed during the startup phase,



after which the currents gradually settle into steady sinusoidal waveforms. The absence of significant current distortion suggests stable magnetic operation with minimal saturation effects in the machine. Balanced stator currents further confirm proper synchronization between the stator rotating magnetic field and the rotor saliency. These current characteristics support efficient reluctance torque production while maintaining smooth and reliable motor operation.

### **Overall Performance Interpretation**

The combined simulation results indicate that the designed synchronous reluctance motor operates with stable torque production, moderate torque ripple, and well-balanced electromagnetic quantities. The flux-barrier rotor design effectively enhances rotor saliency while maintaining a smooth and uniform flux distribution. The observed level of torque ripple is higher than that of an induction motor but significantly lower than that of a BLDC motor, which is consistent with theoretical expectations. These results validate the suitability of the SynRM design for applications that require good torque density with acceptable torque smoothness. Furthermore, the accurate representation of these characteristics in ANSYS Maxwell 2D confirms the reliability of the electromagnetic modeling and the effectiveness of the chosen design parameters.

#### **4.4.2 MAGNETIC FLUX DENSITY DISTRIBUTION**

The magnetic flux density distribution of the Synchronous Reluctance Motor (SynRM) shown in Fig. 4.20 was analyzed using ANSYS Maxwell. The MAG contour plot illustrates how the magnetic flux density is distributed across the stator, rotor, and air-gap regions of the machine.

Simulation results indicate that the maximum magnetic flux density is mainly concentrated in the stator teeth and stator yoke, where the magnetic field becomes stronger as the flux travels through the stator core. The rotor includes specially designed flux barriers that create regions with different magnetic reluctance values. These barriers guide and restrict the magnetic flux in certain directions, forcing it to follow the low-reluctance iron paths.

As a result, the magnetic flux mainly flows through the rotor ribs, which act as preferred paths for flux circulation. This controlled flux path is essential for generating reluctance torque in the motor. The air-gap region exhibits moderate variations in magnetic flux density, indicating the

interaction between the rotating magnetic field produced by the stator and the anisotropic rotor structure.

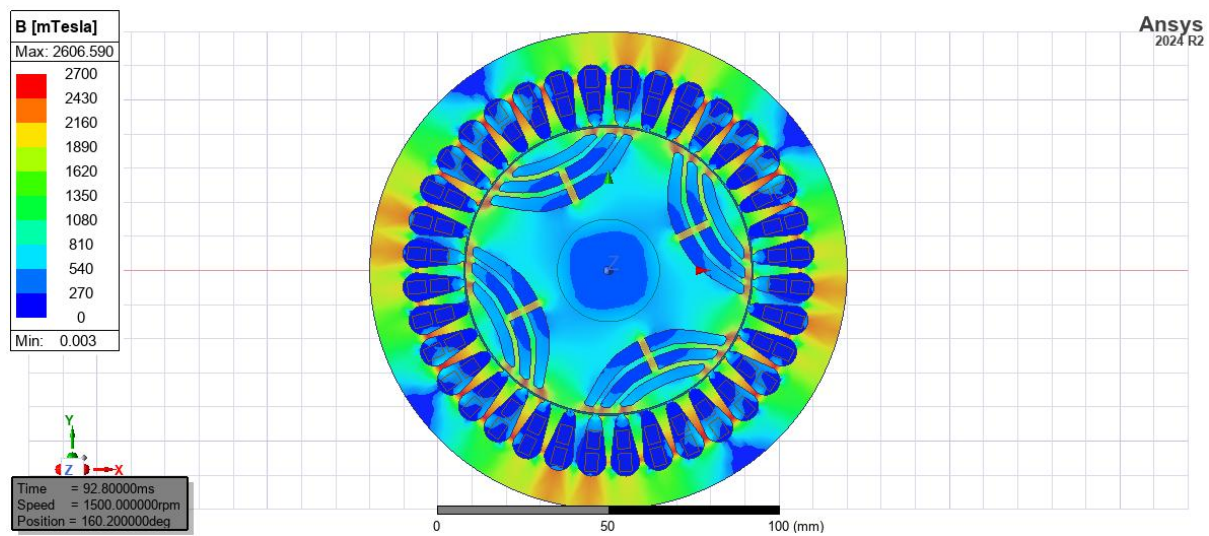


Figure 4.20 Magnetic Flux Density Distribution plot for SynRM

#### 4.4.3 MAGNETIC FLUX LINE DISTRIBUTION

The magnetic flux line plot in Fig 4.21 represents the magnetic vector potential distribution and illustrates the magnetic field paths inside the motor. The flux lines originate from the stator windings, cross the air gap, pass through the rotor iron paths, and return through the stator yoke, forming closed magnetic loops. The presence of rotor flux barriers causes the magnetic field to avoid high-reluctance regions and instead follow the iron ribs within the rotor. This behavior highlights the operating principle of the synchronous reluctance motor, where torque is generated due to the difference in magnetic reluctance along different rotor axes. Compared to conventional motors, the flux lines in the SynRM are influenced by the rotor geometry, resulting in a distinct flux distribution pattern.

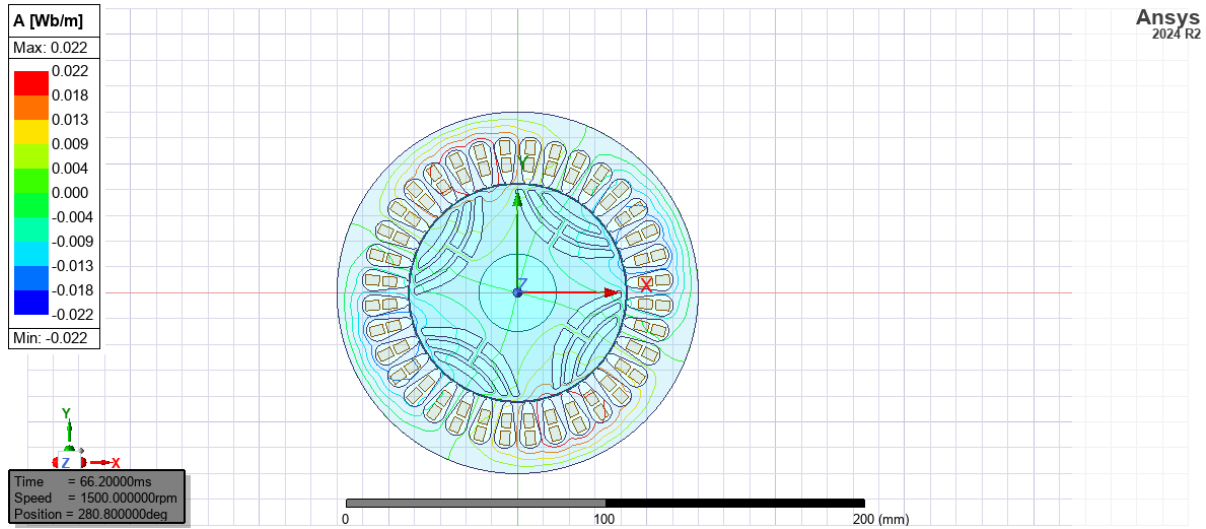


Figure 4.21 Magnetic Flux Line Distribution plot for SynRM

#### 4.4.4 AIR GAP FLUX CHARACTERISTICS

The air gap plays an important role in the operation of the synchronous reluctance motor. The stator windings generate a rotating magnetic field that passes through the air gap and interacts with the rotor structure. Due to the presence of rotor flux barriers, the magnetic reluctance varies depending on the rotor position relative to the stator magnetic field.

This variation causes the rotor to align with the direction of minimum magnetic reluctance, which results in the production of reluctance torque. The air-gap flux distribution shows moderate variation around the circumference because of the rotor barrier geometry. These variations may contribute to torque ripple during machine operation.

#### 4.4.5 OBSERVATIONS

From the magnetic field analysis of the synchronous reluctance motor, it can be observed that the magnetic flux is mainly concentrated in the stator teeth and stator yoke, indicating that these regions carry the majority of the magnetic flux. The rotor flux barriers significantly influence the magnetic path by directing the flux through the low-reluctance rotor ribs while restricting it in the high-reluctance barrier regions. The air-gap flux distribution shows slight variations due to the specific geometry of the rotor. Furthermore, the magnetic field lines form closed loops between the stator and rotor through the air gap, establishing a complete magnetic circuit. These observations confirm the reluctance-based operating principle of the synchronous reluctance motor.

## 4.5 TORQUE RIPPLE CALCULATION AND COMPARISON BETWEEN THE DESIGNED SYNCHRONOUS RELUCTANCE MOTOR, INDUCTION MOTOR AND BRUSHLESS DC MOTOR:

### 4.5.1 TORQUE RIPPLE CALCULATION

The torque ripple for each motor is calculated using the torque–time waveform obtained from the simulation results. To ensure accurate evaluation, only the steady-state region of the torque curve is considered because the initial transient period contains fluctuations caused by motor startup dynamics. Including these transient values may introduce errors in the torque ripple estimation. Therefore, the steady-state torque values are used to compute the average torque ( $T_{avg}$ ) using Eq. (1) and the RMS deviation of torque ( $T_{rms\_dev}$ ) using Eq. (2). Finally, the percentage torque ripple is determined using Eq. (3). This approach ensures that the calculated torque ripple represents the actual steady-state performance of the motor and allows a fair comparison between the Induction Motor, Brushless DC Motor, and Synchronous Reluctance Motor.

$$T_{ave} = \frac{1}{N} \sum_{i=1}^N T_i \quad (4.1)$$

$$T_{rms\_dev} = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i - T_{avg})^2} \quad (4.2)$$

$$Torque\ Ripple_{RMS}(\%) = \frac{T_{rms\_dev}}{T_{avg}} \times 100 \quad (4.3)$$

#### A. Induction motor:

The torque ripple of the Induction Motor is calculated from the torque–time waveform obtained from the simulation results. To avoid the influence of transient startup effects, the steady-state region of the torque curve is considered for the calculation (from equations 4.1, 4.2, 4.3). From the dataset, steady-state operation begins at approximately  $t = 0.08$  s. The average torque ( $T_{avg}$ ) is calculated as 3.61 Nm, and the RMS deviation of torque ( $T_{rms\_dev}$ ) is approximately 0.165 Nm. Using these values, the torque ripple is determined as 4.57%, indicating relatively smooth torque characteristics for the induction motor.

From Dataset:

Steady state starts at:  $t \geq 0.08$  s

$$T_{avg} \approx 3.61\text{ Nm}$$

$$T_{rms\_dev} \approx 0.165\text{ Nm}$$

$$Torque\ Ripple_{RMS} = \frac{0.165}{3.61} \times 100 = 4.57\%$$

### B. Brushless DC motor:

The torque ripple of the Brushless DC (BLDC) motor is determined from the torque–time graph obtained from simulation results. The calculation is performed only after the system reaches steady-state operation to eliminate transient startup effects. From the dataset, the steady-state region begins at approximately  $t = 0.006$  s. The average torque ( $T_{avg}$ ) is calculated as 3.3575 Nm, while the RMS deviation of torque ( $T_{rms\_dev}$ ) is approximately 0.515 Nm. Using these values, the torque ripple is calculated (from equations 4.1,4.2,4.3) as 15.34%, which is comparatively higher due to the switching characteristics and discrete magnet pole structure of the BLDC motor.

From Dataset:

Steady state starts at:  $t \geq 0.006$ s

$$T_{avg} \approx 3.3575\text{ Nm}$$

$$T_{rms\_dev} \approx 0.5150\text{ Nm}$$

$$Torque\ Ripple_{RMS} = \frac{0.5150}{3.3575} \times 100 = 15.34\%$$

### C. Synchronous Reluctance Motor:

The torque ripple of the Synchronous Reluctance Motor (SynRM) is also calculated from the torque–time waveform obtained from simulation results. To ensure accuracy, only the steady-state portion of the torque curve is considered, avoiding the transient startup region. From the dataset, steady-state operation begins at approximately  $t = 0.02$  s. The average torque ( $T_{avg}$ ) is calculated as 2.00 Nm, and the RMS deviation of torque ( $T_{rms\_dev}$ ) is approximately 0.15 Nm. Using these values, the torque ripple is calculated (from equations 4.1,4.2,4.3) as 7.5%, indicating moderate torque fluctuation during steady operation.

From Dataset:

Steady state starts at:  $t \geq 0.02$ s

$$T_{avg} \approx 2.00\text{ Nm}$$

$$T_{rms\_dev} \approx 0.15\text{ Nm}$$

$$Torque\ Ripple_{RMS} = \frac{0.15}{2.00} \times 100 = 7.5\%$$

## 4.5.2 TORQUE RIPPLE COMPARISON

Table 4.1. Torque Ripple Comparison

| MOTOR                        | TORQUE RIPPLE % |
|------------------------------|-----------------|
| Synchronous Reluctance Motor | 7.5%            |
| Induction Motor              | 4.57%           |
| Brushless DC Motor           | 15.34%          |

The torque ripple performance of the three motors was evaluated using RMS torque ripple obtained from ANSYS Maxwell simulations under rated operating conditions. The induction motor (IM) exhibited the lowest torque ripple of 4.57%, which is attributed to the smooth rotating magnetic field and the damping effect of rotor currents. This inherent electromagnetic damping minimizes torque pulsations and results in smoother torque production.

The synchronous reluctance motor (SynRM) showed a moderate torque ripple of 7.5%. In SynRM, torque is produced due to magnetic saliency arising from the difference between d-axis and q-axis inductances. Although the absence of permanent magnets eliminates cogging torque, the presence of rotor flux barriers introduces air-gap permeance variation, leading to higher torque ripple compared to the induction motor. However, optimized flux-barrier geometry helps in limiting the ripple within acceptable levels.

The brushless DC motor (BLDC) exhibited the highest torque ripple of 15.34%. This is mainly due to electronic commutation, trapezoidal back-EMF, phase current switching, and cogging torque caused by the interaction between permanent magnets and stator slots. These factors inherently introduce torque pulsations, resulting in significantly higher torque ripple compared to IM and SynRM.

## 4.6 DATASET GENERATION

Latin Hypercube Sampling (LHS) is used to systematically generate combinations of SynRM rotor flux-barrier design parameters within predefined limits. The selected input parameters include barrier height, barrier width, fillet radius, bottom barrier radius, first barrier bottom width, and the number of barriers per pole. For a specified number of samples, each parameter

range is divided into equal probability intervals, and one value is randomly selected from each interval.

These sampled values are then combined to generate unique parameter sets, ensuring uniform coverage of the entire design space. Each parameter combination generated using LHS corresponds to a unique rotor geometry, which is modeled and simulated using finite element analysis in ANSYS Maxwell under rated operating conditions.

From every simulation, key electromagnetic performance parameters such as average torque, torque ripple, and efficiency are obtained. The design parameters and their corresponding simulation results are organized into a structured dataset, which is later used for performance analysis, optimization, and machine-learning-based modeling.

#### **4.6.1 INPUT PARAMETERS**

The dataset generation using Latin Hypercube Sampling is based on rotor geometric parameters that directly influence the electromagnetic behavior of the Synchronous Reluctance Motor (SynRM). These parameters mainly control the d-axis and q-axis magnetic paths, which significantly affect torque production, torque ripple, and overall efficiency.

##### **1. Barrier Height (H) – mm**

Barrier height represents the radial thickness of the flux barrier in the synchronous reluctance motor rotor. It defines how deeply the flux barrier penetrates radially into the rotor core and directly influences the magnetic path available for flux flow inside the rotor.

Barrier height plays a crucial role in controlling the q-axis magnetic flux path. An increase in barrier height increases the reluctance of the q-axis flux path, thereby reducing the q-axis inductance  $L_q$ . Since the d-axis inductance  $L_d$  remains relatively unaffected, the saliency difference  $(L_d - L_q)$  increases. This enhanced saliency directly leads to higher average electromagnetic torque production. Additionally, an optimally chosen barrier height helps in smoothing the flux distribution within the rotor, which reduces air-gap permeance harmonics, torque ripple, and local magnetic saturation. As a result, iron losses are reduced, contributing to improved overall efficiency.

## 2. Barrier Width (W) – mm

Barrier width refers to the circumferential opening of the flux barrier along the rotor periphery. It determines the angular extent of the low-permeability region created by the flux barrier in the rotor.

Barrier width significantly influences the effectiveness of q-axis flux blocking. A wider barrier increases the reluctance along the q-axis, thereby lowering the q-axis permeance and inductance  $L_q$ . This results in an increased saliency ratio, which improves the average torque capability of the motor. Furthermore, an optimal barrier width helps in reducing air-gap permeance harmonics by providing a more uniform magnetic field distribution. This reduction in harmonics lowers torque ripple and associated acoustic noise, while also decreasing iron losses, thereby enhancing motor efficiency.

## 3. Fillet Radius (R) – mm

Fillet radius is the radius provided at the sharp corners of the flux barriers to smoothen the geometry of the rotor. It modifies the local geometry at regions where abrupt changes in flux direction occur.

Sharp corners in the flux barrier geometry cause local magnetic flux concentration, leading to high levels of magnetic saturation and increased iron losses. Increasing the fillet radius smoothenes the magnetic flux path, thereby reducing peak flux density and mitigating saturation effects. Although the impact of fillet radius on average torque is relatively small, it plays a vital role in significantly reducing torque ripple by minimizing local flux disturbances. The reduction in saturation and harmonic flux components also leads to lower iron losses, improving the efficiency and thermal performance of the motor.

## 4. Bottom Barrier Radius ( $R_\beta$ ) – mm

Bottom barrier radius denotes the radial position of the bottom end of the flux barrier measured from the rotor center. It defines how close the barrier extends toward the air-gap region.

The bottom barrier radius determines the balance between d-axis flux conduction and q-axis flux blocking. Proper placement of the barrier ensures efficient guidance of d-axis flux through low-reluctance paths while maintaining sufficient obstruction to q-axis flux. An optimally selected  $R_\beta$  improves magnetic saliency and minimizes unwanted flux leakage paths within the rotor. This results in higher average torque, reduced torque ripple due to smoother air-gap permeance variation, and improved efficiency due to reduced leakage flux and iron losses.



## **5. First Barrier Bottom Width ( $B_0$ ) – mm**

First barrier bottom width refers to the width of the flux barrier at the rotor tooth root, commonly known as the iron bridge between adjacent flux barriers.

$B_0$  plays a dual role by affecting both the mechanical integrity and magnetic performance of the rotor. A sufficiently large bottom width strengthens the iron bridge, preventing excessive mechanical stress and vibration. Magnetically, it reduces saturation in the d-axis flux path by providing adequate iron area for flux conduction. This stabilization of flux flow improves average torque capability and reduces torque oscillations. Moreover, by minimizing saturation and leakage effects, an optimal  $B_0$  contributes to reduced losses and improved efficiency.

## **6. Number of Barriers per Pole ( $N_\beta$ ) – Dimensionless**

Number of barriers per pole represents the count of flux barriers incorporated within each rotor pole of the SynRM.

Increasing the number of barriers per pole enhances the separation between d-axis and q-axis flux paths, thereby improving magnetic saliency. More barriers result in smoother air-gap permeance variation as the rotor rotates, which significantly reduces torque ripple. Improved flux control also leads to better utilization of the rotor core material, lowering harmonic losses and improving efficiency. However, an excessive number of barriers may increase flux leakage, reduce mechanical robustness, and complicate manufacturing, making it essential to select an optimal number of barriers per pole.

## **4.6.2 INPUT PARAMETERS EXTRACTION USING LATIN HYPERCUBE SAMPLING**

Latin Hypercube Sampling (LHS) is employed to systematically extract input parameter combinations for SynRM rotor design within predefined feasible ranges. In this study, a total of 1000 samples are generated to ensure sufficient coverage of the design space. The selected input parameters include barrier height, barrier width, fillet radius, bottom barrier radius, first barrier bottom width, and number of barriers per pole. For each parameter, minimum and maximum limits are defined based on electromagnetic performance requirements, mechanical constraints, and manufacturing feasibility, thereby establishing the permissible design space.

During the LHS process, the range of each input parameter is divided into 1000 equal probability intervals. One value is randomly selected from each interval for every parameter, and the

sampled values are then randomly combined to form 1000 unique input parameter sets. This approach guarantees that the entire range of each design variable is uniformly represented without clustering of samples in any particular region. The resulting set of 1000 input parameter combinations is subsequently used to generate corresponding rotor geometries and perform finite element simulations in ANSYS Maxwell, forming a comprehensive dataset for performance analysis, optimization, and machine learning-based modeling.

#### 4.6.3 RANGE OF INPUT PARAMETERS TAKEN USING LATIN HYPERCUBE SAMPLING

Table 4.2. Range of Input Parameters

| No. | Parameter Name        | Symbol | Range       | Type    | Unit |
|-----|-----------------------|--------|-------------|---------|------|
| 1   | Barrier height        | H_mm   | 1.0 – 6.0   | Float   | mm   |
| 2   | Barrier top width     | W_mm   | 1.0 – 5.0   | Float   | mm   |
| 3   | Fillet radius         | r_mm   | 0.2 – 2.5   | Float   | mm   |
| 4   | Bottom barrier radius | Rb_mm  | 18.0 – 26.0 | Float   | mm   |
| 5   | Barrier base width    | Bo_mm  | 1.0 – 6.0   | Float   | mm   |
| 6   | Barriers per pole     | N_b    | 2 – 4       | Integer | –    |

The ranges of the rotor flux-barrier parameters were selected based on a combination of electromagnetic design considerations, mechanical feasibility, manufacturing constraints, and values reported in existing SynRM literature. The objective was to ensure that all sampled designs are physically realizable, while still providing sufficient variability to capture the influence of each parameter on torque, torque ripple, and efficiency. The chosen limits allow

exploration of both conservative and aggressive rotor geometries without leading to excessive magnetic saturation, mechanical weakness, or unrealistic flux leakage.

The barrier height (1.0–6.0 mm) and barrier top width (1.0–5.0 mm) ranges were selected to vary the degree of q-axis flux blocking. Lower values represent shallow or narrow barriers with weaker saliency, while higher values correspond to stronger q-axis flux obstruction, which increases saliency but may introduce saturation if excessive. These limits ensure a balance between torque enhancement and magnetic saturation avoidance. The fillet radius range (0.2–2.5 mm) was chosen to study the effect of corner smoothing on flux concentration; smaller radii represent sharp corners that can cause local saturation, while larger radii provide smoother flux paths and reduced torque ripple.

The bottom barrier radius (18.0–26.0 mm) range was selected to control the radial placement of the flux barrier relative to the air-gap. This range allows investigation of both shallow and deeper barrier placements, influencing d-axis flux conduction, leakage flux, and air-gap permeance variation. The barrier base width (1.0–6.0 mm) was chosen to ensure sufficient mechanical strength of the rotor tooth while allowing variation in magnetic saturation levels at the iron bridge. Finally, the number of barriers per pole (2–4) was selected as a discrete integer parameter since it strongly affects saliency and torque ripple; fewer barriers result in higher ripple, while more barriers improve flux separation but increase manufacturing complexity. Overall, these ranges are chosen to provide a comprehensive yet realistic design space, making them suitable for Latin Hypercube Sampling-based dataset generation and subsequent optimization or machine learning analysis.

#### **4.6.4 OUTPUT PARAMETERS**

The output parameters are obtained from finite element simulations performed in ANSYS Maxwell for each rotor geometry generated using Latin Hypercube Sampling. These parameters are used to evaluate the electromagnetic performance and torque quality of the synchronous reluctance motor.

##### **A. Average Electromagnetic Torque ( $T_{avg}$ )- Nm**

Average torque represents the mean value of the electromagnetic torque produced by the motor over one electrical cycle during steady-state operation.

Average torque is a primary performance indicator of the motor and reflects its load-carrying capability. In a SynRM, average torque depends on the magnetic saliency difference ( $L_d - L_q$ ) which is controlled by the rotor flux-barrier geometry. Higher average torque indicates improved utilization of magnetic saliency and better electromagnetic performance.

### B. Torque Ripple (RMS) – %

Torque ripple represents the fluctuation of electromagnetic torque around its average value. In this work, torque ripple is evaluated using an RMS-based formulation.

$$T_{rms\_dev} = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i - T_{avg})^2} \quad (4.2)$$

$$Torque\ Ripple_{RMS}(\%) = \frac{T_{rms\_dev}}{T_{avg}} \times 100 \quad (4.3)$$

Torque ripple is a critical parameter affecting vibration, acoustic noise, and mechanical stress. Lower torque ripple indicates smoother torque production and improved dynamic performance, which is essential for reliable and quiet motor operation.

### C. Electromagnetic Efficiency

Efficiency represents the ratio of output mechanical power to input electrical power.

$$\eta = \frac{P_{out}}{P_{in}} \text{ where } P_{out} = T_{avg} \times \omega \quad (4.4)$$

Efficiency quantifies how effectively the motor converts electrical energy into mechanical energy. Rotor geometries that minimize iron losses, leakage flux, and magnetic saturation result in higher efficiency, making this parameter important for overall motor performance evaluation.

## 4.6.5 OUTPUT PARAMETERS EXTRACTION USING 2D-MAXWELL FEA SIMULATION

The electromagnetic performance of each SynRM rotor geometry generated using Latin Hypercube Sampling is evaluated using ANSYS Maxwell 2D transient simulation. After defining the motor geometry, material properties, excitation conditions, and boundary conditions, transient electromagnetic analysis is performed under rated operating conditions. The simulation computes the time-varying electromagnetic torque based on Maxwell's stress tensor method, which accurately captures the interaction between the stator magnetic field and the rotor saliency.

The average electromagnetic torque is obtained from the transient torque waveform after the motor reaches steady-state operation. The torque signal is monitored over one or more electrical cycles, and its mean value is calculated using the post-processing tools available in Maxwell 2D (Results → Create Report → Transient → Torque). To ensure accuracy, the initial transient portion of the waveform is excluded, and only steady-state torque data are considered. The torque ripple is then evaluated using an RMS-based definition, where the deviation of instantaneous torque values from the average torque is computed over the selected steady-state interval. This approach effectively quantifies torque pulsations caused by air-gap permeance harmonics and magnetic saturation effects.

The electromagnetic efficiency is determined by computing the ratio of output mechanical power to input electrical power. The output power is calculated using the extracted average torque and rotor speed, while the input power is obtained from the electrical input quantities defined in the simulation. ANSYS Maxwell automatically accounts for electromagnetic losses such as iron loss and copper loss during post-processing, enabling accurate efficiency estimation. The extracted output parameters—average torque, torque ripple, and efficiency—are compiled for all LHS-generated designs to form a comprehensive dataset used for performance evaluation, optimization, and machine learning analysis.

## **4.7 SUMMARY**

In this section, the key rotor design parameters of the Synchronous Reluctance Motor (SynRM) were identified and their importance in influencing the electromagnetic performance of the motor was discussed. Parameters such as barrier height, barrier top width, fillet radius, bottom barrier radius, barrier base width, and number of barriers per pole were selected as optimization variables because they directly affect magnetic flux distribution, reluctance variation between the d-axis and q-axis, torque production, and torque ripple. To systematically generate

combinations of these parameters within their feasible ranges, Latin Hypercube Sampling (LHS) was employed. This method ensures efficient coverage of the design space while reducing the number of simulations required. A dataset of parameter combinations was generated based on the defined ranges, which was later used for finite element simulations and machine learning model training in the subsequent optimization process.

# IMPLEMENTATION OF RANDOM FOREST AND GENETIC ALGORITHM

## 5.1 INTRODUCTION

In this chapter, the optimization process of the Synchronous Reluctance Motor Rotor design, as well as its validation, will be presented, utilizing a hybrid framework that combines a Random Forest model with a Genetic Algorithm. Initially, the performance of the Random Forest model will be examined regarding its ability to efficiently predict the performance of the Synchronous Reluctance Motor, regarding certain performance indicators, such as average torque as well as torque ripple. In the next step, the importance of each of the parameters regarding the performance of the Synchronous Reluctance Motor will be examined, aiming to identify the parameters with the most significant impact. Finally, a Genetic Algorithm will be utilized, based on the performance of the RF model, aiming to find the optimal parameters of the Synchronous Reluctance Motor Rotor design, which will result in increased average torque while at the same time minimizing torque ripple. The results of the optimization process will be validated through time-domain finite element simulation, using the ANSYS Maxwell software, while the results of the RF model will be compared with the simulation results.

## 5.2 RANDOM FOREST BASED SURROGATE MODEL

### 5.2.1 PURPOSE OF USING RANDOM FOREST

Finite Element Analysis (FEA) offers an accurate method of electromagnetic evaluation of the SynRM. However, it is computationally intensive to run FEA simulations for various design iterations. To address this issue, a Random Forest (RF) regression method acts as a surrogate to efficiently compute the characteristics of the SynRM. This method learns the non-linear relationship between the SynRM rotor design parameters and the SynRM performance outputs.

### 5.2.2 LIBRARIES AND TOOLS USED

The implementation process is done using various data science and machine learning libraries based on the Python programming language. Libraries such as Pandas and NumPy are used to

process data in an efficient manner. Scikit-learn is also used in the implementation process to train the model. Finally, the model is visualized using the Matplotlib library to carry out graphical analysis when needed. This shows that all the libraries used in the implementation process are efficient in their own way and can carry out the implementation process in a reliable and accurate manner.

### 5.2.3 DATASET DESCRIPTION

**Input Features:** The input features are the main parameters of the Synchronous Reluctance Motor (SynRM) rotor geometry, which affect the magnetic flux path in the motor. These input features include barrier height, barrier width, fillet radius, bottom barrier radius, barrier offset, and the number of barriers per pole. These input features affect the flux barriers in the SynRM, which control the reluctance difference between the d-axis and q-axis. The selection of input features is critical in controlling the magnetic flux, torque-producing capabilities, and torque ripple in the SynRM.

**Output Targets:** The output targets refer to the performance characteristics obtained from the finite element simulation using ANSYS Maxwell. The output targets in this study include average torque and torque ripple percentage. Average torque refers to the performance characteristics of the motor, which produces torque. The torque ripple refers to the variation in torque, which occurs when the motor operates. The variation in torque, also known as torque ripple, affects the smooth performance of the motor.

### 5.2.4 RANDOM FOREST MODEL TRAINING – HYPERPARAMETERS AND DESCRIPTION

Two separate Random Forest Regression models were developed to predict Average Torque (Tavg) and Torque Ripple. The dataset was split into training and testing sets using an 80:20 ratio with a fixed random state to ensure reproducibility. Both models were trained using the same input feature set to maintain consistency.



Table 5.1. Train–Test Split Configuration

| Parameter      | Value           | Description                                          |
|----------------|-----------------|------------------------------------------------------|
| Training Data  | 80%             | Used for learning patterns from input features       |
| Testing Data   | 20%             | Used for validating model performance                |
| Random State   | 42              | Ensures consistent data splitting across runs        |
| Input Features | X               | Independent variables used for prediction            |
| Output Targets | Tavg,<br>Ripple | Continuous variables predicted using separate models |

Table 5.2. Random Forest Regressor Hyperparameters

| Hyperparameter    | Value | Description                                                             |
|-------------------|-------|-------------------------------------------------------------------------|
| n_estimators      | 400   | Number of decision trees in the forest; higher value improves stability |
| max_depth         | 30    | Maximum depth of each tree to control model complexity                  |
| max_features      | 2     | Number of features considered at each split to reduce overfitting       |
| min_samples_split | 2     | Minimum samples required to split an internal node                      |
| min_samples_leaf  | 1     | Minimum samples required at a leaf node                                 |
| random_state      | 42    | Ensures reproducible model training                                     |

### 5.2.5 RANDOM FOREST MODEL VALIDATION AND PERFORMANCE EVALUATION

The effectiveness of the trained Random Forest surrogate models is evaluated using common statistical performance metrics, including the coefficient of determination ( $R^2$ ), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Separate surrogate models are developed for predicting average torque and torque ripple.

Table 5.3. Random Forest Performance Evaluation

| Output Parameter                | $R^2$  | RMSE   | MAE    |
|---------------------------------|--------|--------|--------|
| Average Torque<br>( $T_{avg}$ ) | 0.9833 | 0.7286 | 0.3480 |
| Torque Ripple (%)               | 0.9901 | 0.0920 | 0.0667 |

These values give a quantitative estimate of the accuracy and reliability of the prediction results by comparing the predicted output with the respective simulation results obtained from ANSYS Maxwell. The Random Forest models show high accuracy in predicting average torque and torque ripple values. The  $R^2$  values obtained are higher than 0.98, indicating a strong correlation between the predicted values and the simulation results. Moreover, the low values of the Root Mean Square Error (RMSE) and MAE values indicate that the prediction error is minimal. Therefore, the Random Forest models trained on the simulation results can replace the finite element simulations in the optimization process. The Mean Absolute Error (MAE) is a value indicating the average absolute difference between the predicted values and the respective simulation results. In the case of average torque prediction, the MAE value is approximately 0.35 Nm, indicating that the model predicts the average torque value with a small deviation in relation to the nominal value. Similarly, in the case of torque ripple prediction, the Random Forest model shows a very low MAE value, indicating high accuracy in predicting the torque ripple value.

### 5.2.6 FEATURE IMPORTANCE

Feature importance analysis was conducted on the trained Random Forest models to analyze the relative importance of the rotor's geometric parameters in determining the average torque and torque ripple. For the case of average torque prediction (Figure 5.1), it is observed that the barrier height ( $H_{mm}$ ) is the most important parameter, which shows the sensitivity of torque production to changes in barrier height. The barrier offset ( $B0_{mm}$ ) and barrier width ( $W_{mm}$ )

are the next important parameters, which emphasize the importance of barrier design in controlling the magnetic flux path in the rotor. The parameters, such as fillet radius ( $R_{mm}$ ) and bottom barrier radius ( $Rb_{mm}$ ), have moderate importance, while the number of barriers per pole has the least importance in determining the average torque. For the case of torque ripple prediction (Figure 5.2), the barrier width ( $W_{mm}$ ) is observed to have the highest importance, which shows the importance of barrier width in controlling the torque ripple. The barrier height ( $H_{mm}$ ) is the next important parameter, followed by the barrier offset ( $B0_{mm}$ ). The other parameters, such as fillet radius, bottom barrier radius, and the number of barriers per pole, have relatively less importance in determining the torque ripple.

If one compares the above models, it is observed that the barrier height ( $H_{mm}$ ) is mainly affecting the average torque, while the barrier width ( $W_{mm}$ ) has a higher impact on reducing the torque ripple. Even though  $H_{mm}$  and  $W_{mm}$  are key factors in both models, their importance is different depending on the objective function to be optimized.

Overall, the comparative feature importance analysis has provided some physical insights into the design sensitivity of synchronous reluctance motor rotors, which is helpful in choosing the key parameters to be optimized.

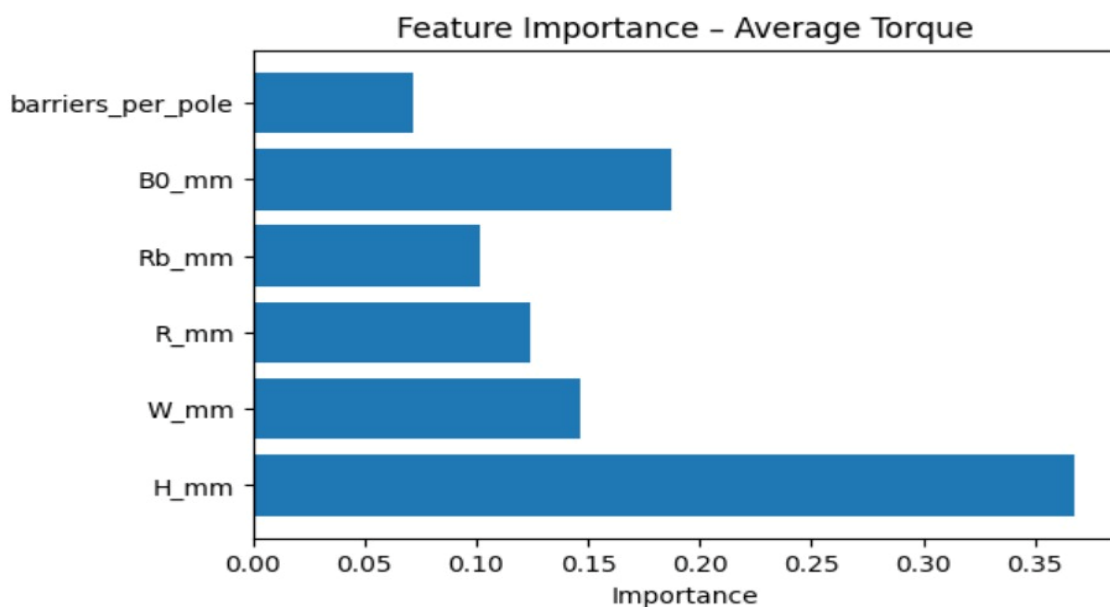


Figure 5.1 Feature Importance – Average Torque (Random Forest Regressor)

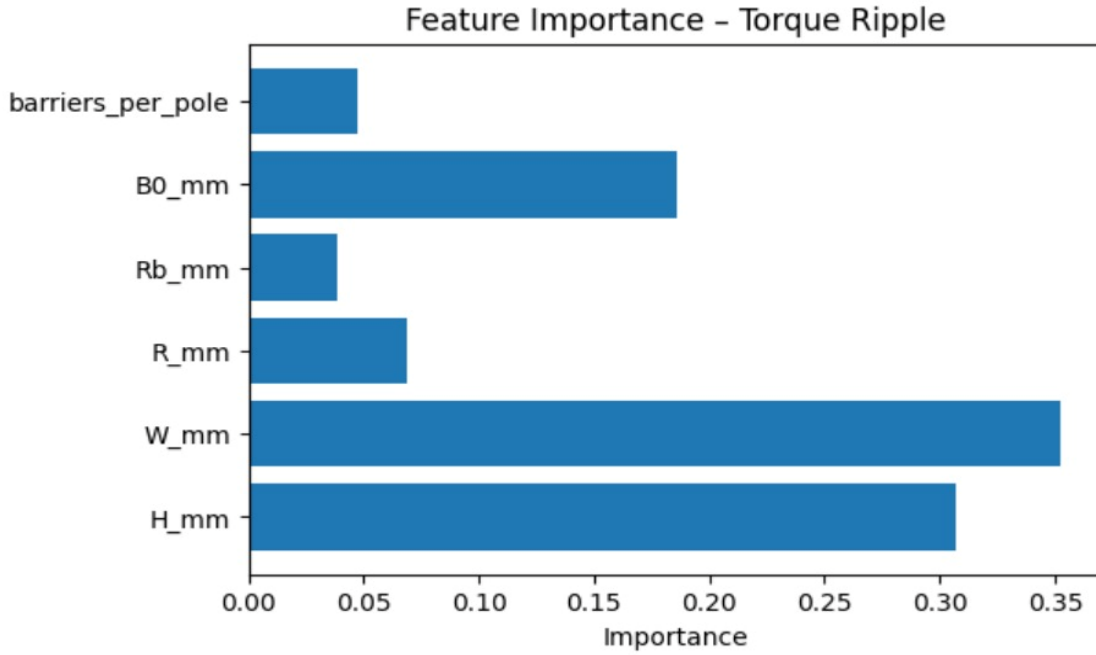


Figure 5.2 Feature Importance – Torque Ripple (Random Forest Regressor)

## 5.3 GENETIC ALGORITHM BASED OPTIMIZATION

### 5.3.1 GENETIC ALGORITHM BASED OPTIMIZATION OF SYNRM

After training the surrogate models using the Random Forest algorithm, a Genetic Algorithm (GA) is utilized to find the optimal rotor geometry of the SynRM. The GA algorithm is used because it can efficiently solve nonlinear optimization problems with multiple parameters. The GA algorithm uses the RF surrogate models to quickly calculate the fitness function for each design candidate, thus eliminating the need to perform simulations.

### 5.3.2 DESIGN VARIABLES AND SEARCH SPACE

The design variables chosen for optimization using the Genetic Algorithm (GA) optimization tool are the critical parameters of the Rotor Geometry of the Synchronous Reluctance Motor (SynRM) which affect its electromagnetic performance. These parameters include the barrier height (H), barrier width (W), fillet radius (R), bottom barrier radius (Rb), barrier offset (B0), and the number of barriers per pole (Nb). These parameters affect the flux barriers created in the rotor, which in turn affect the flux distribution and the reluctance difference between the d-axis and q-axis. By varying the design parameters between the lower and upper bounds obtained from

the dataset created using Maxwell simulations, a search space is created. The Genetic Algorithm optimization tool searches the space and finds the optimum combination of the Rotor Geometry parameters which maximize average torque and minimize torque ripple..

### 5.3.3 FITNESS FUNCTION AND CONSTRAINTS

The objective of the optimization is to maximize average torque while minimizing torque ripple. The fitness function is formulated as:

$$\text{Fitness} = |\text{Tavg} - 4 \times \text{Torque Ripple}| \quad (5.1)$$

A penalty-based constraint is applied to strictly limit torque ripple. If the predicted torque ripple exceeds 5.3%, a large penalty is imposed on the fitness value, forcing the GA to reject such solutions.

This constraint ensures that only low-ripple designs suitable for EV traction applications are retained during the GA search.

### 5.3.4 GENETIC ALGORITHM HYPERPARAMETERS

The Genetic Algorithm is implemented with the following parameters:

Table 5.4. Genetic Algorithm Hyperparameters

| Parameter             | Value                |
|-----------------------|----------------------|
| Population size       | 40                   |
| Number of generations | 50                   |
| Selection method      | Tournament selection |
| Crossover             | Arithmetic crossover |
| Mutation rate         | 0.1                  |

The GA iteratively evolves the population through selection, crossover, and mutation operations until convergence is achieved.

### 5.3.5 OPTIMIZATION PROCEDURE

The overall process of optimization using the Genetic Algorithm (GA) technique involves the initialization of a random population within specified design constraints. The predicted average value of torque and torque ripple for each individual solution are obtained using the trained Random Forest (RF) regression models. The fitness function is then utilized to evaluate the performance of each individual solution based on the predicted outputs. The tournament selection operator is employed to

select parent solutions. This operator ensures that parents with higher fitness have a higher probability of being selected. Crossover and mutation operators are utilized to produce offspring solutions using the parent solutions. The population is then updated with offspring solutions. This process is repeatedly performed until convergence criteria are satisfied. Finally, the optimum rotor design is obtained as a solution that corresponds to the maximum fitness value.

### **5.3.6 FITNESS FUNCTION FORMULATION AND GENETIC OPERATIONS**

The rotor dataset is loaded to define the bounds of each design variable. The bounds of the design variables are set to be within the physically valid space. Each candidate solution is considered to be a complete geometry of the rotor, represented as a vector of geometric parameters. The fitness function is created using the RF-predicted average torque and torque ripple values. The objective is to maximize the average torque while heavily penalizing the average torque ripple. A hard constraint is applied to heavily penalize solutions that exceed a certain ripple value.

Random candidate solutions within the bounds of the design variables are created to generate the initial population. Tournament selection is applied to select the solutions for crossover using the fitness values of the solutions. The parameters of the solutions are averaged to implement crossover. Mutation is applied to the solutions with a fixed probability.

### **5.3.7 EVOLUTIONARY PROCESS AND OPTIMAL DESIGN EXTRACTION**

The Genetic Algorithm is run for a specified number of generations. During this time, the population is updated by selection, crossover, and mutation. The best fitness value and solution are monitored at all times to ensure convergence. Once the optimization process is complete, the best fitness value and corresponding solution represent the optimal solution. The trained Random Forest model is then used to predict the average torque and torque ripple of the optimized rotor design.

### **5.3.8 OPTIMIZED ROTOR DESIGN RESULTS**

The Genetic Algorithm integrated with Random Forest surrogate models successfully converged to an optimized rotor geometry that achieves a balance between high average torque and low torque ripple. The optimal design parameters obtained at the end of the evolutionary process are summarized as follows: the barrier height is 1.8528 mm, the barrier width is 2.2893 mm, the fillet radius is 1.0173 mm, the bottom barrier radius is 22.4368 mm, the barrier offset is 4.5854 mm, and the number of barriers per pole is three.

Using the trained Random Forest models, the performance of the optimized rotor design was predicted. The optimized configuration yields a predicted average torque of 18.59 Nm, indicating effective torque production. Simultaneously, the predicted torque ripple is limited to 5.12, which lies close to the lower bound enforced in the fitness function, demonstrating successful ripple minimization.

These results confirm that the GA-based optimization framework effectively navigated the design space to identify a rotor geometry that satisfies the dual objectives of torque maximization and ripple reduction. The obtained optimized design validates the suitability of combining machine learning surrogate models with evolutionary optimization for efficient and reliable rotor design optimization.

Table 5.5. Optimised Rotor Design Parameters

| Parameter                   | Symbol | Optimized Value | Unit |
|-----------------------------|--------|-----------------|------|
| Barrier Height              | H_mm   | 1.8528          | mm   |
| Barrier Width               | W_mm   | 2.2893          | mm   |
| Fillet Radius               | R_mm   | 1.0173          | mm   |
| Bottom Barrier Radius       | Rb_mm  | 22.4368         | mm   |
| Barrier Offset              | B0_mm  | 4.5854          | mm   |
| Number of Barriers per Pole | —      | 3               | —    |

### 5.3.9 VALIDATION OF OPTIMIZED ROTOR DESIGN USING TIME-DOMAIN SIMULATION

The optimized rotor geometric parameters obtained from the GA–Random Forest framework were further validated using detailed electromagnetic time-domain simulation. The torque waveform obtained from the steady-state simulation was analyzed to compute average torque,

RMS torque, torque deviation, and torque ripple using standard definitions.

The average torque ( $T_{avg}$ ) was calculated as the mean value of the instantaneous torque samples over one steady-state cycle. The RMS torque ( $T_{rms}$ ) was computed to represent the effective torque value, while the torque deviation ( $\sigma$ ) was obtained as the standard deviation of torque about its average value. Torque ripple was evaluated as the ratio of torque deviation to average torque and expressed as a percentage.

Based on the simulation results, the average torque was found to be 17.96 Nm, while the RMS torque was 17.99 Nm, indicating minimal fluctuation around the mean value. The torque deviation was 0.94 Nm, leading to a torque ripple of 0.052 (5.2%). The torque waveform confirms stable steady-state operation with low ripple content after the transient region.

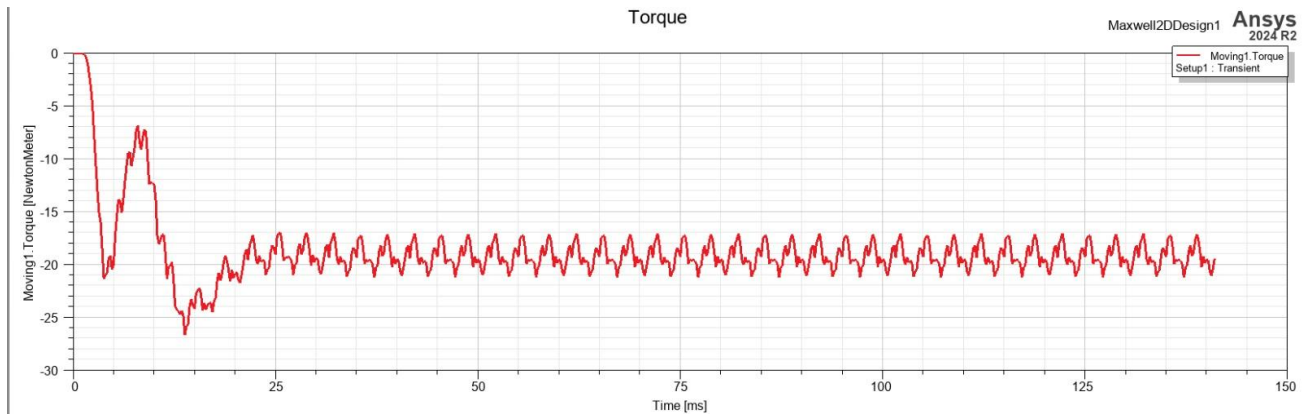


Figure 5.3. Torque vs Time graph for Optimized Motor Model

### 5.3.10 TORQUE RIPPLE CALCULATION:

$$T_{rms} = \sqrt{\frac{1}{N} \sum T_i^2} \quad (5.2)$$

From Dataset:

Steady state starts at:  $t \geq 0.025$  sec



From Equations 4.1,4.2,4.3 and 5.1

$$T_{avg} = -17.96 \text{ Nm}$$

$$T_{rms} = 17.99 \text{ Nm}$$

$$T_{rms\_dev} = 0.94 \text{ Nmz}$$

$$\text{Torque Ripple} = 0.94/17.96 = 0.052$$

$$\text{Torque Ripple \%} = 5.2\%$$

### 5.3.11 COMPARISON BETWEEN RANDOM FOREST PREDICTION AND SIMULATION RESULTS

The Random Forest surrogate models predicted an average torque of 18.589 Nm and a torque ripple of 5.12% for the optimized rotor geometry. When compared with the simulation results, a close agreement is observed, validating the accuracy of the machine learning–based prediction framework.

Table 5.6. Comparison Between Random Forest Prediction and FEM Based Simulation Results

| Performance Metric  | Random Forest Prediction | Simulation Result |
|---------------------|--------------------------|-------------------|
| Average Torque (Nm) | 18.59                    | 17.96             |
| Torque Ripple (%)   | 5.12                     | 5.20              |

The performance of the optimized rotor design obtained through the GA–Random Forest framework was validated using ANSYS Maxwell 2D simulation, and a close agreement was observed between the predicted and simulated results. The Random Forest model predicted an average torque of 18.59 Nm, while the Maxwell 2D simulation produced an average torque of 17.96 Nm, indicating a small deviation that can be attributed to electromagnetic nonlinearities and numerical discretization inherent in finite-element analysis. Similarly, the predicted torque ripple of 5.12% closely matches the simulated torque ripple of 5.2%, confirming the accuracy of the surrogate prediction model in capturing torque fluctuation characteristics.

The close correlation between the machine learning predictions and the detailed electromagnetic

simulation validates the effectiveness of the proposed GA–RF optimization approach. The results demonstrate that Random Forest models can reliably estimate torque performance during the optimization stage, significantly reducing dependence on repeated computationally expensive simulations. Overall, the comparison confirms that the optimized rotor design achieves both high torque output and low torque ripple, making the proposed framework suitable for efficient and accurate synchronous reluctance motor rotor design optimization.

## **5.4 SUMMARY**

In this section, the optimization and validation of the Synchronous Reluctance Motor (SynRM) Rotor Design have been presented using a hybrid of a Random Forest (RF) algorithm and a Genetic Algorithm (GA). The RF algorithm was first employed for an accurate prediction of the important performance parameters of the motor, which include the average torque and torque ripple. The importance of each feature was also analyzed, which helped identify the important design parameters of the motor. The RF algorithm was then used as a performance prediction tool, and a Genetic Algorithm was employed to find the optimal design of the motor that maximizes torque while minimizing torque ripple. The optimized design of the motor was validated using time-domain finite element analysis, which was performed using the ANSYS Maxwell software. The results of the RF algorithm were compared with the simulation results, which showed a close correlation between them, thus proving the reliability of the proposed algorithm for the design of the Synchronous Reluctance Motor Rotor.

## **Chapter 6**

# **VALIDATION OF OPTIMIZED SYNCHRONOUS RELUCTANCE MOTOR USING FINITE ELEMENT SIMULATION**

## **6.1 INTRODUCTION**

In this chapter, the design of the optimized Synchronous Reluctance Motor (SynRM) with its electromagnetic performance analysis, based on the optimized design parameters of the motor's

rotor geometry obtained through the optimization process, will be discussed. The optimized design parameters of the motor's rotor geometry, as presented in the previous chapter, will be used as a basis for designing the optimized Synchronous Reluctance Motor. The design of the motor will first be created using ANSYS RMxpert, which will provide the necessary parameters of the motor's design, including its geometry, materials, and operating conditions. The design will then be imported into Maxwell 2D for simulation.

Moreover, transient electromagnetic simulation results are obtained, which provide an analysis of the dynamic and steady-state performance of the optimized motor design. Key performance parameters, including torque-time curves, phase currents, flux linkages, induced voltages, magnetic flux density contour plots, and flux line plots, are obtained from the simulation results. The magnetic flux density contour plots and flux line plots provide an analysis of the magnetic flux distribution in the stator, rotor, and air-gap regions of the optimized motor design. The simulation results provide an analysis of the effectiveness of the optimized design of the SynRM motor for electric vehicle applications..

## **6.2 RMXPERT DESIGN OF OPTIMIZED SYNCHRONOUS RELUCTANCE MOTOR.**

Using the optimized rotor parameters obtained from the Genetic Algorithm–Random Forest (GA–RF) optimization process, the optimized Synchronous Reluctance Motor (SynRM) is developed in ANSYS RMxpert (figure 6.1). The optimized rotor geometry parameters are incorporated into the motor design to construct the improved rotor structure. RMxpert is used to define the motor dimensions, material properties, and operating conditions, allowing the performance characteristics of the optimized SynRM to be evaluated. This model represents the finalized motor configuration obtained after the optimization process.

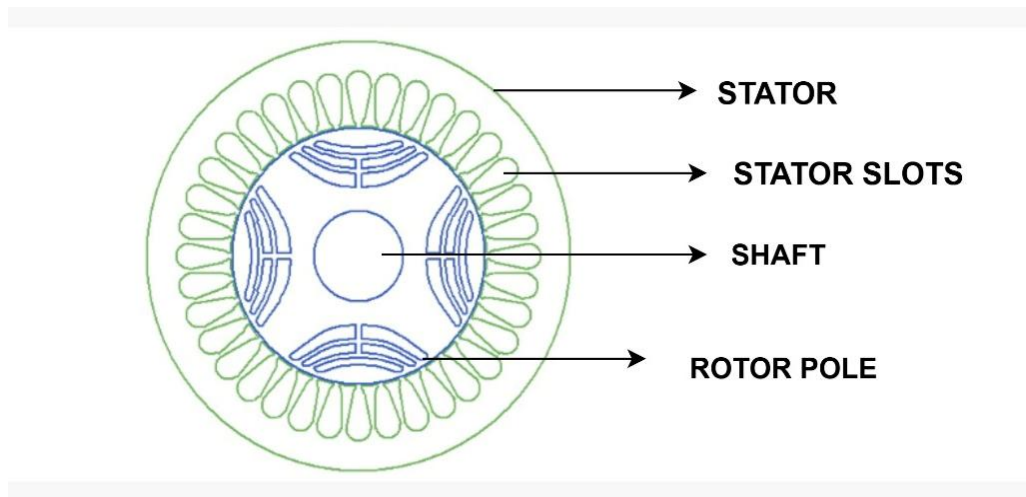


Figure 6.1 Rmxprt Design of Optimized SynRM Motor

Table 6.1 Rotor parameters of SynRM Motor

| Name            | Value   | Unit | Evaluated Value | Description                      |
|-----------------|---------|------|-----------------|----------------------------------|
| Outer Diameter  | 84      | mm   | 84              | Outer diameter of the iron core  |
| Inner Diameter  | 30      | mm   | 30              | Inner diameter of the iron core  |
| Length          | 90      | mm   | 90              | Length of the iron core          |
| Stacking Factor | 0.95    |      |                 | Stacking factor of the iron core |
| Steel Type      | M19_24G |      |                 | Steel type of the iron core      |
| Pole Type       | 4       |      |                 | Pole type of the iron core       |

The rotor parameters used for the optimized Synchronous Reluctance Motor are listed in Table 6.1. These parameters define the basic rotor geometry of the machine, including the outer diameter, inner diameter, and core length. The stacking factor and steel type are also specified to represent the magnetic material properties of the rotor core. These parameters are defined in ANSYS RMXprt to establish the structural configuration of the rotor before performing electromagnetic analysis.

Table 6.2 Stator properties of SynRM Motor

|                       | <b>Value</b> | <b>Unit</b> | <b>Evaluation</b> | <b>Description</b>                                      |
|-----------------------|--------------|-------------|-------------------|---------------------------------------------------------|
| Outer Diameter        | 140          | mm          | 140 mm            | Outer diameter of the iron core                         |
| Inner Diameter        | 85           | mm          | 85 mm             | Inner diameter of iron core                             |
| Length                | 90           | mm          | 90 mm             | Length of the iron core                                 |
| Stacking Factor       | 0.95         |             |                   | Stacking factor of the iron core                        |
| Steel Type            | M19_24G      |             |                   | Steel type of the iron core                             |
| Press Board Thickness | 0            | mm          |                   | One side thickness of the press boards at two core ends |
| Magnetic Press Board  |              |             |                   | Pole press board is made of magnetic material           |
| Skew Width            | 0            | deg         | 0 deg             | Skew width measured in degrees                          |

The stator design parameters of the optimized SynRM are presented in Table 6.2. These parameters include the outer diameter, inner diameter, and core length of the stator along with the stacking factor and electrical steel type used for the stator core. Additional parameters such as press board thickness, magnetic press board, and skew width are also specified to accurately represent the stator construction in the RMxpert model. These parameters ensure proper electromagnetic interaction between the stator windings and the rotor magnetic circuit.

Table 6.3. Optimised Rotor Design Parameters

| Parameter                      | Symbol | Optimized Value | Unit |
|--------------------------------|--------|-----------------|------|
| Barrier Height                 | H_mm   | 1.8528          | mm   |
| Barrier Width                  | W_mm   | 2.2893          | mm   |
| Fillet Radius                  | R_mm   | 1.0173          | mm   |
| Bottom Barrier Radius          | Rb_mm  | 22.4368         | mm   |
| Barrier Offset                 | B0_mm  | 4.5854          | mm   |
| Number of Barriers<br>per Pole | —      | 3               | —    |

The optimized rotor geometry parameters obtained from the RF–GA optimization process are summarized in Table 6.3. These parameters include barrier height, barrier width, fillet radius, bottom barrier radius, barrier offset, and the number of barriers per pole. These variables directly influence the magnetic flux path and the reluctance difference between the direct axis (d-axis) and quadrature axis (q-axis), which is responsible for torque production in the Synchronous Reluctance Motor. The optimized values obtained from the optimization process are incorporated into the motor design to improve average torque while reducing torque ripple.

### 6.3 2D-MAXWELL DESIGN OF OPTIMIZED SYNRM MOTOR

The optimized Synchronous Reluctance Motor (SynRM) model created in ANSYS RMxpert is exported to ANSYS Maxwell, which is used to perform in-depth electromagnetic analysis using the finite element method (FEM). The Maxwell model includes a cross-sectional representation of the motor, which includes the stator core, stator windings, rotor core, air gap, and flux barriers on the rotor. The properties of the materials, dimensions, and working conditions specified in the analytical model created in ANSYS RMxpert are automatically imported into the Maxwell model.

The transient electromagnetic analysis is carried out on the optimized motor model created in Maxwell. The performance characteristics, such as electromagnetic torque, phase currents, flux linkages, and

magnetic flux density, can be calculated using the finite element simulation. The performance characteristics calculated using the finite element simulation can be more accurately determined compared to analytical calculations, as it takes into account the magnetic saturation of the motor.

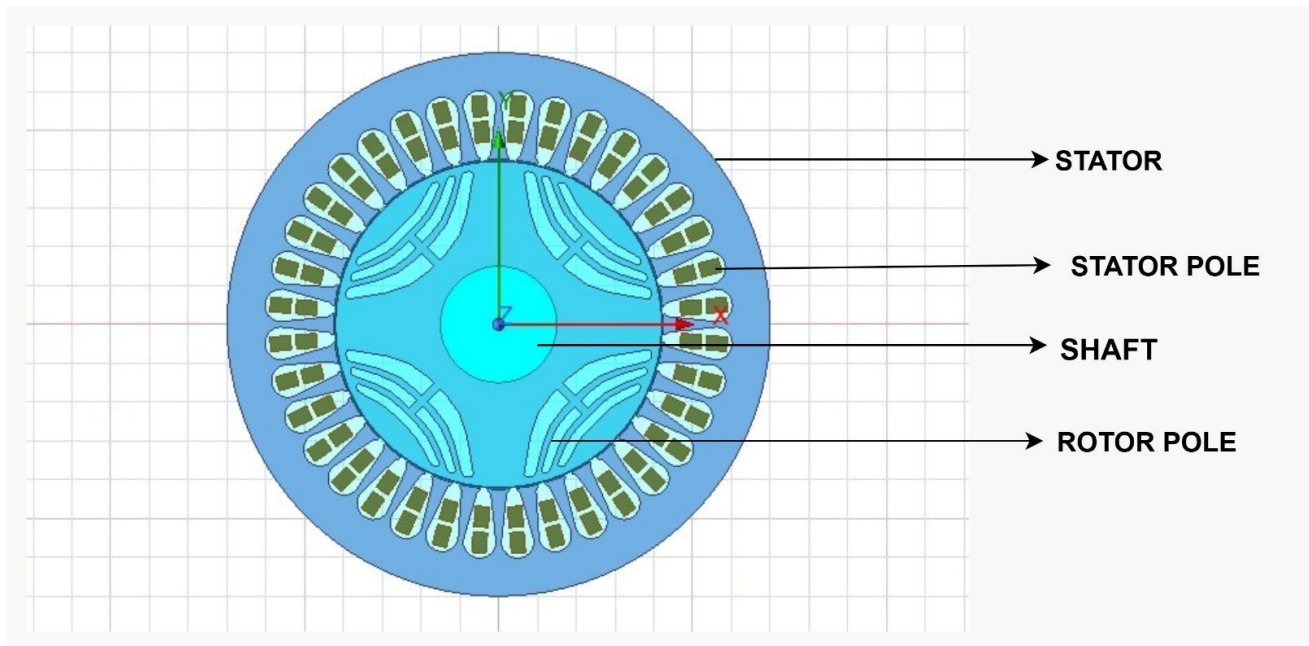


Figure 6.2 2D-Maxwell Design of Synrm Motor

### 6.3.1 ANALYSIS AND RESULTS:

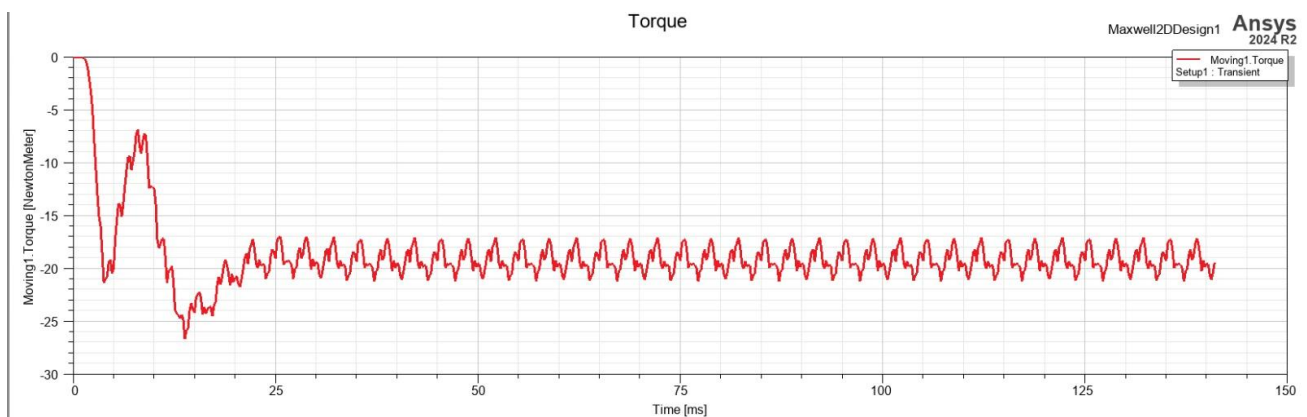


Figure 6.3 Torque vs Time graph

The The waveform of the electromagnetic torque of the optimized SynRM during the transient simulation process is depicted in the "Torque vs. Time" graph. At the initial stages of the motor's operation, there are oscillations of the electromagnetic torque due to the creation of the magnetic field and the interaction of the rotating stator's magnetic field with the rotor's magnetic circuit. As the motor reaches the steady-state period of operation, the waveform of the electromagnetic

torque becomes smooth, oscillating within a specific range of the average torque values. After the initial period of the motor's operation, the oscillations of the electromagnetic torque are relatively minor.

The smaller change in the magnitude of the torque fluctuation proves the efficiency of the optimized geometry of the rotor in reducing the torque ripple. The stable performance of the electromagnetic torque proves that the optimized SynRM configuration is efficient in producing electromagnetic torque for operation.

## **6.4 Optimized Synchronous Reluctance Motor (SynRM) Magnetic Field Analysis**

### **6.4.1 MAGNETIC FLUX DENSITY DISTRIBUTION**

The optimized Synchronous Reluctance Motor (SynRM) design shown in Fig. 6.7 was analyzed using ANSYS Maxwell to evaluate improvements in magnetic flux distribution. The MAG contour plot illustrates the distribution of magnetic flux density within the stator, rotor, and air-gap regions. Simulation results indicate that the highest magnetic flux density is mainly concentrated in the stator teeth and stator yoke, which act as the primary magnetic paths. Compared with the initial SynRM design, the optimized rotor geometry improves the guidance of magnetic flux through the rotor iron paths.

The rotor includes redesigned flux barriers that increase the difference in magnetic reluctance between the direct axis (d-axis) and quadrature axis (q-axis). These barriers restrict flux flow through high-reluctance regions while directing the magnetic field through the rotor ribs. This improved flux guidance enhances the ability of the motor to produce reluctance torque. Furthermore, the air-gap region shows a more uniform magnetic flux density distribution compared with the initial design, indicating improved magnetic interaction between the stator and rotor.



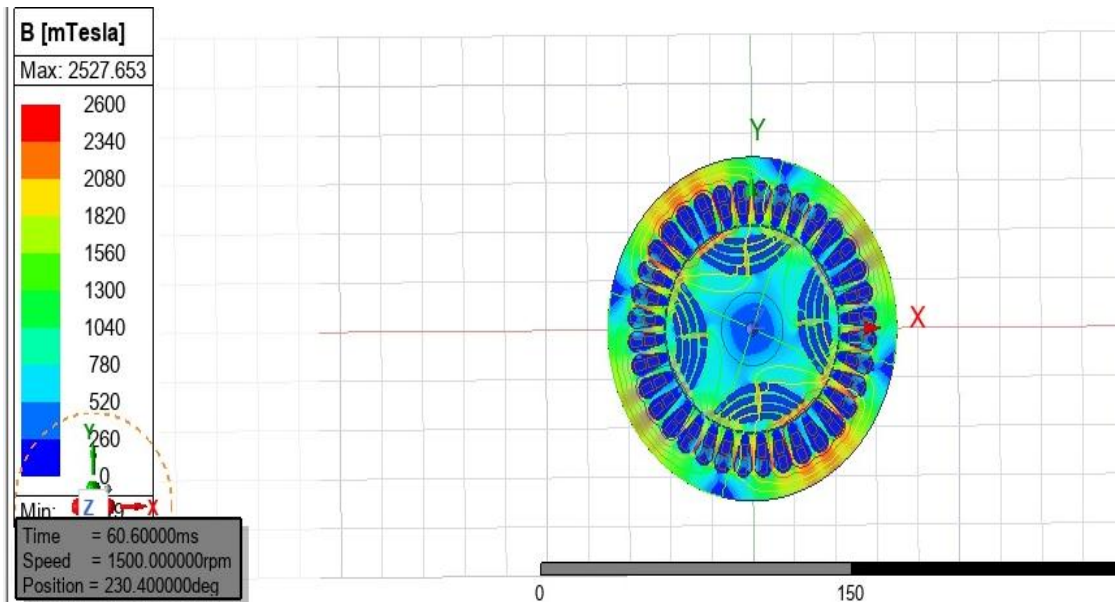


Figure 6.4 Magnetic Flux Density Distribution plot for Optimized SynRM

#### 6.4.2 MAGNETIC FLUX LINE DISTRIBUTION

The magnetic flux line plot in Figure 6.8 illustrates the magnetic vector potential distribution in the motor structure. The magnetic flux lines are seen to originate from the stator windings, traverse the air gap, and then penetrate into the rotor iron paths before finally returning to the stator yoke. In the optimized motor structure, it can be seen that the magnetic flux lines are more smooth and better defined in the rotor ribs. This is due to the better geometry of the rotor structure, which reduces leakage in the motor and ensures that the magnetic flux is mainly confined to those paths where reluctance is low.

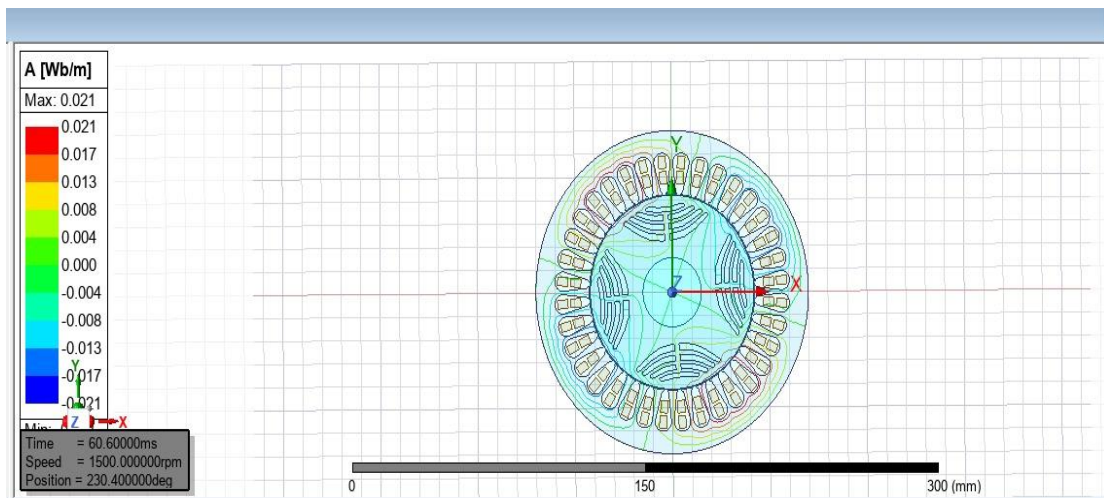


Figure 6.5 Magnetic Flux Line Distribution plot for Optimized SynRM

### **6.4.3 AIR GAP FLUX CHARACTERISTICS**

The region where interaction occurs between the rotating magnetic field of the stator and the rotor structure is called the air gap. In the optimized SynRM motor structure, it can be noticed that the structure of the air gap flux distribution is more uniform compared to the initial motor structure. A more uniform structure in the air gap flux distribution helps to produce smooth electromagnetic forces while minimizing torque pulsations in the motor.

### **6.4.4 OBSERVATIONS**

From the electromagnetic analysis of the optimized synchronous reluctance motor, it can be observed that the flux barrier design in the rotor flux effectively directs the magnetic flux in the rotor ribs. This ensures a controlled magnetic path. The magnetic flux leakage in the cavities of the rotor is minimized compared to the initial design. This shows that the magnetic flux is well utilized. The flux distribution in the air gap is also more uniform. This would ensure improved interaction between the stator and the rotor. The difference between direct axis and quadrature axis reluctance is also maximized. This would ensure improved production of reluctance torque. All these observations show that the optimized synchronous reluctance motor would perform well.

## **6.5 SUMMARY**

In this section, the optimized Synchronous Reluctance Motor (SynRM) design will be created using ANSYS RMxprt. In this design, the motor's geometry, material properties, and operating conditions will be specified based on the optimized rotor design parameters. After creating the design, the created design will be imported into ANSYS Maxwell. Using Maxwell, the electromagnetic performance of the optimized SynRM design will be simulated. The simulation will be performed to assess the magnetic flux density distribution and behavior of the optimized SynRM design, particularly within the stator, rotor, and air gap.

It is observed from the magnetic field analysis of the optimized SynRM design that the flux barriers in the rotor effectively enable the flow of the magnetic flux in the rotor ribs while controlling the flow in the high reluctance areas. This enhances the saliency between the d-axis and the q-axis, which is utilized to produce the reluctance torque in the motor. The magnetic flux density is mainly concentrated in the teeth and yoke of the stator, which is the major component in the magnetic circuit.

In addition to that, the optimized rotor design minimizes flux leakage through the rotor cavities. It also optimizes the flux distribution in the air gap. This helps to facilitate the interaction between the stator's rotating magnetic field and the electromagnetic structure of the rotor. This interaction enables the production of torque capabilities that are more efficient for the motor. Based on the above-mentioned aspects, the optimized SynRM design proves to be efficient in the utilization of the magnetic circuit.

## 7. CONCLUSION

In this project, a complete electromagnetic performance analysis of three different electric motors, namely Induction Motor (IM), Brushless DC Motor (BLDC), and Synchronous Reluctance Motor (SynRM), has been performed using ANSYS RMxprt and ANSYS Maxwell. The transient finite element analysis has been employed to analyze the performance parameters of the motors, including average torque, torque ripple, flux linkages, induced voltage, and stator current. The performance analysis showed that the Induction Motor has a torque ripple of 4.57%, the Brushless DC Motor has a torque ripple of 15.34%, and the initial performance of the Synchronous Reluctance Motor has a torque ripple of 7.5%. The Synchronous Reluctance Motor, among the three motors, has a promising potential due to its magnet-free rotor structure.

To further improve the SynRM's performance, an optimization tool, which is a combination of machine learning and evolutionary optimization, has been developed. The design configurations of the SynRM's rotor have been systematically sampled by Latin Hypercube Sampling (LHS), and each design configuration has been analyzed by finite element analysis in ANSYS Maxwell. The sampled dataset has been utilized to train a Random Forest regression model, which is able to predict the important motor performance characteristics, like average torque and torque ripple. The trained model has been linked with a Genetic Algorithm (GA) to perform multi-objective optimization on the design parameters of the SynRM's rotor.

The optimization algorithm successfully identified a new and better rotor structure, which further enhanced the electromagnetic characteristics of the SynRM. The optimized SynRM produced an average torque of 17.96 Nm, along with a torque ripple of 5.2%, whereas the RF algorithm predicted a torque ripple of 5.12%. The close prediction of the RF algorithm further validates the reliability of the proposed optimization algorithm. The optimized SynRM not only produced better torque characteristics but also ensured a reduced torque ripple, thus making it a viable and cost-effective solution in the context of contemporary electric vehicles.

# REFERENCES

- [1] P. Khude *et al.*, “Comparative analysis of 5 H.P SynRM and FeA-SynRM for EV applications,” *IEEE Trans. Ind. Appl.*, 2024.
- [2] H. Kwon and J. Lim, “Combined Random Forest and Genetic Algorithm for optimal design of PMA-SynRM for electric vehicles,” *IEEE Access*, 2023.
- [3] A. Gore and P. Ugale, “Design and comparative analysis of PMSM, BLDC, SynRM, and PMA-SynRM motors for two-wheeler EV application,” in *Proc. IEEE Int. Transp. Electrific. Conf. (ITEC)*, 2022.
- [4] K. Jędryczka *et al.*, “A design on rotor flux barrier for line start SynRM for enhancing efficiency,” *Eksploracja i Niezawodność*, 2022.
- [5] A. Sheikh *et al.*, “Performance comparison of FOC and DTC techniques for SynRM in EV applications,” in *Proc. IEEE Int. Conf. Electr. Mach. Syst. (ICEMS)*, 2025.
- [6] M. Abou-ElSoud *et al.*, “SynRM torque ripple reduction using feedback cascaded P-I-I controller,” in *Proc. IEEE Middle East Power Conf. (MEPCON)*, 2022.
- [7] J. Park *et al.*, “Design and development of SynRM for EV with emphasis on high torque density,” *IEEE Trans. Energy Convers.*, 2021.
- [8] S. Lee and H. Kim, “Multi-objective optimization of SynRM for low noise and high efficiency in EVs,” *IEEE Trans. Magn.*, 2023.
- [9] Y. Zhao *et al.*, “Loss minimization in SynRM using flux control strategy,” in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, 2022.
- [10] T. Nguyen *et al.*, “Comparative study of PMSM and SynRM drives in compact electric vehicles,” *IEEE Access*, 2024.
- [11] A. Yildiz and M. Polat, “Analysis of synchronous reluctance motor using ANSYS Maxwell,” 2023.
- [12] J. Zhang, F. Xing, L. Kang, and C. Qin, “Design and multi-objective optimization for improving torque performance of a permanent magnet-assisted synchronous reluctance motor,” *Appl. Sci.*, vol. 14, no. 12, p. 5253, Jun. 2024.
- [13] F. Xing, J. Zhang, F. Zuo, and Y. Gao, “Optimization of an asymmetric-rotor permanent magnet-assisted synchronous reluctance motor for improved anti-demagnetization performance,” *Appl. Sci.*, vol. 14, no. 23, p. 11233, Dec. 2024.
- [14] X. Li, Y. Chen, and W. Zhao, “Multi-physics and multi-objective design of an axial flux permanent magnet-assisted synchronous reluctance motor for electric vehicle applications,” *Machines*, vol. 13, no. 7, p. 555, Jul. 2025.
- [15] F. Xing, J. Zhang, M. Zhang, and C. Qin, “Design and multi-objective optimization of an asymmetric-rotor permanent-magnet-assisted synchronous reluctance motor for improved torque performance,” *Appl. Sci.*, vol. 14, no. 15, p. 6734, Aug. 2024.
- [16] A. A. Arkadan and N. Al-Aawar, “Electromagnetic–AI-based design optimization of SynRM drives,” *ACES J.*, vol. 36, no. 11, pp. 1407–1411, Dec. 2021.
- [17] V. Rajini, V. S. Nagarajan, K. Harikrishnan, and M. L. Kolhe, “Electromagnetic design, sensitivity analysis, optimization and multiphysics capability of rare-earth-free synchronous reluctance motor for electric trike vehicle,” *AIMS Energy*, vol. 12, no. 5, pp. 1027–1053, 2024.
- [18] S. Lee, J. Park, and H. Kim, “Design optimization of ferrite-assisted synchronous reluctance motor using multi-objective differential evolution algorithm,” *COMPEL: Int. J. Comput. Math. Electr. Electron. Eng.*, vol. 39, no. 5, pp. 1289–1301, 2023.
- [19] Y. Zhao, M. Chen, and K. Liu, “Response surface method for optimization of synchronous reluctance motor rotor,” *Appl. Sci.*, vol. 10, no. 10, p. 897, 2023.
- [20] P. Colombo, G. Pellegrino, and F. Cupertino, “Synchronous reluctance machine

geometry optimization through a genetic algorithm-based technique,” *IET Electr. Power Appl.*, vol. 12, no. 4, pp. 495–503, 2023.

# ev mtr

*by* Resmi R

---

**Submission date:** 23-Mar-2026 03:38PM (UTC+0530)

**Submission ID:** 2910589177

**File name:** plagiarism.docx (6.32M)

**Word count:** 16756

**Character count:** 98881

## ORIGINALITY REPORT

8%

SIMILARITY INDEX

6%

INTERNET SOURCES

8%

PUBLICATIONS

3%

STUDENT PAPERS

## PRIMARY SOURCES

|   |                                                                                                                                                                      |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1 | <a href="http://www.mdpi.com">www.mdpi.com</a><br>Internet Source                                                                                                    | 1%  |
| 2 | <a href="http://www.jmag-international.com">www.jmag-international.com</a><br>Internet Source                                                                        | <1% |
| 3 | <a href="http://repository.tudelft.nl">repository.tudelft.nl</a><br>Internet Source                                                                                  | <1% |
| 4 | Juha Pyrhönen, Tapani Jokinen, Valéria Hrabovcová. "Design of Rotating Electrical Machines", Wiley, 2013<br>Publication                                              | <1% |
| 5 | Jacek F. Gieras. "Permanent Magnet Motor Technology - Design and Applications, Third Edition", CRC Press, 2019<br>Publication                                        | <1% |
| 6 | <a href="http://www.scribd.com">www.scribd.com</a><br>Internet Source                                                                                                | <1% |
| 7 | Berker Bilgin, James Weisheng Jiang, Ali Emadi. "Switched Reluctance Motor Drives - Fundamentals to Applications", CRC Press, 2019<br>Publication                    | <1% |
| 8 | <a href="http://hdl.handle.net">hdl.handle.net</a><br>Internet Source                                                                                                | <1% |
| 9 | Kovvuri Akhil Reddy, R. Manikandan, S. Ravivarman, Hayitov Abdulla Nurmatovich, Bekzod Madaminov, K. Karthick. "Intelligent Control of BLDC Motors Using AdaptivePID | <1% |



|    |                                                                                                                                                                                                                                     |      |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 10 | <a href="http://www.science.gov">www.science.gov</a><br>Internet Source                                                                                                                                                             | <1 % |
| 11 | Malamas, Ilias. "Μηχανική Μάθηση Σε Ρηχά Νερά: Επίλυση Των Εξισώσεων Ρηχών Υδάτων Χρησιμοποιώντας Φυσιογνωστικά Νευρωνικά Δίκτυα", Technical University of Crete (Greece)<br>Publication                                            | <1 % |
| 12 | <a href="http://www.bartleby.com">www.bartleby.com</a><br>Internet Source                                                                                                                                                           | <1 % |
| 13 | Maedeh Sadat Mirazimi, Arash Kiyoumarsi. "Magnetic Field Analysis of SynRel and PMASynRel Machines With Hyperbolic Flux Barriers Using Conformal Mapping", IEEE Transactions on Transportation Electrification, 2020<br>Publication | <1 % |
| 14 | <a href="http://journals.riverpublishers.com">journals.riverpublishers.com</a><br>Internet Source                                                                                                                                   | <1 % |
| 15 | <a href="http://pdffox.com">pdffox.com</a><br>Internet Source                                                                                                                                                                       | <1 % |
| 16 | <a href="http://www.dejazzzer.com">www.dejazzzer.com</a><br>Internet Source                                                                                                                                                         | <1 % |
| 17 | T. Matsuo, T.A. Lipo. "Rotor design optimization of synchronous reluctance machine", IEEE Transactions on Energy Conversion, 1994<br>Publication                                                                                    | <1 % |
| 18 | K. V. Sambasivarao, Anasuya Sesha Roopa Devi Bhima. "Artificial Intelligence, Computational Intelligence, and Inclusive                                                                                                             | <1 % |

Technologies - Proceedings of International Conference on Artificial Intelligence, Computational Intelligence, and Inclusive Technologies (ICRAIC2IT – 2025)", CRC Press, 2026

Publication

|    |                                                                                                                                                                                                             |      |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 19 | www.emeraldinsight.com<br>Internet Source                                                                                                                                                                   | <1 % |
| 20 | "The Proceedings of the 12th Frontier Academic Forum of Electrical Engineering (FAFEE2025)", Springer Science and Business Media LLC, 2026<br>Publication                                                   | <1 % |
| 21 | Submitted to AlHussein Technical University<br>Student Paper                                                                                                                                                | <1 % |
| 22 | Submitted to Anna University<br>Student Paper                                                                                                                                                               | <1 % |
| 23 | Submitted to University of Hertfordshire<br>Student Paper                                                                                                                                                   | <1 % |
| 24 | Submitted to Nottingham Trent University<br>Student Paper                                                                                                                                                   | <1 % |
| 25 | Wei Tong. "Mechanical Design and Manufacturing of Electric Motors", CRC Press, 2022<br>Publication                                                                                                          | <1 % |
| 26 | publish.kne-publishing.com<br>Internet Source                                                                                                                                                               | <1 % |
| 27 | Aswin Uvaraj Ganesan, Lenin Natesan Chokkalingam. "Influence of rotor cage resistance in torque ripple reduction for line start synchronous machines", IET Electric Power Applications, 2019<br>Publication | <1 % |

|    |                                                                                                                                                                                                                                 |      |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 28 | Frédéric, Cawez. "Development and Characterization of Camelid Antibody Fragments (Nanobodies) as Theranostic Agents Against the $\beta$ -Lactamases TEM-1, NDM-1 and CMY-2", Université de Liège (Belgium), 2024<br>Publication | <1 % |
| 29 | Submitted to University of SWAT<br>Student Paper                                                                                                                                                                                | <1 % |
| 30 | <a href="http://www.1.ise-iq.org">www.1.ise-iq.org</a><br>Internet Source                                                                                                                                                       | <1 % |
| 31 | <a href="http://www.preprints.org">www.preprints.org</a><br>Internet Source                                                                                                                                                     | <1 % |
| 32 | Aigbe, Uyiosa Osagie. "Groundwater Purification Using Functionalised Magnetic Nanoparticles (Electromagnetic Separation)", University of South Africa (South Africa)<br>Publication                                             | <1 % |
| 33 | Nijun Li, Xu Cheng, Haiyan Guo, Zhenyang Wu. "Recognizing human interactions by genetic algorithm-based random forest spatio-temporal correlation", Pattern Analysis and Applications, 2015<br>Publication                      | <1 % |
| 34 | Submitted to SASTRA University<br>Student Paper                                                                                                                                                                                 | <1 % |
| 35 | <a href="http://theses.lib.polyu.edu.hk">theses.lib.polyu.edu.hk</a><br>Internet Source                                                                                                                                         | <1 % |
| 36 | <a href="http://vbook.pub">vbook.pub</a><br>Internet Source                                                                                                                                                                     | <1 % |
| 37 | Huthaifa Mohammad Flieh, Timothy Slininger, Robert D. Lorenz, Shao-Chuan Chien, Li-Hsing Ku. "Flux Weakening Surface Mounted Permanent Magnet Servo Motor Design with                                                           | <1 % |

Enhanced Self-Sensing Properties", 2019 IEEE  
Energy Conversion Congress and Exposition  
(ECCE), 2019  
Publication

|    |                                                                                                                                                                                                                                 |      |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 38 | businessdocbox.com<br>Internet Source                                                                                                                                                                                           | <1 % |
| 39 | ebin.pub<br>Internet Source                                                                                                                                                                                                     | <1 % |
| 40 | Submitted to International Islamic University<br>Malaysia<br>Student Paper                                                                                                                                                      | <1 % |
| 41 | Paulraj T, Yeddula Pedda Obulesu.<br>"Optimization of Torque Performance in<br>SynRM with Strategic Placement of NdFeB<br>Permanent Magnets and Novel Flux Barrier<br>Shape", Engineering Research Express, 2025<br>Publication | <1 % |
| 42 | Submitted to Universitas Brawijaya<br>Student Paper                                                                                                                                                                             | <1 % |
| 43 | Submitted to University of Edinburgh<br>Student Paper                                                                                                                                                                           | <1 % |
| 44 | lib.buet.ac.bd:8080<br>Internet Source                                                                                                                                                                                          | <1 % |
| 45 | "The Proceedings of the 11th Frontier<br>Academic Forum of Electrical Engineering<br>(FAFEE2024)", Springer Science and Business<br>Media LLC, 2025<br>Publication                                                              | <1 % |
| 46 | Submitted to Amrita Vishwa Vidyapeetham<br>Student Paper                                                                                                                                                                        | <1 % |
| 47 | Leandro Augusto Grandin Pereira, Jurandir<br>Itizo Yanagihara. "Sensitivity analysis and<br>optimization to reduce dry weight and<br>footprint of FPSO processing plants in a high                                              | <1 % |

CO2 oil field", Computers & Chemical  
Engineering, 2022

Publication

48

Nicolae-Florin Jurca, Razvan-Alexandru Inte,  
Dan-Cristian Popa, Bogdan Varaticeanu, Paul  
Minciunescu, Claudia Martis. "Electromagnetic  
and Mechanical Analysis of a Modular Outer  
Rotor Synchronous Reluctance Machine for  
Light Propulsion Vehicles", IEEE Transactions  
on Transportation Electrification, 2021

Publication

<1 %

49

Submitted to University of Technology,  
Sydney

Student Paper

<1 %

50

Submitted to University of Western Sydney

Student Paper

<1 %

51

Ye, Chong. "Mechanical Characterization and  
Use-Condition Reliability Assessment of  
Wearable Textile Electronics.", Georgia  
Institute of Technology

Publication

<1 %

Exclude quotes

On

Exclude matches

< 14 words

Exclude bibliography

On

## 24% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

### Detection Groups



**19 AI-generated only 22%**

Likely AI-generated text from a large-language model.



**2 AI-generated text that was AI-paraphrased 2%**

Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

#### Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

### Frequently Asked Questions

#### How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (\*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

#### What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



# CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS

Title of the Project: **AI And FEA Based Design and Analysis Of Efficient EV Motor**

**Project Member(s) Name & Register Number :**

Dharani Malar V – CB.EN.U4ELC22009

Karthigha S – CB.EN.U4ELC22020

**Guide Name & Designation:** Dr. Resmi S, Assistant Professor

| SDGs | Justification to the Contribution                                                                   |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | Improves EV motor efficiency using optimized synchronous reluctance motors, reducing energy losses. |
| 9    | Applies AI, ML, and FEA for advanced motor design and innovation.                                   |
| 11   | Enhances EV efficiency, reducing fossil fuel dependence and supporting sustainable transport.       |
| 12   | Uses magnet-free motors, minimizing reliance on rare-earth materials.                               |
| 13   | Reduces energy consumption and lowers greenhouse gas emissions.                                     |

